3D-Central Force Problems I

Read: C-TDL, pages 643-660 for next lecture.

Every step toward greater complexity is classical mechanics plus a tiny bit of something new.

All 2-Body, 3-D problems can be reduced to
* a 2-D angular part that is exactly and universally soluble
* a 1-D radial part that is system-specific and soluble by 1-D techniques in which you are now expert

what is it? how do we use it?

Next 3 lectures: [Correspondence Principle; Commutation Rules] ⟶ all matrix elements without actually doing any integrals

Roadmap

1. define radial momentum $\mathbf{p}_r = \mathbf{r}^{-1}\left(\mathbf{q}\cdot\mathbf{q} - i\hbar\right)$

2. define orbital angular momentum $\vec{\mathbf{L}} = \vec{\mathbf{q}} \times \vec{\mathbf{p}}$

general definition of angular momentum and of "vector operators"

$$\left(\text{also } \mathbf{L}\times\mathbf{L} = i\hbar\mathbf{L} \text{ and } \left[\mathbf{L}_i, \mathbf{L}_j\right] = i\hbar\sum_k \varepsilon_{ijk}\mathbf{L}_k\right)$$

3. separate $\mathbf{p}^2$ into radial and angular terms: $\mathbf{p}^2 = \mathbf{p}_r^2 + \mathbf{r}^{-2}\mathbf{L}^2$

4. find Complete Set of Commuting Observables (CSCO) that is useful for "block-diagonalizing" **H**

   $\left[\mathbf{H}, \mathbf{L}^2\right] = \left[\mathbf{H}, \mathbf{L}_i\right] = \left[\mathbf{L}^2, \mathbf{L}_i\right] = 0$ $\quad \mathbf{H}, \mathbf{L}^2, \mathbf{L}_i$ CSCO

   $|L, M_L\rangle$ universal basis set

5. separate radial part of **H**: $\mathbf{H}_\ell(r) = \dfrac{\mathbf{p}_r^2}{2\mu} + \underbrace{V(\mathbf{r}) + \dfrac{\hbar^2\ell(\ell+1)}{2\mu\mathbf{r}^2}}_{V_\ell(\mathbf{r})}$ effective radial potential

Recover a 1-D Schrödinger Equation

6. ALL Matrix Elements of Angular Momentum Components May be Derived from Commutation Rules.

7. Spherical Tensor Classification of **all** operators.

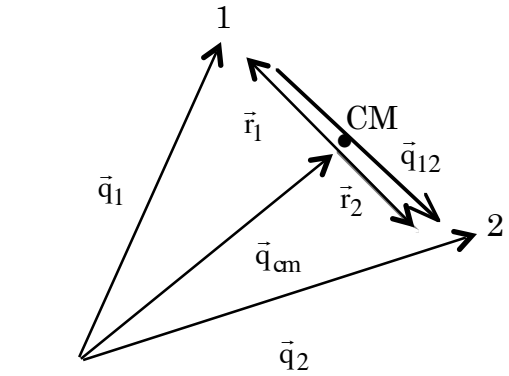
   ⇓

8. Wigner-Eckart Theorem → all angular matrix elements of all operators.

I hate differential operators. Replace them by exclusively using simple Commutation Rule based Operator Algebra.

Lots of derivations are based on classical VECTOR ANALYSIS — much of that will be set aside as NON-LECTURE

Central Force Problems: 2 bodies where interaction force is along the vector $\vec{q}_1 - \vec{q}_2$



$$\vec{q}_2 = \vec{q}_1 + \vec{q}_{12}$$

$$\vec{q}_{12} = \vec{q}_2 - \vec{q}_1 = \hat{i}(x_2 - x_1) + \hat{j}(y_2 - y_1) + \hat{k}(z_2 - z_1)$$

$$r \equiv |\vec{q}_{12}| = \left[(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2\right]^{1/2}$$

also Center of Mass (CM) Coordinate system

$$\vec{r}_1 = \vec{q}_1 - \vec{q}_{cm} \qquad \left[|r_1|/r = m_2/M\right]$$

$$\vec{r}_2 = \vec{q}_2 - \vec{q}_{cm} \qquad \left[|r_2|/r = m_1/M\right]$$

$$\mathbf{H} = \mathbf{H}_{\text{translation}} + \mathbf{H}_{\text{center of mass}}$$

$\mathbf{H}_{\text{translation}}$: free translation of C of M of system of mass $M = m_1 + m_2$

$\mathbf{H}_{\text{center of mass}}$: motion of fictitious particle of mass $\mu = \frac{m_1 m_2}{m_1 + m_2}$ in coordinate system with origin at C of M (CTDL page 713)

LAB $$\hat{\mathbf{H}}_{\text{translation}} = \frac{\mathbf{p}_{trans}^2}{2(m_1 + m_2)} + \underbrace{V}_{\text{constant}}$$

free translation of system with respect to lab (not interesting)

BODY $$\hat{\mathbf{H}}_{CM} = \frac{1}{2\mu}\mathbf{p}_{cm}^2 + \underbrace{V(r)}_{\substack{\text{free rotation} \\ (\text{no } \theta,\phi \\ \text{dependence})}}$$

motion of particle of mass μ with respect to origin at center of mass

This is $\vec{p}$ *in* CM frame, not $\vec{p}$ *of* CM

$\vec{p}_{cm}$ is a vector

GOAL IS TO SIMPLIFY $\mathbf{p}_{CM}^2$
because that is only place where the θ , ϕ degrees of freedom appear.

1. Define Radial Component of $\vec{p}_{CM}$

Correspondence Principle: recipe for going from classical to quantum mechanics

* classical mechanics
* Cartesian Coordinates
* symmetrize to avoid failure to satisfy Commutation Rules

** verify that all three derived operators, **p**, $\mathbf{p}_r$, and **L**
- are Hermitian
- satisfy [**q**,**p**] = $i\hbar$

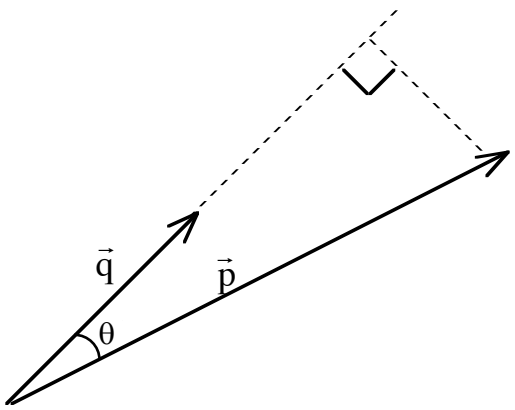
Purpose of this lecture is to walk you through the standard vector analysis and Quantum Mechanical **Correspondence Principle** procedures

$$\vec{q} \equiv \hat{i}x + \hat{j}y + \hat{k}z$$

$$\vec{p} \equiv \hat{i}p_x + \hat{j}p_y + \hat{k}p_z$$

$$r \equiv \left[x^2 + y^2 + z^2\right]^{1/2} = \left[q \cdot q\right]^{1/2} = |q|$$

find radial (i.e. along $\vec{q}$) part of $\vec{p}$



project $\vec{p}$ onto $\vec{q}$

$$q \cdot p = |q||p|\cos\theta$$

$$\cos\underbrace{(q,p)}_{\theta} = \frac{q \cdot p}{|q||p|}$$

radial component of p is obtained by projecting $\vec{p}$ onto $\vec{q}$

$$p_r = |p|\cos\theta = |p|\frac{q \cdot p}{|q||p|} = \frac{q \cdot p}{r}$$

so from standard vector analysis we get $p_r = r^{-1}\vec{q} \cdot \vec{p}$

This is a trial form for $\mathbf{p}_r$, but it is necessary, according to the Correspondence Principle recipe, to symmetrize it.

$$\mathbf{p}_r = \frac{1}{4}\left[\mathbf{r}^{-1}(\mathbf{q}\cdot\mathbf{p}+\mathbf{p}\cdot\mathbf{q})+(\mathbf{q}\cdot\mathbf{p}+\mathbf{p}\cdot\mathbf{q})\mathbf{r}^{-1}\right]$$

This expression arranges the terms in all possible orders!

This will be simplified to **almost** what one expected from CM. The only surprise must be multiplied by $\hbar$. That's QM!

NONLECTURE (except for Eq. (4))
SIMPLIFY ABOVE Definition to $\mathbf{p_r} = \mathbf{r}^{-1}(\mathbf{q}\cdot\mathbf{p} - i\hbar)$ ($\mathbf{r}$ is not a vector)

$[\vec{\mathbf{q}},\vec{\mathbf{p}}]$ is a vector commutator — be careful

$$[\vec{\mathbf{q}},\vec{\mathbf{p}}] = [\mathbf{x},\mathbf{p}_x]+[\mathbf{y},\mathbf{p}_y]+[\mathbf{z},\mathbf{p}_z] = 3i\hbar$$

$$\therefore \mathbf{p}\cdot\mathbf{q} = \mathbf{q}\cdot\mathbf{p} - [\vec{\mathbf{q}},\vec{\mathbf{p}}] \quad \text{because} [\vec{\mathbf{q}},\vec{\mathbf{p}}] = \vec{\mathbf{q}}\cdot\vec{\mathbf{p}} - \vec{\mathbf{p}}\cdot\vec{\mathbf{q}}$$

$$\mathbf{p}_r = \frac{1}{4}\left[\mathbf{r}^{-1}\left(2\mathbf{q}\cdot\mathbf{p} - \overbrace{[\vec{\mathbf{q}},\vec{\mathbf{p}}]}^{3i\hbar}\right) + \left(2\mathbf{q}\cdot\mathbf{p} - [\vec{\mathbf{q}},\vec{\mathbf{p}}]\right)\mathbf{r}^{-1}\right] \tag{1}$$

$$= \frac{1}{4}\left[\underbrace{\mathbf{r}^{-1}4\mathbf{q}\cdot\mathbf{p} - \mathbf{r}^{-1}2\mathbf{q}\cdot\mathbf{p}}_{\text{add and subtract } 2\mathbf{r}^{-1}\mathbf{q}\cdot\mathbf{p}} + 2\mathbf{q}\cdot\mathbf{p}\mathbf{r}^{-1} - 6i\hbar\mathbf{r}^{-1}\right] \tag{2}$$

$$= \mathbf{r}^{-1}\mathbf{q}\cdot\mathbf{p} - \frac{3}{2}i\hbar\mathbf{r}^{-1} + \frac{1}{2}\left[\mathbf{q}\cdot\mathbf{p},\mathbf{r}^{-1}\right] \tag{3}$$

LEMMA: need a more general Commutation Rule for which $\left[\mathbf{q}\cdot\mathbf{p},\mathbf{r}^{-1}\right]$ is a special case

1st simplify: $[\mathrm{f}(\mathbf{r}),\mathbf{q}\cdot\mathbf{p}] = \mathbf{q}\cdot[\mathrm{f}(\mathbf{r}),\vec{\mathbf{p}}] + \overbrace{[\mathrm{f}(\mathbf{r}),\vec{\mathbf{q}}]}^{0}\cdot\vec{\mathbf{p}}$

but, from 1-D, we could have shown

$$\left[f(\mathbf{x}),\mathbf{p}\right]\phi = f(\mathbf{x})\frac{\hbar}{i}\frac{\partial}{\partial \mathbf{x}}\phi - \frac{\hbar}{i}\frac{\partial}{\partial \mathbf{x}}\left(f(\mathbf{x})\phi\right)$$

$$= \frac{\hbar}{i}\left[f(\mathbf{x})\overbrace{\phi' - f'\phi - f\phi'}^{\text{cancel}}\right] = i\hbar f'(\mathbf{x})\phi$$

$$\boxed{\left[f(\mathbf{x}),\mathbf{p}\right] = i\hbar\frac{\partial \mathbf{f}}{\partial \mathbf{x}} \qquad \text{for 1-D}}$$

Thus, in 3-D, the chain rule gives, for the vector commutator:

$$\left[f(\mathbf{r}),\vec{\mathbf{p}}\right] = i\hbar\left[\hat{i}\,\frac{\partial \mathrm{f}}{\partial \mathbf{r}}\frac{\partial \mathbf{r}}{\partial \mathbf{x}} + \hat{j}\,\frac{\partial f}{\partial \mathbf{r}}\frac{\partial \mathbf{r}}{\partial \mathbf{y}} + \hat{k}\,\frac{\partial f}{\partial \mathbf{r}}\frac{\partial \mathbf{r}}{\partial \mathbf{z}}\right]$$

evaluate these first

$$\frac{\partial \mathbf{r}}{\partial \mathbf{x}} = \frac{\partial}{\partial \mathbf{x}}\left[\mathbf{x}^2+\mathbf{y}^2+\mathbf{z}^2\right]^{1/2} = \mathbf{x}\left[\mathbf{x}^2+\mathbf{y}^2+\mathbf{z}^2\right]^{-1/2} = \mathbf{x}/\mathbf{r}$$

etc. for $\frac{\partial \mathbf{r}}{\partial \mathbf{y}}$ & $\frac{\partial \mathbf{r}}{\partial \mathbf{z}}$

$$\text{Thus } \left[f(\mathbf{r}),\vec{\mathbf{p}}\right] = i\hbar\frac{\partial \mathbf{f}}{\partial \mathbf{r}}\left[\hat{i}\,\frac{\mathbf{x}}{\mathbf{r}} + \hat{j}\,\frac{\mathbf{y}}{\mathbf{r}} + \hat{k}\,\frac{\mathbf{z}}{\mathbf{r}}\right] = i\hbar\frac{\partial \mathbf{f}}{\partial \mathbf{r}}\frac{\vec{\mathbf{q}}}{\mathbf{r}}$$

$$\left[f(\mathbf{r}),\vec{\mathbf{q}}\cdot\vec{\mathbf{p}}\right] = \mathbf{q}\cdot\left[f(\mathbf{r}),\mathbf{p}\right] = i\hbar\frac{\partial \mathbf{f}}{\partial \mathbf{r}}\left(\frac{\mathbf{x}^2+\mathbf{y}^2+\mathbf{z}^2}{\mathbf{r}}\right) = i\hbar\frac{\partial \mathbf{f}}{\partial \mathbf{r}}\mathbf{r}$$

$$\boxed{\left[f(\mathbf{r}),\mathbf{q}\cdot\mathbf{p}\right] = i\hbar\frac{\partial \mathbf{f}}{\partial \mathbf{r}}\mathbf{r}}$$ this is a scalar, not a vector, equation (4)

But we wanted to evaluate the commutation rule for f(**r**) = $\mathbf{r}^{-1}$

$$\left[\mathbf{r}^{-1},\mathbf{q}\cdot\mathbf{p}\right] = i\hbar\frac{\partial}{\partial \mathrm{r}}\left(\frac{1}{\mathrm{r}}\right)\mathrm{r} = -i\hbar\mathbf{r}^{-1} \qquad (5)$$

plug this result into (3)

$$\mathbf{p}_\mathbf{r} = \mathbf{r}^{-1}\mathbf{q}\cdot\mathbf{p} - \frac{3}{2}i\hbar\mathbf{r}^{-1} + \frac{1}{2}\left(i\hbar\mathbf{r}^{-1}\right) = r^{-1}\left(q\cdot p - i\hbar\right)$$

RESUME HERE

$$\boxed{\mathbf{p}_\mathbf{r} = \mathbf{r}^{-1}\left(\mathbf{q}\cdot\mathbf{p} - i\hbar\right)} \qquad (6)$$

This is the compact but non-symmetric result we got starting with a carefully symmetrized starting point – as required by Correspondence Principle.

* This result is identical to the result obtained from standard vector analysis IN THE LIMIT OF $\hbar \to 0$.

Still must do 2 things: show that $[\mathbf{r},\mathbf{p}_\mathbf{r}] = i\hbar$
show that $\mathbf{p}_\mathbf{r}$ is Hermitian

$$\left[\mathbf{r},\mathbf{p}_\mathbf{r}\right] = \left[\mathbf{r},\mathbf{r}^{-1}\left(\mathbf{q}\cdot\mathbf{p} - i\hbar\right)\right]$$

$$= \mathbf{r}^{-1}\left[\mathbf{r},\mathbf{q}\cdot\mathbf{p}\right] - \mathbf{r}^{-1}\underbrace{\left[\mathbf{r},i\hbar\right]}_{0} + \underbrace{\left[\mathbf{r},\mathbf{r}^{-1}\right]}_{0}\left(\mathbf{q}\cdot\mathbf{p} - i\hbar\right)$$

$$= \mathbf{r}^{-1}\left[\mathbf{r},\mathbf{q}\cdot\mathbf{p}\right] \quad \text{Use Eq. (4) to get}$$

$$\left[\mathbf{r},\mathbf{q}\cdot\mathbf{p}\right] = i\hbar\mathbf{r} \quad \text{using the non-lecture result: } \left[f(\mathbf{r}),\mathbf{q}\cdot\mathbf{p}\right] = i\hbar\frac{\partial f}{\partial \mathbf{r}}\mathbf{r}$$

* $$\therefore \left[\mathbf{r},\mathbf{p}_\mathbf{r}\right] = i\hbar$$

* we do not need to confirm that $\mathbf{p}_\mathbf{r}$ is Hermitian because it was constructed from a symmetrized form which is guaranteed to be Hermitian. Why is this guaranteed?

Correspondence Principle!

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2. Verify that the Classical Definition of Angular Momentum is Appropriate for QM.

$$\vec{L} = \vec{q}\times\vec{p} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ p_x & p_y & p_z \end{vmatrix} \tag{7}$$

We will see that this definition of an angular momentum is acceptable as far as the correspondence principle is concerned, but it is not sufficiently general.

NONLECTURE

What about symmetrizing $\vec{\mathrm{L}}$?

$$\mathrm{L}_\mathrm{x} = yp_z - zp_y = p_z y - p_y z$$

$$= -\left(\vec{p}\times\vec{q}\right)_x$$

$$\therefore p\times q = -L$$

PRODUCTS OF NON-CONJUGATE QUANTITIES

$\mathbf{q}\times\mathbf{p}+\mathbf{p}\times\mathbf{q}=0$ symmetrization is impossible!

$\mathbf{q}\times\mathbf{p}-\mathbf{p}\times\mathbf{q}=2\vec{\mathbf{L}}$ antisymmetrization is unnecessary!

But is $\vec{\mathbf{L}}$ Hermitian as defined?

BE CAREFUL: $(\mathbf{q}\times\mathbf{p})^\dagger \neq \mathbf{p}^\dagger\times\mathbf{q}^\dagger$!

go back to definition of vector cross product

$$\mathbf{L}_x = \mathbf{y}\mathbf{p}_z - \mathbf{z}\mathbf{p}_y$$

$$\mathbf{L}_x^\dagger = \mathbf{p}_z^\dagger\mathbf{y}^\dagger - \mathbf{p}_y^\dagger\mathbf{z}^\dagger = \mathbf{p}_z\mathbf{y} - \mathbf{p}_y\mathbf{z} = \mathbf{y}\mathbf{p}_z - \mathbf{z}\mathbf{p}_y = \mathbf{L}_x$$

(derived using fact that $\mathbf{p}$ and $\mathbf{q}$ are Hermitian)

$\therefore \vec{\mathbf{L}}$ is Hermitian as defined .

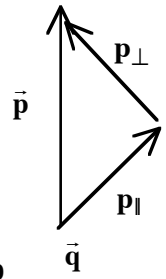
RESUME

3A. Separate $\mathbf{p}^2$ into radial and angular terms.

This is a transformation definition using different operators

GOAL: $\mathbf{p}^2 = \mathbf{p}_\mathbf{r}^2 + \mathbf{r}^{-2}\mathbf{L}^2$ (8)

vector analysis $\vec{\mathbf{p}} = \vec{\mathbf{p}}_\parallel + \vec{\mathbf{p}}_\perp$ ($\parallel$ and $\perp$ with respect to $\vec{\mathbf{q}}$)



part of $\vec{\mathbf{p}}$ points along $\vec{\mathbf{q}}$: $\mathbf{p}_\parallel$

Classically $$\vec{\mathbf{p}} = \mathbf{r}^{-2}\left[\vec{\mathbf{q}}(\mathbf{q}\cdot\mathbf{p}) - \vec{\mathbf{q}}\times(\mathbf{q}\times\mathbf{p})\right] \quad (9)$$

scalar projection on $\vec{q}$; $\vec{\mathbf{L}}$; $\perp$ to q,p plane

component of $\vec{\mathbf{p}}$ $\parallel$ to $\vec{\mathbf{q}}$

component in $\vec{\mathbf{q}},\vec{\mathbf{p}}$ plane which is $\perp$ to $\vec{\mathbf{q}}$ (is the sign correct?)

* Right Hand rule for $\vec{\mathbf{q}}\times(\vec{\mathbf{q}}\times\vec{\mathbf{p}})$ gives component mostly opposite to $\vec{\mathbf{p}}$, hence the minus sign

* $\mathbf{r}^{-2}$ is needed in both terms to remain dimensionally correct

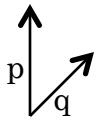
talk through this vector identity

1st term ($\mathbf{p}_{\|}$): $\vec{q}\cdot\vec{p} = |\vec{q}||\vec{p}|\cos\theta$

$\vec{q}/|q| =$ unit vector along $\vec{q}$

$\vec{p}/|p| =$ unit vector along $\vec{p}$

2nd term ($\mathbf{p}_{\perp}$): $\vec{q}\times\vec{p}$ points $\perp$ up out of paper



thumb: $\vec{q}$ | finger: $\vec{p}$ | palm: $\perp$ out of paper

$\overset{\text{thumb}}{\vec{q}} \times \underbrace{\vec{q}\times\vec{p}}_{\text{finger}}$ is in plane of paper in opposite direction of $p_{\perp}$, hence minus sign.

Is it necessary to symmetrize Eq. (9)? Find out below.

NONLECTURE

Examine Eq. (9) for QM consistency

x component

$$p_x = r^{-2}\left[x\left(xp_x + yp_y + zp_z\right) - \left(yL_z - zL_y\right)\right]$$

but $\quad yL_z - zL_y = y\left(xp_y - yp_x\right) + z\left(xp_z - zp_x\right)$

$$p_x = r^{-2}\left[\left(x^2 + y^2 + z^2\right)p_x + \cancel{(xy - yx)}^{0}p_y + \cancel{(xz - zx)}^{0}p_z\right] = p_x$$

similarly for p_y, p_z

Symmetrize? No, because the 2 parts of $\vec{p}$ are already shown to be Hermitian.

RESUME

3B. Evaluate $\mathbf{p}\cdot\mathbf{p}$. Use Eq. (9)

$$\mathbf{p}^2 = \vec{\mathbf{p}}\mathbf{r}^{-2}\left[\mathbf{q}(\mathbf{q}\cdot\mathbf{p}) - \mathbf{q}\times(\mathbf{q}\times\mathbf{p})\right] \tag{10}$$

$$\left[\text{goal is } \mathbf{p}^2 = \mathbf{p}_\mathbf{r}^2 + \mathbf{r}^{-2}\mathbf{L}^2\right]$$

commute $\vec{\mathbf{p}}$ through $\mathbf{r}^{-2}$ to be able to take advantage of classical vector triple product

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NONLECTURE

$$\left[\vec{\mathbf{p}},\mathbf{r}^{-2}\right] = -i\hbar\left[\hat{i}\frac{\partial}{\partial\mathbf{x}}\mathbf{r}^{-2} + \hat{j}\frac{\partial}{\partial\mathbf{y}}\mathbf{r}^{-2} + \hat{k}\frac{\partial}{\partial\mathbf{z}}\mathbf{r}^{-2}\right] \text{ using } \vec{\mathbf{p}} = \frac{\hbar}{i}\left[\hat{i}\frac{\partial}{\partial\mathbf{x}}\hat{j}\frac{\partial}{\partial\mathbf{y}}\hat{k}\frac{\partial}{\partial\mathbf{z}}\right]$$

$$= 2i\hbar\mathbf{r}^{-4}\vec{\mathbf{q}}$$

$$\left[\text{Recall } \left[\mathrm{f}(\mathbf{x}),\mathbf{p}_\mathrm{x}\right] = i\hbar\frac{\partial\mathbf{f}}{\partial\mathbf{x}}\right]$$

because $\frac{\partial}{\partial\mathbf{x}}\mathbf{r}^{-2} = -2\mathbf{r}^{-3}\frac{\partial\mathbf{r}}{\partial\mathbf{x}} = -2\mathbf{r}^{-3}\left(\frac{1}{2}\right)\frac{2\mathbf{x}}{\mathbf{r}} = -2\mathbf{x}/\mathbf{r}^4$

$$\left[\vec{\mathbf{p}},\mathbf{r}^{-2}\right] = \vec{\mathbf{p}}\mathbf{r}^{-2} - \mathbf{r}^{-2}\vec{\mathbf{p}} = 2i\hbar\mathbf{r}^{-4}\vec{\mathbf{q}}$$

thus $\quad \vec{\mathbf{p}}\mathbf{r}^{-2} = \mathbf{r}^{-2}\left(\vec{\mathbf{p}} + 2i\hbar\mathbf{r}^{-4}\vec{\mathbf{q}}\right)$ (11)

now insert Equation (11) into Equation (10), we get

$$\mathbf{p}^2 = r^{-2}\left(\vec{\mathbf{p}} + 2i\hbar\mathbf{r}^{-2}\vec{\mathbf{q}}\right)\left[\vec{\mathbf{q}}\cdot(\vec{\mathbf{q}}\cdot\vec{\mathbf{p}}) - \vec{\mathbf{q}}\times(\vec{\mathbf{q}}\times\vec{\mathbf{p}})\right] \tag{12}$$

multiply this out and get 4 terms.

$$\mathbf{p}^2 = \mathbf{r}^{-2}(\mathbf{p}\cdot\mathbf{q})(\mathbf{q}\cdot\mathbf{p}) - \mathbf{r}^{-2}\mathbf{p}\cdot\left[\mathbf{q}\times(\mathbf{q}\times\mathbf{p})\right] + \mathbf{r}^{-2}(2\mathrm{i}\hbar)\mathbf{r}^{-2}(\mathbf{q}\cdot\mathbf{q})(\mathbf{q}\cdot\mathbf{p}) - \mathbf{r}^{-2}(2\mathrm{i}\hbar)\mathbf{r}^{-2}\mathbf{q}\cdot\left[\mathbf{q}\times(\mathbf{q}\times\mathbf{p})\right]$$

I **II** **III** **IV**

$$\left.\begin{aligned}\mathbf{I} &= \mathbf{r}^{-2}(\mathbf{q}\cdot\mathbf{p} - 3\mathrm{i}\hbar)(\mathbf{q}\cdot\mathbf{p})\\ \mathbf{III} &= \mathbf{r}^{-2}(2\mathrm{i}\hbar)(\mathbf{q}\cdot\mathbf{p})\end{aligned}\right\}\mathbf{r}^{-2}(\mathbf{q}\cdot\mathbf{p} - \mathrm{i}\hbar)(\mathbf{q}\cdot\mathbf{p}) = \mathbf{r}^{-1}\mathbf{p}_\mathbf{r}(\mathbf{q}\cdot\mathbf{p})$$

$\mathbf{rp}_\mathbf{r} + \mathrm{i}\hbar$

$$\mathbf{II} = -\mathbf{r}^{-2}(\mathbf{p}\times\mathbf{q})\cdot(\mathbf{q}\times\mathbf{p}) = -\mathbf{r}^{-2}\left(-\mathbf{L}^2\right) = \mathbf{r}^{-2}\mathbf{L}^2$$

$$\mathbf{IV} = -\mathbf{r}^4(2\mathrm{i}\hbar)(\mathbf{q}\times\mathbf{q})^0\cdot(\mathbf{q}\times\mathbf{p})$$

$$\mathbf{p}^2 = \mathbf{r}^{-1}\mathbf{p}_\mathbf{r}(\mathbf{rp}_\mathrm{r} + \mathrm{i}\hbar) + \mathbf{r}^{-2}\mathbf{L}^2 = \mathbf{r}^{-1}\left[\mathbf{rp}_\mathbf{r} - \mathrm{i}\hbar\right]\mathbf{p}_\mathbf{r} + \mathbf{r}^{-1}\mathbf{p}_\mathbf{r}\mathrm{i}\hbar + \mathbf{r}^{-2}\mathbf{L}^2 = \mathbf{p}_\mathbf{r}^2 + \mathbf{r}^{-2}\mathbf{L}^2 \tag{13}$$

$\mathbf{rp}_\mathbf{r}$-$[\mathbf{r},\mathbf{p}_\mathbf{r}]$

We have achieved our goal.

RESUME

This $\mathbf{p}^2 = \mathbf{p}_\mathbf{r}^2 + \mathbf{r}^{-2}\mathbf{L}^2$ equation

is a very useful and simple form for $\mathbf{p}^2$ – separated into additive radial and angular terms! Whenever $\mathbf{H}$ can be separated into additive radial and angular terms, then the eigenvectors can be factored into radial and angular parts.

SUMMARY

$\mathbf{p}_\mathbf{r} = \mathbf{r}^{-1}(\mathbf{q}\cdot\mathbf{p} - i\hbar)$ radial momentum

$\mathbf{p}^2 = \mathbf{p}_\mathbf{r}^2 + \mathbf{r}^{-2}\mathbf{L}^2$ separation of radial and angular terms

$\mathbf{H} = \dfrac{\mathbf{p}_\mathbf{r}^2}{2\mu} + \left[\dfrac{\mathbf{L}^2}{2\mu\mathbf{r}^2} + V(\mathbf{r})\right]$ Separation of $\mathbf{H}$ into radial and angular terms

eventually $V_\ell(\mathbf{r}) = \dfrac{\hbar^2\ell(\ell+1)}{2\mu\mathbf{r}^2} + V(\mathbf{r})$ a sum of a "centrifugal" repulsive term and a radial potential energy term

Next Lecture: properties of $\mathbf{L}_i, \mathbf{L}^2 \longrightarrow$ Complete Set of Commuting Observables (CSCO)

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3D-Central Force Problems II. Levi-Civita: ε_{ijk}.

Last time: * $[\mathbf{x},\mathbf{p}] = i\hbar \rightarrow$ use to obtain vector commutation rules: generalize from 1-D to 3-D
* we have conjugate position and momentum components in Cartesian coordinates

Correspondence Principle Recipe
- Cartesian coordinates and vector analysis
- Symmetrize (make it Hermitian)
- classical mechanics in $\hbar \rightarrow 0$ limit

Derived key results:

$$\left[f(\mathbf{x}), \mathbf{p}_x \right] = i\hbar \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \Rightarrow \left[f(\mathbf{r}), \mathbf{p}_x \right] = i\hbar \frac{\partial \mathbf{f}}{\partial \mathbf{r}} \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}\mathbf{x}} = i\hbar \frac{\partial \mathbf{f}}{\partial \mathbf{r}} \frac{\mathbf{x}}{\mathbf{r}}$$

$$\left[f(\mathbf{r}), \mathbf{q}\cdot\mathbf{p} \right] = i\hbar \frac{\partial \mathbf{f}}{\partial \mathbf{r}} \mathbf{r} \quad \text{based on} \quad \frac{\partial \mathbf{f}}{\partial \mathbf{r}} \left(\frac{\partial \mathbf{r}}{\partial \mathbf{x}} + \frac{\partial \mathbf{r}}{\partial \mathbf{y}} + \frac{\partial \mathbf{r}}{\partial \mathbf{z}} \right) \text{ and } \mathbf{r} = \left[\mathbf{x}^2 + \mathbf{y}^2 + \mathbf{z}^2 \right]^{1/2}$$

$$*\mathbf{p}_r = \mathbf{r}^{-1} \left(\mathbf{q}\cdot\mathbf{p} - i\hbar \right) \leftarrow$$ (came from symmetrization in Cartesian coordinates)

$$*\mathbf{p}^2 = \mathbf{p}_r^2 + \mathbf{r}^{-2}\mathbf{L}^2 \leftarrow$$ separated $\mathbf{p}_{\|}$ from $\mathbf{p}_{\perp}$

operator algebra gave simple separation of variables

$$*\mathbf{L} = \mathbf{q} \times \mathbf{p}$$

not necessary (or possible) to symmetrize

$$*\mathbf{H} = \frac{\mathbf{p}_r^2}{2\mu} + \left[\frac{\mathbf{L}^2}{2\mu\mathbf{r}^2} + V(\mathbf{r}) \right]$$

need to show we can ignore the order of L^2 and r^2

$V_\ell(r)$ radial effective potential

We do not yet know anything about the eigenstates and the eigenvalues and eigenstates of $\mathbf{L}^2$ and $\mathbf{L}_i$.

TODAY [purpose is mostly to practice commutation rule [,], and angular momentum algebras]

* Obtain angular Momentum Commutation Rules → Block diagonalize **H**
* ε_{ijk} Levi-Civita Antisymmetric Tensor
  useful in derivations, vector commutators, and remembering stuff.

Next Lecture: Begin derivation of all angular momentum matrix elements starting from the Commutation Rule definitions of angular momentum components.

revised 17 August 2020 @ 10:25 AM

GOALS

1. $\left[\mathbf{L}_i, f(r)\right]=0$ any scalar function of scalar r.
2. $\left[\mathbf{L}_i, \mathbf{p}_\mathbf{r}\right]=0$ difficult - need to use ε_{ijk}!
3. $\left[\mathbf{L}_i, \mathbf{p}_\mathbf{r}^2\right]=0$
4. $\left[\mathbf{L}_i, \mathbf{L}^2\right]=0$ $\left(\text{but } \left[\mathbf{L}_\text{i}, \mathbf{L}_\text{j}^2\right]\neq 0!\right)$
5. *C.S.C.O.* $\mathbf{H}, \mathbf{L}^2, \mathbf{L}_i \rightarrow$ enable block diagonalization of $\mathbf{H}$

$\mathbf{L}^2$ and $\mathbf{L}_i$ block-diagonalize $\mathbf{H}$ according to different eigenvalues of $\mathbf{L}^2$ and $\mathbf{L}_i$. Items 1-4 are chosen to show that all terms in $\mathbf{H}$ commute with $\mathbf{L}^2$ and $\mathbf{L}_\text{i}$

$\mathbf{L}_i$: choose $\mathbf{L}_z$ for example

1. $$\left[\mathbf{L}_\text{Z}, \text{f}(\mathbf{r})\right]=\left[\mathbf{x}\mathbf{p}_\text{y}-\mathbf{y}\mathbf{p}_\text{x}, \text{f}(\mathbf{r})\right]=\mathbf{x}\left[\mathbf{p}_\text{y}, \text{f}\right]+[\mathbf{x}, \text{f}]\mathbf{p}_\text{y}-\mathbf{y}\left[\mathbf{p}_\text{x}, \text{f}\right]-[\mathbf{y}, \text{f}]\mathbf{p}_\text{x}$$

$[\mathbf{x}, \text{f}]=0, \quad [\mathbf{y}, \text{f}]=0$ because $\left[\vec{\mathbf{q}}, \text{f}(\mathbf{r})\right]=0\hat{\text{i}}+0\hat{\text{j}}+0\hat{\text{k}}$

recall $\left[\text{f}(\mathbf{r}), \mathbf{p}_x\right]=i\hbar\frac{\partial f}{\partial r}\frac{\partial r}{\partial x}=i\hbar\frac{\partial f}{\partial r}\frac{x}{r}$

$$\left[\mathbf{L}_z, f(\mathbf{r})\right]=-i\hbar\frac{\partial f}{\partial r}\left[x\frac{y}{r}-y\frac{x}{r}\right]=0$$

2. $$\left[\mathbf{L}_z, \mathbf{p}_\mathbf{r}\right]=\left[\mathbf{L}_z, \mathbf{r}^{-1}\left(\mathbf{q}\cdot\mathbf{p}-i\hbar\right)\right]=\left[\mathbf{L}_z, \mathbf{r}^{-1}\mathbf{q}\cdot\mathbf{p}\right]$$ (already know that $[\mathbf{L}_z, \mathbf{r}^{-1}i\hbar] = 0$)

$$=\underbrace{\left[\mathbf{L}_z, \mathbf{r}^{-1}\right]}_{0}\mathbf{q}\cdot\mathbf{p}+\mathbf{r}^{-1}\left[\mathbf{L}_z, \mathbf{q}\cdot\mathbf{p}\right]$$

$\frac{1}{r}$ is $f(\mathbf{r})$ and we just showed this commutation rule = 0

$\left[\mathbf{L}_z, \mathbf{q}\cdot\mathbf{p}\right]=\mathbf{q}\cdot\left[\mathbf{L}_z, \vec{\mathbf{p}}\right]+\left[\mathbf{L}_z, \vec{\mathbf{q}}\right]\cdot\mathbf{p}$ two vector commutators on RHS

Note that vector $\vec{\mathbf{q}}$ is a not scalar f($\mathbf{r}$)!

need to define special notational trick to evaluate these DIFFICULT COMMUTATORS

Either $A' = A$ or $\mathbf{H}_{AA'} = 0$. If $A = A'$, it is still possible to find linear combination of different eigenstates of $\mathbf{A}$ (with same-A eigenvalues of $\mathbf{A}$) that diagonalizes the associated block of $\mathbf{H}$.

$$\left[\mathbf{H},\mathbf{A}\right]=0$$

$$0=\left\langle A\middle|\left[\mathbf{H},\mathbf{A}\right]\middle|A'\right\rangle=\left\langle A\middle|\left[-\mathbf{AH}+\mathbf{HA}\right]\middle|A'\right\rangle=(-A+A')\mathbf{H}_{AA'}$$

so either $A=A'$ or $\mathbf{H}_{AA'}=0$

Now for something very special and useful

Levi-Civita Symbol	ε_{ijk}
cyclic order	$\varepsilon_{xyz}=\varepsilon_{yzx}=\varepsilon_{zxy}=+1$
adjacent interchange	$\varepsilon_{yxz}=\varepsilon_{zyx}=\varepsilon_{xzy}=-1$ (anti-cyclic order)
2 repeated indices	$\varepsilon_{xxy}=$ etc. $=0$

I claim $\left[\mathbf{L}_i,\mathbf{p}_j\right]=i\hbar\sum_k\varepsilon_{ijk}\mathbf{p}_k$. This will become the *definition* of a "vector operator" □ *with respect to* **L**! [[vector = 1st rank tensor]

Nonlecture: Verify claim for 1 of 3 × 3 = 9 possible cases
let i = x, j = y

Do This!

$$\left[\mathbf{L}_x,\mathbf{p}_y\right]=\left[\mathbf{yp}_z-\mathbf{zp}_y,\mathbf{p}_y\right]=\mathbf{y}\left[\mathbf{p}_z,\mathbf{p}_y\right]+\left[\mathbf{y},\mathbf{p}_y\right]\mathbf{p}_z-\mathbf{z}\left[\mathbf{p}_y,\mathbf{p}_y\right]-\left[\mathbf{z},\mathbf{p}_y\right]\mathbf{p}_y$$
$$=0+i\hbar\mathbf{p}_z-0-0$$

Now check this using ε_{ijk}

$$\left[\mathbf{L}_x,\mathbf{p}_y\right]=i\hbar\sum_k\varepsilon_{xyk}\mathbf{p}_k=i\hbar\left[\cancel{\varepsilon_{xyx}}^{0}\mathbf{p}_x+\cancel{\varepsilon_{xyy}}^{0}\mathbf{p}_y+\varepsilon_{xyz}\mathbf{p}_z\right]$$
$$=i\hbar\mathbf{p}_z. \qquad \text{OK}$$

All other 8 cases go similarly. Feel the power of ε_{ijk}!

Other important Commutation Rules

$$\left.\begin{aligned}\left[\mathbf{L_i},\mathbf{p_j}\right]&=i\hbar\sum_k\varepsilon_{ijk}\mathbf{p}_k\\ \left[\mathbf{L_i},\mathbf{q_j}\right]&=i\hbar\sum_k\varepsilon_{ijk}\mathbf{q}_k\end{aligned}\right\}$$ general definition of a "vector" operator

$\vec{\mathbf{q}}$ and $\vec{\mathbf{p}}$ are examples of vector operators. Classify as vectors *with respect to* **L**!

$$\left[\mathbf{L_i},\mathbf{L_j}\right]=i\hbar\sum_k\varepsilon_{ijk}\mathbf{L}_k$$ general definition of an "angular momentum." Works even for spin where a $\mathbf{q}\times\mathbf{p}$ definition cannot exist. This is the MOST IMPORTANT STEP

All angular momentum matrix elements will be derived next lecture from these commutation rules.

FOR THE READER: VERIFY <u>ONE COMPONENT</u> OF EACH OF THE THREE ABOVE COMMUTATORS

$\left[\mathbf{L}_i, \mathbf{L}_j\right] = i\hbar \sum_k \varepsilon_{ijk} \mathbf{L}_k$ is identical to

$$\mathbf{L} \times \mathbf{L} = i\hbar \mathbf{L}$$

(expect 0! because vector cross product $\left|\vec{A} \times \vec{B}\right| = |A|\,|B| \sin\theta_{AB}$)

$$\mathbf{L} \times \mathbf{L} = \begin{pmatrix} \hat{i} & \hat{j} & \hat{k} \\ \mathbf{L}_x & \mathbf{L}_y & \mathbf{L}_z \\ \mathbf{L}_x & \mathbf{L}_y & \mathbf{L}_z \end{pmatrix} = \hat{i}\left(\mathbf{L}_y\mathbf{L}_z - \mathbf{L}_z\mathbf{L}_y\right) + \hat{j}\left(\mathbf{L}_z\mathbf{L}_x - \mathbf{L}_x\mathbf{L}_z\right) + \hat{k}\left(\mathbf{L}_x\mathbf{L}_y - \mathbf{L}_y\mathbf{L}_x\right)$$

note reversal of x and z terms

$$= i\hbar\left[\hat{i}\mathbf{L}_x + \hat{j}\mathbf{L}_y + \hat{k}\mathbf{L}_z\right] = i\hbar\mathbf{L}$$

This vector cross product definition of **L** is more general than **q** × **p** because there is no way to define spin in **q** × **p** form but **S** × **S** = i$\hbar$**S** is quite meaningful.

Ask: Can one generalize that, if **L** × **L** = i$\hbar$**L** (instead of 0), and the [$\mathbf{L}_i$, $\mathbf{L}_j$] and [$\mathbf{L}_i$, $\mathbf{p}_j$] commutation rules have similar forms, that **L** × **p** = i$\hbar$**p**? NO! Check for yourself!

---

2. Continued. use $\mathbf{p}_r = \mathbf{r}^{-1}\left(\mathbf{q}\cdot\mathbf{p} - i\hbar\right)$

already know that this commutes with $\mathbf{L}_i$

$$\left[\mathbf{L}_z, \mathbf{p}_r\right] = \mathbf{r}^{-1}\vec{\mathbf{q}}\cdot\left[\mathbf{L}_z, \vec{\mathbf{p}}\right] + \mathbf{r}^{-1}\left[\mathbf{L}_z, \vec{\mathbf{q}}\right]\cdot\vec{\mathbf{p}}$$

vector commutators

evaluate the first term

$$\left[\mathbf{L}_i, \vec{\mathbf{p}}\right] = i\hbar \sum_k \left(\hat{i}\varepsilon_{ixk} + \hat{j}\varepsilon_{iyk} + \hat{k}\varepsilon_{izk}\right)\mathbf{p}_k$$

sum of 3 terms

$$\mathbf{q}\cdot\left[\mathbf{L}_i, \vec{\mathbf{p}}\right] = i\hbar \sum_k \left(\mathbf{x}\varepsilon_{ixk} + \mathbf{y}\varepsilon_{iyk} + \mathbf{z}\varepsilon_{izk}\right)\mathbf{p}_k$$

only one of these terms is nonzero (but use simpler form)

call this an index-j sum

$$= i\hbar \sum_{j,k} \varepsilon_{ijk}\mathbf{q}_j\mathbf{p}_k \qquad (1)$$

and evaluate the second term $\left[\mathbf{L}_i,\vec{\mathbf{q}}\right]\cdot\vec{\mathbf{p}}$

$$\left[\mathbf{L}_i,\vec{\mathbf{q}}\right]=i\hbar\sum_k\left[\hat{i}\varepsilon_{ixk}+\hat{j}\varepsilon_{iyk}+\hat{k}\varepsilon_{izk}\right]\mathbf{q}_k$$

$$\left[\mathbf{L}_i,\vec{\mathbf{q}}\right]\cdot\mathbf{p}=i\hbar\sum_k\left[\varepsilon_{ixk}\mathbf{q}_k\mathbf{p}_x+\varepsilon_{iyk}\mathbf{q}_k\mathbf{p}_y+\varepsilon_{izk}\mathbf{q}_k\mathbf{p}_z\right]$$ (a 2nd –index sum)

$$=i\hbar\sum_{j,k}\varepsilon_{ijk}\mathbf{q}_k\mathbf{p}_j=i\hbar\sum_{k,j}\varepsilon_{i\mathrm{kj}}\mathbf{q}_\mathrm{j}\mathbf{p}_\mathrm{k}$$

sum is over j and k, so can permute the k↔j labels

$$=-i\hbar\sum_{k,j}\varepsilon_{ijk}\mathbf{q}_j\mathbf{p}_k \qquad (2)$$

switch order of j and k

putting Eqs. (1) and (2) together

$$\vec{\mathbf{q}}\cdot\left[\mathbf{L}_i,\vec{\mathbf{p}}\right]+\left[\mathbf{L}_i,\vec{\mathbf{q}}\right]\cdot\vec{\mathbf{p}}=i\hbar\sum_{j,k}\left[\varepsilon_{ijk}\mathbf{q}_j\mathbf{p}_k-\varepsilon_{ijk}\mathbf{q}_j\mathbf{p}_k\right]=0!$$

The 2 terms from the $[\mathbf{L}, \mathbf{p}\cdot\mathbf{q}]$ are combined here.

Elegance and power of ε_{ijk} notation!

We have shown, for $\mathbf{p}_r=\mathbf{r}^{-1}(\mathbf{q}\cdot\mathbf{p}-i\hbar)$, that:

* $[\mathbf{L}_\mathrm{i},\mathbf{p}_\mathrm{r}]=0$ for all i

* easy now to show $[\mathbf{L}_\mathrm{i},\mathbf{p}_\mathrm{r}^{\,2}]=0$

Finally $\left[\mathbf{L}_i,\mathbf{L}^2\right]=\sum_j\left[\mathbf{L}_i,\mathbf{L}_j^2\right]=\sum_j\left(\mathbf{L}_j\left[\mathbf{L}_i,\mathbf{L}_j\right]+\left[\mathbf{L}_i,\mathbf{L}_j\right]\mathbf{L}_j\right]$

$$=\sum_j\left[\mathbf{L}_j\left(i\hbar\sum_k\varepsilon_{ijk}\mathbf{L}_k\right)+\left(i\hbar\sum_k\varepsilon_{ijk}\mathbf{L}_k\right)\mathbf{L}_j\right]$$

sum is over j and k, so can permute the j & k indices

same trick: permute j ↔ k indices in second term

Thus $\varepsilon_{\mathrm{ijk}}=-\varepsilon_{ikj}$

$$-\left(i\hbar\sum_k\varepsilon_{ijk}\mathbf{L}_j\right)\mathbf{L}_k$$

$$\left[\mathbf{L}_i,\mathbf{L}^2\right]=0$$

But be careful: $\mathrm{L_i}$ and $\mathrm{L_j}^2$ do not commute even though $\mathrm{L_i}$ and L^2 do commute

$$\left[\mathbf{L}_i,\mathbf{L}_\mathrm{j}^2\right]=\mathbf{L}_j\left[\mathbf{L}_i,\mathbf{L}_j\right]+\left[\mathbf{L}_i,\mathbf{L}_j\right]\mathbf{L}_j=i\hbar\left(\mathbf{L}_j\sum_k\varepsilon_{ijk}\mathbf{L}_k+\left(\sum_k\varepsilon_{ijk}\mathbf{L}_k\right)\mathbf{L}_j\right)\neq 0$$

because this is a sum only over k, can't combine and cancel terms. See detail on next page.

for i=x, j=y

$$\left[\mathbf{L}_x,\mathbf{L}_y^2\right]=\mathbf{L}_y\left[\mathbf{L}_x,\mathbf{L}_y\right]+\left[\mathbf{L}_x,\mathbf{L}_y\right]\mathbf{L}_y=i\hbar\left[\mathbf{L}_y\mathbf{L}_z+\mathbf{L}_z\mathbf{L}_y\right]\neq 0!$$

---

so we have shown

$$\left[\mathbf{L}^2,\mathbf{L}_\mathrm{i}\right]=0$$

$$\left[\mathbf{L}^2,\mathrm{f}(\mathbf{r})\right]=0$$

$$\left[\mathbf{L}_\mathrm{i},\mathrm{f}(\mathbf{r})\right]=0$$

$$\left[\mathbf{L}^2,\mathbf{p}_\mathrm{r}\right]=0$$

$$\left[\mathbf{L}_\mathrm{i},\mathbf{p}_\mathrm{r}\right]=0$$

$\therefore$ $\mathbf{L}^2$, $\mathbf{L}_\mathrm{i}$, $\mathbf{H}$ all commute — **C**omplete **S**et of **M**utually **C**ommuting **O**perators

eigenfunction of $\mathbf{L}^2$ with eigenvalue $\hbar^2\mathrm{L}(\mathrm{L}+1)$

So what does this tell us about $\langle \mathrm{L}|\mathbf{H}|\mathrm{L}'\rangle = ?$ also $\langle M_L|\mathrm{H}|M'_L\rangle$

$L_z|LM_L\rangle=\hbar M_L|LM_L\rangle$

Both $H_{LL'}=0$ and $H_{M_L,M'_L}=0$

BLOCK DIAGONALIZATION OF **H**!

Basis functions $\psi=\underbrace{\chi(\mathrm{r})}_{\text{radial special}}\underbrace{|\mathrm{L}^2,\mathrm{L}_\mathrm{z}\rangle}_{\text{angular universal}}=|\mathrm{nLM_L}\rangle$

- $\mathrm{M_L}$ — eigenfunctions of $\mathbf{L}_\mathrm{z}$
- L — eigenfunctions of $\mathbf{L}^2$
- n — which radial eigenfunction?

Next time I will show, starting from

$$\boxed{\left[\mathbf{L}_i,\mathbf{L}_j\right]=i\hbar\sum_k\varepsilon_{ijk}\mathbf{L}_k}\quad\text{, that}$$

* $\mathbf{L}^2|nLM_L\rangle=\hbar^2\mathrm{L}(\mathrm{L}+1)|nLM_L\rangle \qquad \mathrm{L}=0,1,\ldots$

* $\mathbf{L}_z|nLM_L\rangle=\hbar M_L|nLM_L\rangle \qquad M_L=-L,-L+1,\ldots+L$

also derive all $\mathbf{L}_\mathrm{x}$ and $\mathbf{L}_\mathrm{y}$ matrix elements in $|nLM_L\rangle$ basis set.

Translation and Rotation Operators

We are interested in QM operators that cause translation or rotation of an initially localized state: $|x_0,y_0,z_0\rangle$ or $|\alpha_0,\beta_0,\gamma_0\rangle$ (where α,β,γ are Euler angles that relate the body-fixed axis system to the laboratory-fixed axis system.

Translations are related to $\hat{p}_x,\hat{p}_y,\hat{p}_z$ operators and rotations are related to $\hat{L}_x,\hat{L}_y,\hat{L}_z$ operators. How do we demonstrate these relationships?

Begin by asking what does $e^{-i\hat{p}_x\delta/\hbar}$ do to an initially localized state $|x_0,y_0,z_0\rangle$.

An initially localized state is an eigenfunction of $\hat{x},\hat{y},\hat{z}$ operators

$$\hat{x}\left|x_0,y_0,z_0\right\rangle = x_0\left|x_0,y_0,z_0\right\rangle$$

similarly for $\hat{y}$ and $\hat{z}$

What does $e^{-i\hat{p}_x\delta/\hbar}$ do to $|x_0,y_0,z_0\rangle$? We want to know to what eigenvalue(s) of $\hat{x}$ does $e^{-i\hat{p}_x\delta/\hbar}\left|x_0,y_0,z_0\right\rangle$ belong? We ask for $\hat{x}\left[e^{-i\hat{p}_x\delta/\hbar}\left|x_0,y_0,z_0\right\rangle\right]$ and we use the commutation rule $\left[\hat{x},f\left(\hat{p}_x\right)\right]=i\hbar\dfrac{df}{dp_x}$

$$\hat{x}\left[e^{-i\hat{p}_x\delta/\hbar}\left|x_0,y_0,z_0\right\rangle\right]=\left(e^{-i\hat{p}_x\delta/\hbar}\hat{x}+\left[\hat{x},e^{-i\hat{p}_x\delta/\hbar}\right]\right)\left|x_0,y_0,z_0\right\rangle$$

$$f\left(\hat{p}_x\right)=e^{-i\hat{p}_x\delta/\hbar}$$

$$\frac{d\hat{f}}{dp_x}=\left(-i\delta/\hbar\right)e^{-i\hat{p}_x\delta/\hbar}$$

$$\left[\hat{x},f\left(\hat{p}_x\right)\right]=(i\hbar)\left(-i\delta/\hbar\right)e^{-i\hat{p}_x\delta/\hbar}=\delta e^{-i\hat{p}_x\delta/\hbar}$$

Put it all together:

$$\hat{x}\left[e^{-i\hat{p}_x\delta/\hbar}\left|x_0,y_0,z_0\right\rangle\right]=e^{-i\hat{p}_x\delta/\hbar}\left[x_0+\delta\right]\left|x_0,y_0,z_0\right\rangle$$

$$\hat{x}\left[e^{-i\hat{p}_x\delta/\hbar}\left|x_0,y_0,z_0\right\rangle\right]=\left(x_0+\delta\right)\left[e^{-i\hat{p}_x\delta/\hbar}\left|x_0,y_0,z_0\right\rangle\right]$$

This means that $e^{-i\hat{p}_x\delta/\hbar}\left|x_0,y_0,z_0\right\rangle$ belongs to the $(x_0 + \delta)$ eigenvalue of $\hat{x}$!

$$e^{-i\hat{p}_x\delta/\hbar}\left|x_0,y_0,z_0\right\rangle=\left|x_0+\delta,y_0,z_0\right\rangle$$

So we know how to build an operator that causes translations of a localized state in the x, y, or z directions: $\hat{T}_x,\hat{T}_y,\hat{T}_z$.

But we know that $\left[\hat{p}_i,\hat{p}_j\right]=0$ for all components of linear momentum. This means that for all linear translations, $\left[\hat{T}_i,\hat{T}_j\right]=0$. *The sequence of the linear translations does not matter!* What about rotations of the initially localized state$|\,\alpha_0,\beta_0,\gamma_0\rangle$? What does $e^{-i\phi\hat{L}_z/\hbar}$ do to $|\,\alpha_0,\beta_0,\gamma_0\rangle$?

Consider

$$\hat{\alpha}\left[e^{-i\phi\hat{L}_z/\hbar}\left|\alpha_0,\beta_0,\gamma_0\right\rangle\right].$$

Following an argument similar to that for the translational operators

$$\hat{\alpha}e^{-i\phi\hat{L}_z/\hbar}\left|\alpha_0,\beta_0,\gamma_0\right\rangle=\left(\alpha_0+\phi\right)e^{-i\phi\hat{L}_z/\hbar}\left|\alpha_0,\beta_0,\gamma_0\right\rangle$$

$\hat{\alpha}e^{-i\phi\hat{L}_z/\hbar}\left|\alpha_0,\beta_0,\gamma_0\right\rangle$ belongs to the $\alpha_0 + \phi$ eigenvalue of $\hat{\alpha}$.

Now show something beautiful: that infinitesimal rotations about different axes do not commute!

$$e^{-i\phi\hat{L}_z/\hbar}e^{-i\theta\hat{L}_y/\hbar}\left|\alpha_0,\beta_0,\gamma_0\right\rangle=\left(e^{-i\theta\hat{L}_y/\hbar}e^{-i\phi\hat{L}_z/\hbar}+\left[e^{-i\phi\hat{L}_z/\hbar},e^{-i\theta\hat{L}_y/\hbar}\right]\right)\left|\alpha_0,\beta_0,\gamma_0\right\rangle$$

Expand the exponentials for infinitesimal θ,ϕ: 1st two terms in the power series expansion of $e^{-i\alpha}$: $1 - i\alpha$.

$$\left[e^{-i\theta\hat{L}_y/\hbar}, e^{-i\phi\hat{L}_z/\hbar}\right] = [1,1] - \left[1, -i\phi\hat{L}_z/\hbar\right] + \left[-i\theta\hat{L}_y/\hbar, 1\right] - \left(-\frac{i}{\hbar}\right)\left[\phi\hat{L}_z, \theta\hat{L}_y\right]$$

$$= 0 + 0 + 0 + \left(-\frac{i}{\hbar}\right)(\phi\theta)i\hbar\hat{L}_x$$

Reversing the order of the rotations about the *y* and *z* axes results in a *non-zero* rotation by $\theta\phi$ about the x-axis!

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Angular Momentum Matrix Elements Derived from Commutation Rules

LAST TIME: * derived all [,]=0 Commutation Rules needed to block diagonalize **H**:

$$\mathbf{H}=\frac{\mathbf{p}_r^2}{2\mu}+\left[\frac{\mathbf{L}^2}{2\mu\mathbf{r}^2}+V(\mathbf{r})\right] \text{ in } \left|nLM_L\right\rangle \text{ basis sets}$$

*ε_{ijk} Levi-Civita antisymmetric tensor — useful properties, especially for derivations involving components of angular momenta

* Commutation Rule DEFINITIONS of Angular Momentum and "Vector" Operators $\left[\mathrm{L}_i,\mathrm{L}_j\right]=i\hbar\sum_k \varepsilon_{ijk}\mathrm{L}_k$

$$\left[\mathrm{L}_i,\mathrm{V}_j\right]=i\hbar\sum_k \varepsilon_{ijk}\mathrm{V}_k$$

Classification of operators: *universality* of angular factors of matrix elements for 3D central force problems.

TODAY: Obtain <u>all</u> angular momentum matrix elements from the commutation rule definition of an angular momentum, without ever looking at a differential operator or a wavefuncton. *Possibilities for phase inconsistencies.* [Similar generalization to derivation for angular parts of matrix elements of all "spherical tensor" operators, $\mathbf{T}_q^{(k)}$.]

1. Define Components of an Angular Momentum using a Commutation Rule.

2. Define the eigenbasis for J^2 and J_z. $|\lambda\mu\rangle$ (we know the eigenbasis must exist, but we start out not knowing anything about it).

3. Show $\lambda \geq \mu$.

4. Raising and lowering operators (like $\mathbf{a}^\dagger$, **a** and $\boldsymbol{x} \pm i\boldsymbol{p}$ for the harmonic oscillator). $\mathbf{J}_\pm|\lambda\mu\rangle$ gives eigenfunction of $\mathbf{J}_z$ that belongs to the $\mu \pm \hbar$ eigenvalue and the eigenfuncton of $\mathbf{J}^2$ that belongs to the λ eigenvalue.

5. Must be at least one μ_{MIN} pair of eigenstates of $\mathbf{J}_z$ such that
   $\mathbf{J}_-(\mathbf{J}_+|\lambda\mu_{MAX}\rangle) = 0$
   $\mathbf{J}_+(\mathbf{J}_-|\lambda\mu_{MIN}\rangle) = 0$
   This leads to: $\hbar\left(\frac{n}{2}\right), \lambda = \hbar^2\frac{n}{2}\left(\frac{n}{2}+1\right)$, and n is a positive integer.

6. Obtain all matrix elements of $\mathbf{J}_x$, $\mathbf{J}_y$, $\mathbf{J}_\pm$, but there remains. phase ambiguity for the non-zero matrix elements.

7. Standard phase choice: "Condon and Shortley".

1. Commutation Rule $\left[\mathrm{J}_i,\mathrm{J}_j\right]=i\hbar\sum_k \varepsilon_{ijk}\mathrm{J}_k$

   This is a general definition of angular momentum (call it **J**, **L**, **S**, anything!). Each angular momentum generates a state space.

2. eigenfunctions of $\mathbf{J}^2$ and $\mathbf{J}_z$ exist (Hermitian operators. Hermiticity is guaranteed by symmetrization.)

   $$\mathbf{J}^2|\lambda\mu\rangle=\lambda|\lambda\mu\rangle$$
   $$\mathbf{J}_z|\lambda\mu\rangle=\mu|\lambda\mu\rangle$$

   but what are the values of λ ,μ?
   $\mathbf{J}^2$ and $\mathbf{J}_z$ are Hermitian, therefore λ ,μ are real

3. find upper and lower bounds for μ in terms of λ : $\lambda \geq \mu^2$

   $$\left\langle\lambda\mu\left|\left(\mathbf{J}^2-\mathbf{J}_z^2\right)\right|\lambda\mu\right\rangle=\lambda-\mu^2$$ Want to show that $\lambda-\mu^2$ is ≥ 0.

   but $\mathbf{J}^2=\mathbf{J}_x^2+\mathbf{J}_y^2+\mathbf{J}_z^2$

   $$\mathbf{J}^2-\mathbf{J}_z^2=\mathbf{J}_x^2+\mathbf{J}_y^2$$
   $$\lambda-\mu^2=\left\langle\lambda\mu\left|\mathbf{J}_x^2+\mathbf{J}_y^2\right|\lambda\mu\right\rangle$$

   completeness

$$\lambda-\mu^2=\sum_{\lambda',\mu'}\left[\left\langle\lambda\mu\left|\mathbf{J}_x\right|\lambda'\mu'\right\rangle\left\langle\lambda'\mu'\left|\mathbf{J}_x\right|\lambda\mu\right\rangle+\left\langle\lambda\mu\left|\mathbf{J}_y\right|\lambda'\mu'\right\rangle\left\langle\lambda'\mu'\left|\mathbf{J}_y\right|\lambda\mu\right\rangle\right]$$

We know that $\mathbf{J}^2$ and $\mathbf{J}_z$ are Hermitian because they were constructed by symmetrization of classical mechanical operators.

Hermitian ($\mathbf{A}=\mathbf{A}^\dagger$ or $A_{ij}=A_{ji}^*$): $\left\langle\lambda'\mu'\left|\mathbf{J}_x\right|\lambda\mu\right\rangle=\left\langle\lambda\mu\left|\mathbf{J}_x\right|\lambda'\mu'\right\rangle^*$

$$\lambda-\mu^2=\sum_{\lambda',\mu'}\left[\left|\left\langle\lambda\mu\left|\mathbf{J}_x\right|\lambda'\mu'\right\rangle\right|^2+\left|\left\langle\lambda\mu\left|\mathbf{J}_y\right|\lambda'\mu'\right\rangle\right|^2\right]\geq 0$$

Thus $\lambda-\mu^2\geq 0$ and $\lambda\geq\mu^2\geq 0$

and from these we get $\mu_{\mathrm{MAX}}\leq\lambda^{1/2},\mu_{\mathrm{MIN}}\geq-\lambda^{1/2}$

4. Raising/Lowering Operators

$\mathbf{J}_\pm \equiv \mathbf{J}_x \pm i\mathbf{J}_y$ $\left(\text{not Hermitian: } \mathbf{J}_+^\dagger = \mathbf{J}_-\right)$ $\left(\text{just like } \mathbf{a}, \mathbf{a}^\dagger\right)$

$$\left[\mathbf{J}_z, \mathbf{J}_\pm\right] = \left[\mathbf{J}_z, \mathbf{J}_x\right] \pm i\left[\mathbf{J}_z, \mathbf{J}_y\right]$$

$$= i\hbar\mathbf{J}_y \pm i\left(-i\hbar\mathbf{J}_x\right) = \pm\hbar\left[\mathbf{J}_x \pm i\mathbf{J}_y\right]$$

$$= \pm\hbar\mathbf{J}_\pm$$

$\mathbf{J}_z\mathbf{J}_\pm = \mathbf{J}_\pm\mathbf{J}_z \pm \hbar\mathbf{J}_\pm$ right multiply by $|\lambda\mu\rangle$

$$\mathbf{J}_z\left(\mathbf{J}_\pm|\lambda\mu\rangle\right) = \mathbf{J}_\pm\left(\mathbf{J}_z|\lambda\mu\rangle\right) \pm \hbar\mathbf{J}_\pm|\lambda\mu\rangle$$

$$= \mathbf{J}_\pm\mu|\lambda\mu\rangle \pm \hbar\mathbf{J}_\pm|\lambda\mu\rangle$$

$$= (\mu \pm \hbar)\left(\mathbf{J}_\pm|\lambda\mu\rangle\right), \text{ which means that}$$

$(\mathbf{J}_\pm|\lambda\mu\rangle)$ is an eigenfunction of $\mathbf{J}_z$ belonging to eigenvalue $\mu \pm \hbar$.
Thus $\mathbf{J}_\pm$ "raises" or "lowers" the $\mathbf{J}_z$ eigenvalue in steps of $\hbar$.

Similar exercise for $\left[\mathbf{J}^2, \mathbf{J}_\pm\right]$ to get effect of $\mathbf{J}_\pm$ on eigenvalue of $\mathbf{J}^2$

$\left[\mathbf{J}^2, \mathbf{J}_\pm\right] = \left[\mathbf{J}^2, \mathbf{J}_x\right] \pm i\left[\mathbf{J}^2, \mathbf{J}_y\right] = 0$ $\left(\text{We already knew that } \left[\mathbf{J}^2, \mathbf{J}_i\right] = 0\right)$

$\mathbf{J}^2\left(\mathbf{J}_\pm|\lambda\mu\rangle\right) = \mathbf{J}_\pm\left(\mathbf{J}^2|\lambda\mu\rangle\right) = \lambda\left(\mathbf{J}_\pm|\lambda\mu\rangle\right)$, which means that

$\left(\mathbf{J}_\pm|\lambda\mu\rangle\right)$ belongs to the same eigenvalue of $\mathbf{J}^2$ as $|\lambda\mu\rangle$

$\mathbf{J}_\pm$ has no effect on λ.

* upper and lower bounds on μ are $\pm\lambda^{1/2}$
* $\mathbf{J}_{\pm}$ raises/lowers μ by steps of $\hbar$

* Since $\mathbf{J}_x = \frac{1}{2}(\mathbf{J}_+ + \mathbf{J}_-)$ and $\mathbf{J}_y = \frac{1}{2i}(\mathbf{J}_+ - \mathbf{J}_-)$,

The only nonzero matrix elements of $\mathbf{J}_i$ in the $|\lambda\mu\rangle$ basis set are those where $\Delta\mu = 0, \pm\hbar$ and $\Delta\lambda = 0$. As for derivation of Harmonic Oscillator matrix elements, we are not assured that *all* values of μ differ in steps of $\hbar$. Divide basis states into sets, where the members of each set are related by integer steps of $\hbar$ in μ.

5. For each set, there are μ_{MIN} and μ_{MAX}: $\lambda \geq \mu^2$

Thus, for each set $\mathbf{J}_+|\lambda\mu_{MAX}\rangle = 0$
$\mathbf{J}_-|\lambda\mu_{MIN}\rangle = 0$

but
$$\mathbf{J}_-\mathbf{J}_+ = \left(\mathbf{J}_x - i\mathbf{J}_y\right)\left(\mathbf{J}_x + i\mathbf{J}_y\right) = \mathbf{J}_x^2 + \mathbf{J}_y^2 + i\mathbf{J}_x\mathbf{J}_y - i\mathbf{J}_y\mathbf{J}_x$$
$$= \mathbf{J}_x^2 + \mathbf{J}_y^2 + i\left[\mathbf{J}_x, \mathbf{J}_y\right]$$
$$= \mathbf{J}_x^2 + \mathbf{J}_y^2 + i\left(i\hbar\mathbf{J}_z\right)$$
$$= \mathbf{J}_x^2 + \mathbf{J}_y^2 - \hbar\mathbf{J}_z$$

but $\mathbf{J}_x^2 + \mathbf{J}_y^2 = \mathbf{J}^2 - \mathbf{J}_z^2$, thus

$$\mathbf{J}_-\mathbf{J}_+ = \mathbf{J}^2 - \mathbf{J}_z^2 - \hbar\mathbf{J}_z$$
$$0 = \mathrm{J}_-\mathrm{J}_+\left|\lambda\mu_{MAX}\right\rangle = \left(\mathrm{J}^2 - \mathrm{J}_z^2 - \hbar\mathrm{J}_z\right)\left|\lambda\mu_{MAX}\right\rangle$$
$$= \left(\lambda - \mu_{MAX}^2 - \hbar\mu_{MAX}\right)\left|\lambda\mu_{MAX}\right\rangle$$

$$\lambda = \mu_{MAX}^2 + \hbar\mu_{MAX}$$

Similarly for μ_{MIN}

$$\mathbf{J}_+\mathbf{J}_-\left|\lambda\mu_{MIN}\right\rangle = 0$$

$$\mathbf{J}_+\mathbf{J}_- = \mathbf{J}^2 - \mathbf{J}_z^2 + \hbar\mathbf{J}_z$$

$$\lambda = \mu_{\text{MIN}}^2 - \hbar\mu_{\text{MIN}}$$

subtract 2 equations for λ

$$0 = \mu_{\text{MAX}}^2 - \mu_{\text{MIN}}^2 + \hbar\left(\mu_{\text{MAX}} + \mu_{\text{MIN}}\right)$$

$$0 = \left(\mu_{\text{MAX}} + \mu_{\text{MIN}}\right)\left(\mu_{\text{MAX}} - \mu_{\text{MIN}} + \hbar\right)$$

now factor this equation

Thus $\mu_{\text{MAX}} = -\mu_{\text{MIN}}$ OR $\mu_{\text{MAX}} = \mu_{\text{MIN}} - \hbar$

(*impossible* because μ_{MAX} cannot be smaller than μ_{MIN})

Thus for each set of $|\lambda\mu\rangle$, μ goes from μ_{MAX} to μ_{MIN} in steps of $\hbar$

$$\mu_{\text{MAX}} = \mu_{\text{MIN}} + n\hbar$$

$$\mu_{\text{MAX}} = \frac{n}{2}\hbar$$

Thus μ is either integer or half integer or both!

Thus there will *at worst* be only two non-communicating sets of $|\lambda\mu\rangle$ because if μ were both integer and 1/2-integer, each set would form a set of μ-values, within which the members would be separated in steps of $\hbar$.

Now, to specify the allowed values of λ:

$$\lambda = \mu_{\text{MAX}}^2 + \hbar\mu_{\text{MAX}} = \left(\frac{n}{2}\hbar\right)^2 + \hbar\left(\frac{n}{2}\hbar\right) = \hbar^2\frac{n}{2}\left(\frac{n}{2}+1\right)$$

let $\frac{n}{2} \equiv j$

$$\mu_{\text{MAX}} = \hbar j$$

$$\mu_{\text{MIN}} = -\hbar j$$

$$\lambda = \hbar^2 j(j+1)$$

!

j either integer or half integer or both

Revised August 17, 2020 8:21 AM

Rename our basis states

$$\mathbf{J}^2|jm\rangle = \hbar^2 j(j+1)|jm\rangle$$
$$\mathbf{J}_z|jm\rangle = \hbar m|jm\rangle$$

valid for all angular momentum operators that are certified as an angular momentum by satisfying the defining commutation rule $\left[A_i, A_j\right] = i\hbar\sum_k \varepsilon_{ijk}A_k$. We can define an $|am_a\rangle$ basis set for any angular momentum operator defined as above. We never need to look at the functional form of the $\{\psi_{am_a}\}$ wavefunctions!

6. $\mathbf{J}_x$, $\mathbf{J}_y$, $\mathbf{J}_\pm$ matrix elements

recall page 23-3, but in new notation

$$|jm\pm1\rangle = N_\pm \mathbf{J}_\pm |jm\rangle \qquad (\mathbf{J}_\pm \text{ raises / lowers } m \text{ by } 1)$$

$N_\pm$: normalization factor (to be determined below)

$$1 = \langle jm\pm1|jm\pm1\rangle = \left(N_\pm \mathrm{J}_\pm|jm\rangle\right)^\dagger\left(N_\pm \mathrm{J}_\pm|jm\rangle\right) = N_\pm^*\langle jm|\mathrm{J}_\mp N_\pm \mathrm{J}_\pm|jm\rangle$$

$$N_\pm^\dagger = N_\pm^*$$
$$\mathrm{J}_\pm^\dagger = \mathrm{J}_\mp \qquad !$$

$$1 = |N_\pm|^2\langle jm|\mathbf{J}_\mp\mathbf{J}_\pm|jm\rangle$$

use this to evaluate matrix elements of $\mathbf{J}_\mp\mathbf{J}_\pm$

$$\mathrm{J}_\mp\mathrm{J}_\pm = \left(\mathrm{J}_x \mp i\mathrm{J}_y\right)\left(\mathrm{J}_x \pm i\mathrm{J}_y\right) = \mathrm{J}_x^2 + \mathrm{J}_y^2 \pm i\left[\mathrm{J}_x, \mathrm{J}_y\right]$$
$$= \mathrm{J}^2 - \mathrm{J}_z^2 \pm i\left(i\hbar\mathrm{J}_z\right) = \mathrm{J}^2 - \mathrm{J}_z^2 \mp \hbar\mathrm{J}_z$$
$$= \mathrm{J}^2 - \mathrm{J}_z\left(\mathrm{J}_z \pm \hbar\right)$$

$$1 = |N_\pm|^2\left[\hbar^2 j(j+1) - \hbar^2\left(m(m\pm1)\right)\right]$$

$$|N_\pm| = \frac{1}{\hbar}\left[j(j+1) - m(m\pm1)\right]^{-1/2} e^{-i\delta_\pm}$$

$e^{-i\delta_\pm}$: arbitrary phase factor that results from taking square root

$$\mathbf{J}_\pm|jm\rangle = \hbar\left[j(j+1) - m(m\pm1)\right]^{1/2}|jm\pm1\rangle e^{-i\delta_\pm}$$

Usual phase choice is $\delta_\pm = 0$ for all j, m:

known as the "Condon and Shortley" phase choice

(sometimes an alternative phase choice is used, $\delta_\pm = \pm\pi/2$, so be careful)

standard phase choice: $\delta_{\pm} = 0$

$$\left\langle j'm' \middle| \mathbf{J}_{\pm} \middle| jm \right\rangle = \hbar \delta_{j'j} \delta_{m'm\pm1} \left[j(j+1) - m(m\pm1) \right]^{1/2}$$

$$\left(\text{or } \hbar \delta_{jj'} \delta_{m'm\pm1} \left[j(j+1) - \underline{\underline{m(m')}} \right]^{1/2} \right)$$

remember matrix elements of **x** and **p** in harmonic oscillator basis set?

Now, since $\mathbf{J}_x = \frac{1}{2}\left(\mathbf{J}_+ + \mathbf{J}_-\right)$

$$\left\langle j'm' \middle| \mathbf{J}_x \middle| jm \right\rangle = \frac{\hbar}{2} \delta_{j'j} \left\{ \delta_{m'm+1} \left[j(j+1) - m(m+1) \right]^{1/2} + \delta_{m'm-1} \left[j(j+1) - m(m-1) \right]^{1/2} \right\}$$

$\mathbf{J}_y = \frac{1}{2i}(\mathbf{J}_+ - \mathbf{J}_-)$

two sign surprises

$$\left\langle j'm' \middle| \mathbf{J}_y \middle| jm \right\rangle = -i \frac{\hbar}{2} \delta_{j'j} \left\{ \delta_{m'm+1} \left[j(j+1) - m(m+1) \right]^{1/2} - \delta_{m'm-1} \left[j(j+1) - m(m-1) \right]^{1/2} \right\}$$

This phase choice leaves all matrix elements of $\mathbf{J}^2$, $\mathbf{J}_x$ and $\mathbf{J}_{\pm}$ real and positive.

[If, instead, you use $\delta_{\pm} = +\pi/2$, this gives $\mathbf{J}_y$ real and $\mathbf{J}_x$,$\mathbf{J}_{\pm}$ imaginary.]

Summary

$$\left\langle j'm' \middle| \mathbf{J}^2 \middle| jm \right\rangle = \delta_{j'j} \delta_{m'm} \hbar^2 j(j+1)$$

$$\left\langle jm \middle| \vec{\mathbf{J}} \middle| jm \right\rangle = \hat{k} \hbar m \qquad \left(\Delta m = 0 \text{ selects } \hat{k} \mathbf{J}_z \right)$$

$$\left\langle jm \pm 1 \middle| \vec{\mathbf{J}} \middle| jm \right\rangle = \left(\hat{i} \mp i\hat{j} \right) \frac{\hbar}{2} \left[j(j+1) - m(m\pm1) \right]^{1/2}$$

$$\hat{i}\mathbf{J}_x + \hat{j}\mathbf{J}_y = \frac{1}{2}\hat{i}\left(\mathbf{J}_+ + \mathbf{J}_-\right) + \hat{j}\frac{1}{2i}\left(\mathbf{J}_+ - \mathbf{J}_-\right)$$

$$= \frac{1}{2}\mathbf{J}_+\left(\hat{i} - i\hat{j}\right) + \frac{1}{2}\mathbf{J}_-\left(\hat{i} + i\hat{j}\right)$$

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5.73 Quantum Mechanics I
Fall 2018

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J Matrices

Last time:

starting with $\left[\mathrm{J}_i,\mathrm{J}_j\right]=i\hbar\sum_k\varepsilon_{ijk}\mathrm{J}_k$ DEFINITION OF AN ANGULAR MOMENTUM !

$$\mathrm{J}^2\left|jm\right\rangle=\hbar^2 j(j+1)\left|jm\right\rangle$$

$$\mathrm{J}_z\left|jm\right\rangle=\hbar m\left|jm\right\rangle$$

$$\mathrm{J}_\pm=\mathrm{J}_x\pm i\mathrm{J}_y$$

$$\mathrm{J}_\pm\left|jm\right\rangle=\hbar\left[j(j+1)-m(m\pm1)\right]^{1/2}\left|jm\pm1\right\rangle$$

nonzero matrix elements and "Condon Shortley" phase choice

$$\left\langle j'm'\right|\mathrm{J}^2\left|jm\right\rangle=\hbar^2 j(j+1)\delta_{j'j}\delta_{m'm}$$

$$\left\langle j'm'\right|\mathrm{J}_z\left|jm\right\rangle=\hbar m\delta_{j'j}\delta_{m'm}$$

$$\left\langle j'm'\right|\mathrm{J}_\pm\left|jm\right\rangle=\hbar\left[j(j+1)-mm'\right]^{1/2}\delta_{j'j}\delta_{m'm\pm1}$$

$\left(\mathbf{J}^2,\mathbf{J}_z,\mathbf{J}_x,\mathbf{J}_y,\mathbf{J}_+,\mathbf{J}_-\right)$ all stay within the same j quantum number

all matrix elements of $\mathbf{J}^2,\mathbf{J}_z,\mathbf{J}_x,\mathbf{J}_\pm$ are real and positive (only those of $\mathbf{J}_y$ are imaginary)

TODAY:

1. What do the matrices look like for $J=0,\frac{1}{2},1$?
2. many operators are expressed as an angular momentum times a constant: Zeeman and density matrix examples
3. other operators involve things like $\vec{q}$ or products of two angular momenta

Stark effect

Wigner-Eckart Theorem

* classify operators by commutation rule
* matrix elements in convenient basis sets
* transform between inconvenient and convenient basis sets.

$[\mathbf{p}_x, \mathbf{p}_y] = 0$

A student in 1999 suggested that he could find *f(x,y)* such that

$$\frac{\partial^2 f}{\partial x \partial y} \neq \frac{\partial^2 f}{\partial y \partial x}$$

Thus $[\mathbf{p}_x, \mathbf{p}_y] \neq 0$!

This is possible, but *f(x,y)* would have to have a form that excludes it as an acceptable $\Psi(x,y)$. Typically, such an *f(x,y)* would have to be discontinuous or have discontinuous first derivatives. For all well behaved *V(x,y)*, $\Psi(x,y)$ will have continuous first derivatives. The *f(x,y)* used to prove a commutation rule must be acceptable as a quantum mechanical wavefunction, $\Psi(x,y)$. This is a good thing because (see Lecture #22, page 22-7 or the Angular Momentum Handout)

$$e^{-ia\mathbf{p}_x/\hbar}\left|x_1\right\rangle = \left|x_1 + a\right\rangle$$

$e^{-iap_x/\hbar}$ generates a linear translation of $+a$ in x direction.

linear translations commute (but rotations do not)

This is the basis for (or a consequence of) $\left[\mathbf{p}_i, \mathbf{p}_j\right] = 0$

$$\left[\mathbf{J}_i, \mathbf{J}_j\right] = i\hbar \sum_k \varepsilon_{ijk} \mathbf{J}_k$$

Nonlecture

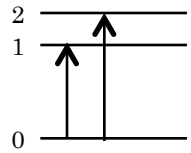
prepare (excite) **E**
evolve $e^{-i\mathbf{H}t/\hbar}$
detect **D**

e.g. basis set $|0\rangle,|1\rangle,|2\rangle$

This is artificial because it assumes that all population is moved from state 0 and transferred to states 1 and 2.

excite: $\mathbf{E}|0\rangle = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$

The "excitation matrix", **E**, creates equal amplitudes in two excited eigenstates:



$$\rho(0) = \mathbf{E}|0\rangle\langle 0|\mathbf{E}^\dagger$$

evolve: If we are in the eigenbasis of **H**

$$e^{-i\mathbf{H}t/\hbar}\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} ae^{-iE_at/\hbar} \\ be^{-iE_bt/\hbar} \\ ce^{-iE_ct/\hbar} \end{pmatrix}$$

(translation in time)

but if we are not in the eigenbasis of **H**, need $\left[\mathbf{T}e^{-i\mathbf{T}^\dagger\mathbf{H}\mathbf{T}t/\hbar}\mathbf{T}^\dagger\right] = \mathrm{U}(t,0)$

$$\rho(t) = \mathrm{T}e^{-i\mathrm{T}^\dagger\mathrm{H}\mathrm{T}t/\hbar}\mathrm{T}^\dagger\mathrm{E}|0\rangle\langle 0|\underbrace{\mathrm{E}^\dagger\mathrm{T}e^{+i\mathrm{T}^\dagger\mathrm{H}\mathrm{T}t/\hbar}\mathrm{T}^\dagger}$$

$$= \mathrm{U}(t,0)\rho(0)\mathrm{U}^\dagger(t,0)$$

$\rho(0)$ in eigenbasis of **H**

This can be confusing because it is not clear what basis ρ(0) is expressed in and what the appropriate **U** is for that basis set.

detect: **D** the "detection matrix"

$$\langle\mathbf{D}\rangle_t = \mathrm{Trace}(\rho\mathbf{D})$$

Building Blocks!

Many QM operators have the form $f\left(\vec{\mathrm{J}}\right)$

e.g. Zeeman effect $\mathrm{H}^{\text{Zeeman}} = -\gamma\vec{B}\cdot\vec{\mathrm{J}}$ $\left(\vec{B}\text{ is magnetic field}\right)$

Others have the form $f\left(\vec{\mathrm{q}}\right)$

e.g. Stark effect $\mathrm{H}^{\text{Stark}} = e\vec{\varepsilon}\cdot\vec{\mathrm{q}}$ $\left(\vec{\varepsilon}\text{ is electric field}\right)$

Others have the form of $f(\mathbf{J}_1,\mathbf{J}_2)$

e.g. spin - orbit $\mathbf{H}^{SO} = a\mathbf{L}\cdot\mathbf{S}$

We are going to want to be able to write matrix representations of these classes of operators. We are going to discover in this lecture that all of these operators have matrix representations that may be expressed as linear combinations of angular momentum matrices.

Let us begin by writing matrices for $\mathbf{J}^2$, $\mathbf{J}_z$, $\mathbf{J}_x$, $\mathbf{J}_y$, $\mathbf{J}_+$, $\mathbf{J}_-$.

$\mathbf{J}_+ = \frac{1}{2}\left(\mathbf{J}_x + i\mathbf{J}_y\right)$

j = 0 only basis state is $|jm\rangle = |00\rangle$

1×1 matrix

$$\mathbf{J}^2|00\rangle = \hbar^2\begin{pmatrix} 0 \end{pmatrix}$$

same for all components of $\mathbf{J}$

j = 1/2 $\left|\frac{1}{2}\frac{1}{2}\right\rangle$ and $\left|\frac{1}{2}-\frac{1}{2}\right\rangle$ 2×2 matrices

$$\mathrm{J}^2\left|\frac{1}{2}\frac{1}{2}\right\rangle = \hbar^2\frac{1}{2}\left(\frac{1}{2}+1\right)\left|\frac{1}{2}\frac{1}{2}\right\rangle$$

$$\mathbf{J}^{2(1/2)} = \frac{3}{4}\hbar^2\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\mathbf{J}_z^{(1/2)} = \frac{1}{2}\hbar\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\mathbf{J}_+^{(1/2)} = \hbar\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

e.g.

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \text{m = +1/2}$$

$$J_+^{(1/2)}\left|\frac{1}{2}\frac{1}{2}\right\rangle = 0$$

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{m = –1/2}$$

$$J_+^{(1/2)}\left|\frac{1}{2}-\frac{1}{2}\right\rangle = \hbar\left|\frac{1}{2}\frac{1}{2}\right\rangle$$

$$\mathbf{J}_-^{(1/2)} = \hbar\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$\mathbf{J}_x^{(1/2)} = \frac{1}{2}\left(\mathbf{J}_+ + \mathbf{J}_-\right) = \frac{1}{2}\hbar\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\mathbf{J}_y^{(1/2)} = \frac{1}{2i}\left(\mathbf{J}_+ - \mathbf{J}_-\right) = -\frac{i}{2}\hbar\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$$

verify that $\boldsymbol{J}^2 = \boldsymbol{J}_x^2 + \boldsymbol{J}_y^2 + \boldsymbol{J}_z^2$

$$\left(\mathrm{J}^{(1/2)}\right)^2 = \frac{3\hbar^2}{4}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\mathbf{J}_x^2 = \frac{\hbar^2}{4}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{\hbar^2}{4}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\mathbf{J}_y^2 = \frac{\hbar^2}{4}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \frac{\hbar^2}{4}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

An amazing amount of insight gained from this complete set of 2 × 2 matrices

$$\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\boldsymbol{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \rightarrow \mathbf{J}_x^{(1/2)}$$

$$\boldsymbol{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \rightarrow \mathbf{J}_y^{(1/2)}$$

$$\boldsymbol{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \rightarrow \mathbf{J}_z^{(1/2)}$$

3 matrices with eigenvalues ±1

CTDL, pages 417-454
1. Pauli Matrices
2. Diagonalization of 2 × 2
3. Geometric interpretation of 2 × 2 $\boldsymbol{\rho}$ in terms of fictitious spin 1/2
4. spin 1/2 $\boldsymbol{\rho}$
5. magnetic resonance

What is $[\boldsymbol{\sigma}_x, \boldsymbol{\sigma}_y] = ?$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$$

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}$$

$$\left(\left[\boldsymbol{\sigma}_x, \boldsymbol{\sigma}_y\right]\right) = 2\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = 2i\boldsymbol{\sigma}_z$$

Surprise? Why the 2?

$$\text{arbitrary } \mathbf{M} = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix} = \frac{m_{11}+m_{22}}{2}\mathbf{I} + \frac{m_{11}-m_{22}}{2}\boldsymbol{\sigma}_z + \frac{m_{12}+m_{21}}{2}\boldsymbol{\sigma}_x + i\frac{m_{12}-m_{21}}{2}\boldsymbol{\sigma}_y$$

$$\mathbf{M} = a_0\mathbf{I} + \vec{a}\cdot\vec{\boldsymbol{\sigma}}$$

a_0: scalar part of $\mathbf{M}$ — $\vec{a}$: vector part of $\mathbf{M}$

$$a_0 = \frac{1}{2}Tr(\mathbf{M})$$

Center of Gravity

$$\vec{a} = \frac{1}{2}Tr(\mathbf{M}\boldsymbol{\sigma})$$

$\mathbf{M} \leftrightarrow \boldsymbol{\rho}$

$$a_x = \frac{1}{2}\mathrm{Tr}(\mathbf{M}\boldsymbol{\sigma}_x)$$

$$a_y = \frac{1}{2}\mathrm{Tr}(\mathbf{M}\boldsymbol{\sigma}_y)$$

$$a_z = \frac{1}{2}\mathrm{Tr}(\mathbf{M}\boldsymbol{\sigma}_z)$$

Information in 2 by 2 $\boldsymbol{\rho}$ is repackaged into a 3 component vector. Visualization of dynamics!

This provides a basis for taking apart the dynamics of an arbitrary 2 × 2 $\boldsymbol{\rho}$ into dynamics of *x, y, z* fictitious spin-1/2 components. Beat the *S = 1/2* Zeeman problem to death and use it as basis for understanding the dynamics in *any* 2 × 2 space.

We have done J = 0, J = ½, now we do J = 1.

J = 1 A set of 3 × 3 matrices

$$\mathbf{J}^{2(1)} = 2\hbar^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\mathbf{J}_z^{(1)} = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$\mathbf{J}_+^{(1)} = 2^{1/2}\hbar \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\mathbf{J}_-^{(1)} = 2^{1/2}\hbar \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\mathbf{J}_x^{(1)} = 2^{-1/2}\hbar \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\mathbf{J}_y^{(1)} = 2^{-1/2}\hbar \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}$$

An example:

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\mathrm{J}_+^{(1)} \left| 11 \right\rangle = 0$$

For a 2 × 2 problem (e.g. J = 1/2), we need 4 independent 2 × 2 matrices (because there are 4 elements in a 2 × 2 matrix) in order to represent an arbitrary 2 × 2 matrix.

For the 3×3 problem, we need 9 independent 3 × 3 matrices:

$\left(x, y, z, x^2, y^2, z^2, xy, xz, yz\right)$ (because there are 9 elements in a 3 × 3 matrix)

[actually one scalar (I), three vector, five second-rank tensor]

$s + p + d = 9$

[for 2 × 2 it was $s + p = 4$].

Can you write out each of the $\mathrm{J}^{(3/2)}$ matrices (16 4 × 4 matrices)?

Nine 3 × 3 basis matrices for **J** = 1 is not nearly as nice as the 4 basis matrices for the 2 × 2 **J** = 1/2 problem. But this set of nine basis matrices turns out to be everything that is needed to "understand" and picture spin = 1 systems.

---

similarly for *j = 3/2, 2*, etc.

There are 2 lovely consequences of being able to take an arbitrary matrix and rewrite it as sum of **J** matrices.

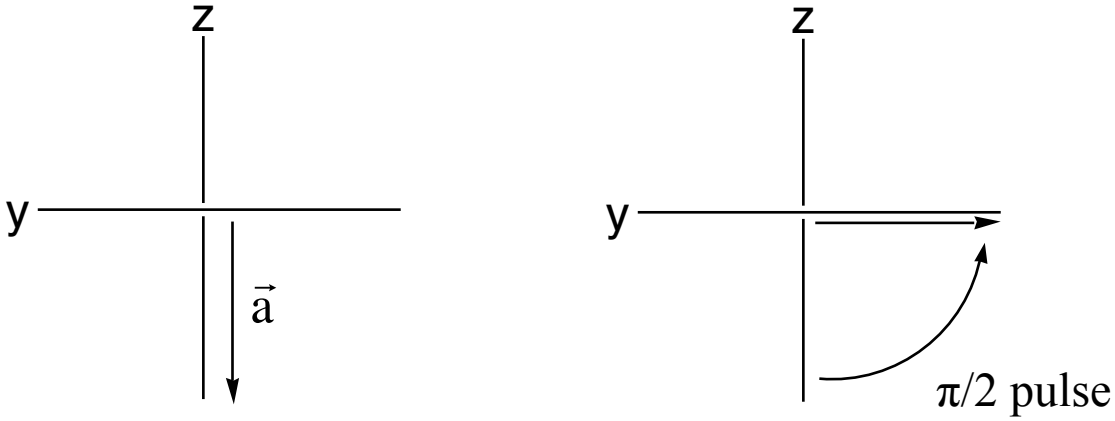
1. If **M** is the matrix of an operator – a term in the Hamiltonian – then it is clear that this operator may be re-expressed as a sum of operators, each of which behaves exactly like a (combination of) component(s) of **J** – evaluated in the $|\,jm\rangle$ basis set.

$$\mathbf{M}^{(j)} = a_0\mathbf{I} + \sum_{i,j} \overbrace{a_{1ij}}^{a_{1i}\mathbf{J}_i + b_{2ij}\mathbf{J}_i\mathbf{J}_j} \mathbf{J}_i\mathbf{J}_j + \sum_{i,j,k} c_{3ijk}\mathbf{J}_i,\mathbf{J}_j,\mathbf{J}_k + \ldots$$

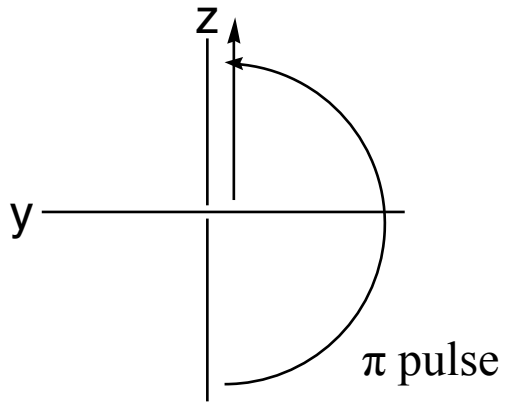
Cartesian vs. spherical tensor forms

This is the basis for classification of operators into $T^{(k)}_m$ (*k*-rank tensor, *m*-component) and the Wigner– Eckart Theorem for evaluation of matrix elements.

2. especially for 2-level systems, if **M** = $\boldsymbol{\rho}$ and $\vec{a}$ is defined from **M** as on page 24-6, then we have a vector picture that enables us too understand preparation, evolution, and detection steps:

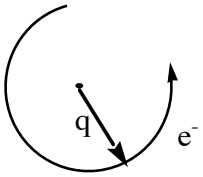


evolution of vector, fictitious B-fields



Now let's do some J = 1 examples

Zeeman effect for an $\ell = 1$ (p orbital) state



current on a circular wire

$\vec{\mathbf{L}} = \vec{\mathbf{q}} \times \vec{\mathbf{p}}$ (up out of page)

$\vec{\boldsymbol{\mu}}_L \propto -\vec{\mathbf{q}} \times \vec{\mathbf{p}}$ (down into page)

$\vec{\boldsymbol{\mu}}_L \propto -\vec{\mathbf{L}}$ $\quad \vec{\boldsymbol{\mu}}_L \equiv -\gamma\vec{\mathbf{L}}$

right hand rule

field strength

classical energy $\quad \mathrm{E} = \vec{\mathrm{B}} \cdot \boldsymbol{\mu}_{\mathrm{L}} = \mathrm{B}_z \hat{\mathrm{k}} \cdot \left(-\gamma \vec{\mathbf{L}}\right) = -\gamma \mathrm{B}_z \mathbf{L}_z$

field exclusively along z

for $L = 1$ system: $\quad \mathbf{H}^{\text{Zeeman}} = -\gamma B_z \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

<u>case (1)</u> $\quad$ Let $\Psi(0) = \left| LM_L \right\rangle = \left| \; 1 \quad 1 \; \right\rangle$

$$\rho = \left|\Psi\right\rangle\left\langle\Psi\right| = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$E_{LM_L} = E_{11} = \mathrm{Trace}(\rho \mathbf{H})$$

$$= -\hbar\gamma B_z \mathrm{Tr}\left[\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \right]$$

$$= -\hbar\gamma B_z$$

$$\boldsymbol{\rho} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

What about? $E_{10} = \mathrm{T}race\,(\boldsymbol{\rho}\mathbf{H})=0$

$E_{1-1} = +\gamma B_z \hbar$ no motion of E

<u>case (2)</u>: Let $\Psi(0) = 2^{-1/2}\left(|11\rangle + |10\rangle\right)$

$$\Psi(t) = 2^{-1/2}\left[|11\rangle e^{-iE_{11}t/\hbar} + |10\rangle e^{-i0t/\hbar}\right]$$

$$\boldsymbol{\rho}(t) = \frac{1}{2}\left(|11\rangle\langle 11| + |10\rangle\langle 10| + |11\rangle\langle 10| e^{-iE_{11}t/\hbar} + |10\rangle\langle 11| e^{+iE_{11}t/\hbar}\right)$$

$$\boldsymbol{\rho}(t) = \frac{1}{2}\begin{pmatrix} 1 & e^{-i\omega_{11}t} & 0 \\ e^{i\omega_{11}t} & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

zeroes at locations of coherence in $\boldsymbol{\rho}$

$$\mathbf{H}(B \| +z) = -\gamma B_z \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$E(t) = \langle \mathbf{H} \rangle = \mathrm{Trace}(\mathbf{H}\boldsymbol{\rho}) = -\frac{1}{2}\gamma B_z \hbar (1)$ no time evolution of E

Looked at 2 cases:

1. pure state $|11\rangle, B \| z$ $E = -\gamma B_z \hbar$
2. mixed state $2^{-1/2}\left(|11\rangle + |10\rangle\right), B \perp z$ $E = -\frac{1}{2}\gamma B_z \hbar$

mixed state always gives time-independent $\langle E \rangle$ in **H** is time-independent
NMR: oscillating B_x, B_y, cw B_z. Many wonderful things happen!

Stark Effect: Electric field

classical $E \propto \vec{\varepsilon} \cdot \left(\vec{q}_{e^-} - \vec{q}_{p^+} \right) \approx \varepsilon_z z$

so we will need matrix elements of x, y, z in $|\mathrm{jm}\rangle$ basis set. How?

Based on $\left[\mathrm{z}, \mathrm{L}_j \right] = -i\hbar \sum_k \varepsilon_{zjk} \mathrm{q}_k$ from Vector operator definition (Lecture #23) — later

Other angular momenta

1. $\boldsymbol{\ell}$ electron orbital angular momentum
2. $\mathbf{s}$ electron spin
3. $\mathbf{I}$ nuclear spin

These separate angular momenta interact with each other

spin-orbit: $\zeta(r)\, \boldsymbol{\ell} \cdot \mathbf{s}$

Zeeman: $-\gamma B_z \left(\ell_z + g_s \mathbf{s}_z + g_I \mathbf{I}_z \right)$

hyperfine: a $\mathbf{I} \bullet \mathbf{s}$

We use coupled and uncoupled basis sets: $|\ell m_\ell\rangle |s m_s\rangle \leftrightarrow |j \ell s m_j\rangle$
to evaluate all matrix elements of these multiple-operator terms.

case (3): $\Psi(0) = 2^{-1/2}(|11\rangle + |10\rangle)$

but **H** is for $\vec{B} \| x$

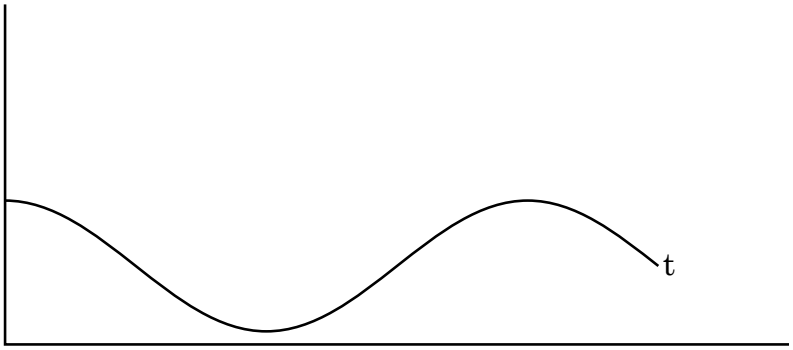
a one at the location of a coherence in ρ

$$\mathrm{H} = -\gamma B_x \hbar 2^{-1/2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Something subtle is intentionally wrong here. Can you find it?

$$E(t) = \mathrm{Tr}(\mathbf{H}\rho) = -\frac{1}{2}\gamma B_x \hbar \left[e^{+i\omega_{11}t} + e^{-i\omega_{11}t} + 0 \right]$$
$$= -\gamma B_x \hbar \cos\omega_{11} t$$

$E(t) =$



Put in t for t-dependent basis set, which is not the eigenbasis set.

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5.73 Quantum Mechanics I
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$H^{SO} + H^{Zeeman}$
Coupled vs. Uncoupled Basis Sets

Last time:

matrices for $\mathbf{J}_2$, $\mathbf{J}_+$, $\mathbf{J}_-$, $\mathbf{J}_z$, $\mathbf{J}_x$, $\mathbf{J}_y$ in $|jm_j\rangle$ basis for J = 0, ½, 1

Pauli spin ½ matrices

arbitrary 2 × 2 matrix $M = a_0 I + \vec{a}_1 \cdot \vec{\sigma}$ decomposed as scalar plus vector.

When **M** is $\boldsymbol{\rho}$ → visualization via fictitious vector in fictitious B-field.

When **M** is a term in **H** → idea that arbitrary operator can be decomposed as a sum of the terms that behave like components of J = 0, J = 1, J = 2… This leads to *spherical tensor algebra*.

types of operators

$a\mathrm{J}$	e.g. magnetic moment (a is a known constant or a function of r)
$\vec{\mathrm{q}}$	how to evaluate matrix elements (e.g. Stark Effect)
$\mathrm{J}_1 \cdot \mathrm{J}_2$	e.g. Spin-Orbit

Special simplification of Trace (**AH**)

For example

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{H} = \begin{pmatrix} H_{11} & H_{12} & H_{13} \\ H_{21} & H_{22} & H_{23} \\ H_{31} & H_{32} & H_{33} \end{pmatrix}$$

$$\mathbf{AH} = \begin{pmatrix} H_{21} & H_{22} & H_{23} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

simpler

extreme simplification!

$$\mathrm{Trace}(\mathbf{AH}) = H_{21}$$

$\mathbf{A}_{12}$ picks out only $\mathbf{H}_{21}$, $\mathbf{A}_{21}$ picks out only $\mathbf{H}_{12}$.

Extreme labor saving trick!

TODAY:

1. $\mathbf{H}^{SO} + \mathbf{H}^{Zeeman}$ as illustrative
2. Dimension of two basis sets, $|\, JLSM_J\rangle$ and $|\, LM_LSM_S\rangle$, is the same
3. matrix elements of $\mathbf{H}^{SO}$ in both basis sets
4. matrix elements of $\mathbf{H}^{Zeeman}$ in both basis sets
5. ladder operators and orthogonality for transformation between basis sets. Necessary to be able to evaluate matrix elements of $\mathbf{H}^{Zeeman}$ in "coupled basis". Why? Because coupled basis set does not explicitly reveal the effects of $\mathbf{L}_z$ or $\mathbf{S}_z$.

Nos. 3, 4 and 5 will be repeated in Lecture #26.

Suppose we have 2 kinds of angular momenta, which can be coupled to each other to form a *total* angular momentum.

$\left.\begin{array}{ll}\vec{L} & \text{orbital} \\ \vec{S} & \text{spin}\end{array}\right\}$ operate on different coordinates or in different vector spaces

$\vec{J}=\vec{L}+\vec{S}$ total

The components of **L,S,** and **J** each follow the standard angular momentum definition commutation rule, but, in addition

$$\left[\vec{L},\vec{S}\right]=0 \quad , \quad \left[J_i,L_j\right]=i\hbar\sum_k \varepsilon_{ijk}L_k$$

$$\left[J_i,S_j\right]=i\hbar\sum_k \varepsilon_{ijk}S_k.$$

These commutation rules specify that **L** and **S** act like vectors with respect to **J** but as scalars with respect to each other.

$$\vec{\mathbf{J}} \rightarrow \left| jm_j \right\rangle$$
$$\vec{\mathbf{L}} \rightarrow \left| \ell m_\ell \right\rangle$$
$$\vec{\mathbf{S}} \rightarrow \left| sm_s \right\rangle$$

Coupled $|j\ell sm_j\rangle$ vs. uncoupled $|\ell m_\ell\rangle|sm_s\rangle$ representations.

* matrix elements of certain operators are more convenient in one basis set than the other
* a unitary transformation between basis sets must exist
* limiting cases for energy level patterns (and Zeeman tuning rates and intensities for transitions into eigenstates) → assignment; determination of key parameters, structure, and dynamics

matrix elements of ℓ and $\mathbf{s}$ will each give a factor of $\hbar$

1.
$$\mathrm{H}^{SO} = \xi(r)\,\ell\cdot\mathrm{s} \equiv \frac{\zeta_{n\ell}}{\hbar}\,\ell\cdot\mathrm{s}$$

each will give a factor of $\hbar$

anomalous g - value of e^-

$$\mathrm{H}^{Zeeman} = -\gamma B_z\left(\ell_z + 2\mathrm{s}_z\right) \equiv -\left(\omega_0\right)\left(\ell_z + 2\mathrm{s}_z\right)$$

($\zeta_{n\ell}$ and ω_0 are in units of rad/s)

* evaluate matrix elements in both basis sets
* look at energy levels and their Zeeman tuning rate in *high field* $\left|\gamma B_z\right| \gg \zeta_{n\ell}$ limit
* and in *low field* $\left|\gamma B_z\right| \ll \zeta_{n\ell}$ limit

Notation: { lower case for $1e^-$ atom angular momenta; upper case for many - e^- angular momenta

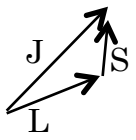
<u>two different CSCOs</u>

a)	$\mathrm{H}^{\mathrm{elect}}, \mathrm{J}^2, \mathrm{J}_z, \mathrm{L}^2, \mathrm{S}^2$	coupled basis	recall tensor product states and "entanglement"
	$\left\|nJLSM_{\mathrm{J}}\right\rangle$	(can't be factored)	
b)	$\mathrm{H}^{\mathrm{elect}}, \mathrm{L}^2, \mathrm{L}_z, \mathrm{S}^2, \mathrm{S}_z$	uncoupled basis	
	$\left\|nLM_L\right\rangle\left\|SM_S\right\rangle$	(*explicitly* factored)	

2. Coupled and Uncoupled Basis Sets have the same dimension

COUPLED $\vec{J} = \vec{L} + \vec{S}$ $\quad |L-S| \le J \le L+S$

each J has $2J+1$ M_J's



every term in this sum has 2L + 1 and there are 2S + 1 of them. The second term shows that the S,S–1,…–S terms in sum all cancel.

$J =$		Dimension	
	$L+S$	$2(L+S)+1$	Every allowed value of J contributes $2L$ + 1 to sum. How many allowed values of J are there?
	$L+S-1$	$2(L+S-1)+1$	
	$L+S-2$	$2(L+S-2)+1$	
	…..	……	If L > S, there are 2S + 1 terms in sum.
	$\lvert L-S\rvert$	$2(\lvert L-S\rvert)+1$	

$$(2S+1)(2L+1) + \underbrace{2[S+(S-1)+\cdots(-S)]}_{=0} = \underbrace{(2S+1)(2L+1)}_{\Uparrow}$$

total dimension of basis set for specified L and S

UNCOUPLED $\underbrace{LM_L}_{2L+1}\underbrace{SM_S}_{2S+1}$ total dimension $(2L+1)(2S+1)$ again

There is a term by term correspondence between the 2 basis sets $\therefore$ a transformation must exist:

Coupled basis state in terms of uncoupled basis states:

$$\left|JLSM_J\right\rangle = \sum_{M_L} a_{M_L} \left|LM_L\right\rangle \left|S \underbrace{M_S = M_J - M_L}_{\text{constraint}}\right\rangle$$

Trade J, M_J for M_L, M_S, but $M_J = M_L + M_S$.

Going in the opposite direction: express uncoupled basis state in terms of coupled basis states:

$$\text{OR}\quad \left|LM_L\right\rangle\left|SM_S\right\rangle = \sum_{J=|L-S|}^{L+S} b_J \left|JLS\underbrace{M_J = M_L + M_S}_{\text{constraint}}\right\rangle$$

3. Matrix elements of $H^{SO} = \frac{\zeta_{n\ell}}{\hbar}\ell\cdot s$

A. Coupled Representation

$$\vec{\mathbf{J}} = \vec{\mathbf{L}} + \vec{\mathbf{S}} \qquad \mathbf{J}^2 = \mathbf{L}^2 + \mathbf{S}^2 + 2\mathbf{L}\cdot\mathbf{S}$$

L and S commute because they operate in different vector spaces

$$\mathbf{L}\cdot\mathbf{S} = \frac{\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2}{2}$$

(useful trick!)

$$\left\langle J'L'S'M_J'\left|\mathbf{L}\cdot\mathbf{S}\right|JLSM_J\right\rangle = \left(\hbar^2/2\right)\left[J(J+1) - L(L+1) - S(S+1)\right]\delta_{J'J}\delta_{L'L}\delta_{S'S}\delta_{M_J'M_J}$$

an entirely diagonal matrix.

B. Uncoupled Representation: work out all of the matrix elements.

$\mathbf{L}\cdot\mathbf{S} = \mathbf{L}_z\mathbf{S}_z + \tfrac{1}{2}(\mathbf{L}_+\mathbf{S}_- + \mathbf{L}_-\mathbf{S}_+)$: because $\mathbf{L}_+\mathbf{S}_- + \mathbf{L}_-\mathbf{S}_+ = (\mathbf{L}_x + i\mathbf{L}_y)(\mathbf{S}_x - i\mathbf{S}_y)$

diagonal off-diagonal $+ (\mathbf{L}_x - i\mathbf{L}_y)(\mathbf{S}_x + i\mathbf{S}_y) = 2(\mathbf{L}_x\mathbf{S}_x + \mathbf{L}_y\mathbf{S}_y)$

$$\left\langle L'M_L'S'M_S'\left|\mathrm{L}\cdot\mathrm{S}\right|LM_LSM_S\right\rangle = \hbar^2\delta_{L'L}\delta_{S'S}\times$$

can't change L can't change S

$$\left\{\left[M_LM_S\delta_{M_L'M_L}\delta_{M_S'M_S}\right] + \frac{1}{2}\left[L(L+1) - M_L'M_L\right]^{1/2}\times\right.$$

$$\left.\left[S(S+1) - M_S'M_S\right]^{1/2}\delta_{M_L'M_L\pm1}\times\delta_{M_S'M_S\mp1}\right\} \qquad \Delta M_L = -\Delta M_S = 0,\pm1$$

Non-Lecture notes for evaluated matrices

$S = 1/2, \qquad L = 0,1,2 \qquad \boxed{^2S,\ ^2P,\ ^2D \text{ states}}$

NONLECTURE for $\mathbf{H}^{SO}$: **COUPLED** BASIS

$^{2S+1}L_J$

$^2S_{1/2}$ $\quad H^{SO}_{COUPLED} = \frac{\hbar}{2}\zeta_{ns}(0) = 0$ $\quad$ a 1 × 1 matrix with matrix element = 0

$^2P_{1/2\ \&\ 3/2}$ $\quad H^{SO}_{COUPLED} = \frac{\hbar}{2}\zeta_{np}$

$$\left(\begin{array}{cc|cccc} -2 & 0 & 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right)$$

$J = 1/2$ $\quad (2 \times 2)$

$J = 3/2$ $\quad (4 \times 4)$

	L	J	$J(J+1)$	$-L(L+1)$	$-S(S+1)$	=
$\left(^2S_{1/2}\right)$	0	1/2	3/4	0	−3/4	0
$\left(^2P_{1/2}\right)$	1	1/2	3/4	−2	−3/4	−2
$\left(^2P_{3/2}\right)$	1	3/2	15/4	−2	−3/4	+1
$\left(^2D_{3/2}\right)$	2	3/2	15/4	−6	−3/4	−3
$\left(^2D_{5/2}\right)$	2	5/2	35/4	−6	−3/4	+2

$J = 3/2$

$^2D_{\frac{3}{2}\text{ and }\frac{5}{2}}$ $\quad H^{SO}_{COUPLED} = \frac{\hbar}{2}\zeta_{nd}$

$$\left(\begin{array}{cccc|cccccc} -3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{array}\right)$$

(4×4)

(6×6)

$J = 5/2$

center of gravity rule: trace of matrix = 0
(obeyed for all *scalar* terms in **H**)

^{2S+1}L NONLECTURE for H^{SO} : **UNCOUPLED** BASIS

^2S $H^{SO}_{UNCOUPLED} = \hbar\zeta_{ns}(1/2 \cdot 0) = (0)$ (1×1)

^2P $H^{SO}_{UNCOUPLED} = \hbar\zeta_{np} \times$

M_J	M_L	M_S							
$3/2$	1	$1/2$	$1/2$	0	0	0	0	0
$1/2$	1	$-1/2$	0	$-1/2$	$2^{-1/2}$	0	0	0
$1/2$	0	$1/2$	0	$2^{-1/2}$	0	0	0	0
$-1/2$	0	$-1/2$	0	0	0	0	$2^{-1/2}$	0
$-1/2$	-1	$1/2$	0	0	0	$2^{-1/2}$	$-1/2$	0
$-3/2$	-1	$-1/2$	0	0	0	0	0	$1/2$

Each box along main diagonal is for one value of $M_J = M_L + M_S$.

^2D $H^{SO}_{UNCOUPLED} = \hbar\zeta_{nd} \times$

M_J	M_L	M_S										
$5/2$	2	$1/2$	1	0	0	0	0	0	0	0	0	0
$3/2$	2	$-1/2$	0	-1	1	0	0	0	0	0	0	0
$3/2$	1	$1/2$	0	1	$1/2$	0	0	0	0	0	0	0
$1/2$	1	$-1/2$	0	0	0	$-1/2$	$(3/2)^{1/2}$	0	0	0	0	0
$1/2$	0	$1/2$	0	0	0	$(3/2)^{1/2}$	0	0	0	0	0	0
$-1/2$	0	$-1/2$	0	0	0	0	0	0	$(3/2)^{1/2}$	0	0	0
$-1/2$	-1	$1/2$	0	0	0	0	0	$(3/2)^{1/2}$	$-1/2$	0	0	0
$-3/2$	-1	$-1/2$	0	0	0	0	0	0	0	$1/2$	1	0
$-3/2$	-2	$1/2$	0	0	0	0	0	0	0	1	-1	0
$-5/2$	-2	$-1/2$	0	0	0	0	0	0	0	0	0	1

End of Non-Lecture

4. Matrix Elements of $\mathbf{H}^{\text{Zeeman}} = -\gamma B_z(\mathbf{L}_z + 2\mathbf{S}_z)$

   A. Very easy in uncoupled representation

$$\begin{aligned} H^{Zeeman}_{\text{uncoupled}} &= -\gamma B_z \left\langle L'M'_L S'M'_S \middle| L_z + 2S_z \middle| LM_L SM_S \right\rangle \\ &= -\gamma B_z \hbar (M_L + 2M_S)\delta_{L'L}\delta_{S'S}\delta_{M'_L M_L}\delta_{M'_S M_S} \end{aligned}$$

   strictly diagonal

   B. Coupled representation

$$\mathbf{L}_z + 2\mathbf{S}_z = \underbrace{\mathbf{J}_z}_{\text{easy}} + \underbrace{\mathbf{S}_z}_{\text{hard — no clue!}}$$

   can't evaluate matrix elements in coupled representation without a new trick, discussed in item #5

---

5. If we wish to work in *coupled* representation, because it diagonalizes $\mathbf{H}^{SO}$, we need to find the transformation between coupled and uncoupled representations.

$$\left|JLSM_J\right\rangle = \sum_{M_L} a_{M_L} \left|LM_L SM_S = M_J - M_L\right\rangle$$

lengthy procedure: use $J_\pm = L_\pm + S_\pm$ and orthogonality

Always start with an extreme M_L, M_S basis state, where we are assured of a trivial 1 to 1 correspondence between basis sets:

$$M_L = L, \quad M_S = S, \quad M_J = M_L + M_S = L + S, \quad J = L + S$$

$$\left|J = L+S \;\; LSM_J = L+S\right\rangle = \left|LM_L = L \;\; SM_S = S\right\rangle$$

coupled uncoupled

Now the fun begins …

Apply $\mathbf{J}_-$ to both sides of the equation:

$$\mathbf{J}_- \left|\overbrace{L+S}^{J}\ LS\ \overbrace{L+S}^{M_J}\right\rangle = \left(\mathbf{L}_- + \mathbf{S}_-\right)\left|LM_L = L\ SM_S = S\right\rangle$$

$$\begin{bmatrix}(L+S)(L+S+1)\\ -(L+S)(L+S-1)\end{bmatrix}^{1/2}\left|L+S\ \ LS\ \ L+S-1\right\rangle = \left[L(L+1)-L(L-1)\right]^{1/2}\left|LL-1SS\right\rangle + \left[S(S+1)-S(S-1)\right]^{1/2}\left|LLSS-1\right\rangle$$

Thus we have derived an equality between one coupled basis state and a specific linear combination of two uncoupled basis states.

There is only one other orthogonal linear combination that belongs to the same value of $M_L + M_S = M_J$: it must belong to the $\underbrace{\left|L+S-1\right.}_{\text{lowered J}}\left.LS\ L+S-1\right\rangle$ basis state.

NONLECTURE
Work this out for ^{2}P using $\mathbf{J}^- = \mathbf{L}^- + \mathbf{S}^-$

$$\left|JLSM_J\right\rangle = \underbrace{\left|3/2\ \ 1\ \ 1/2\ \ 3/2\right\rangle}_{\text{}} = \left|LM_LSM_S\right\rangle = \underbrace{\left|1\ \ 1\ \ 1/2\ \ 1/2\right\rangle}_{\text{}}$$

$$\left|JLSM_J - 1\right\rangle = \frac{2^{1/2}\left|1\ \ 0\ \ 1/2\ \ 1/2\right\rangle + \left|1\ \ 1\ \ 1/2\ \ -1/2\right\rangle}{3^{1/2}}$$

now use orthogonality:

$$\left|J-1LSM_J-1\right\rangle = \left|1/2\ \ 1\ \ 1/2\ \ 1/2\right\rangle = \frac{-\left|1\ \ 0\ \ 1/2\ \ 1/2\right\rangle + 2^{1/2}\left|1\ \ 1\ \ 1/2\ \ -1/2\right\rangle}{3^{1/2}}.$$

Continue laddering down to get all four J = 3/2 and all two J = 1/2 basis states.

$$\left|3/2\ \ 1\ \ 1/2\ \ -1/2\right\rangle = \left(\frac{2}{3}\right)^{1/2}\left|1\ \ 0\ \ 1/2\ \ -1/2\right\rangle + \left(\frac{1}{3}\right)^{1/2}\left|1\ \ -1\ \ 1/2\ \ 1/2\right\rangle$$

$$\left|3/2\ \ 1\ \ 1/2\ \ -3/2\right\rangle = \left|1\ \ -1\ \ 1/2\ \ -1/2\right\rangle$$

$$\left|1/2\ \ 1\ \ 1/2\ \ 1/2\right\rangle = -\left(\frac{1}{3}\right)^{1/2}\left|1\ \ 0\ \ 1/2\ \ -1/2\right\rangle + \left(\frac{2}{3}\right)^{1/2}\left|1\ \ -1\ \ 1/2\ \ 1/2\right\rangle$$

You work out the transformation for ^{2}D!

Next step will be to evaluate $\mathbf{H}^{SO} + \mathbf{H}^{Zeeman}$ in both coupled and uncoupled basis sets and look for limiting behavior.

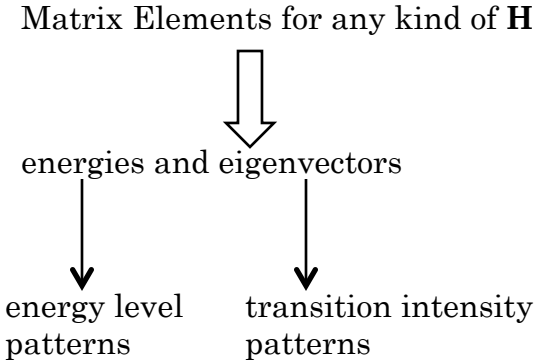
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$H^{SO} + H^{Zeeman}$: Two Incompatible Terms

My goal for 5.73



Non-degenerate Perturbation Theory
Quasi-degenerate Perturbation Theory
Exact diagonalization

$$\rho(t) = \sum_k p_k \left|\psi_k(t)\right\rangle\left\langle\psi_k(t)\right|$$

$$\langle \mathbf{A} \rangle_t = \mathrm{Trace}\left(\mathrm{A}\rho\right)$$

$$i\hbar \frac{\partial \rho}{\partial t} = \left[\mathbf{H}, \rho\right]$$

Dynamics.

Previous Lecture

$\mathbf{H}^{SO} = \xi(r)\boldsymbol{\ell}\cdot\mathbf{s}$ product of 2 angular momenta

$\mathbf{H}^{Zeeman} = -\hbar B_z \gamma\left(\mathbf{L}_z + 2\mathbf{S}_z\right)$ sum of 2 angular momenta

2 incompatible **H**'s
Natural basis sets $|\,nJLSM_J\rangle$ for $\mathbf{H}^{SO}$
$|\,nLM_L\rangle\,|\,SM_S\rangle$ for $\mathbf{H}^{Zeeman}$

Two types of building blocks:

* uncoupled: each angular momentum reports directly to lab frame

* coupled; each angular momentum couples in body frame to a sum-angular momentum which reports to lab frame

There are many angular momenta: decide on most convenient sequences of couplings and uncouplings. 6-j coefficients give transformation between coupling schemes.

For coupled↔uncoupled transformation, need to know that the dimensions of the two basis sets are the same. Unitary transformation must exist because operators are Hermitian.

Use ladders [$\mathbf{J}_\pm = \mathbf{L}_\pm + \mathbf{S}_\pm$] plus orthogonality to work out the transformation, one pair of basis states at a time.

More compact and generally applicable methods will exploit 3-j and 6-j coefficients for transformations between basis states.

Today:

A note before starting: atoms are spherical, space is isotropic, and distortions from angular symmetry and spatial isotropy are treated as small perturbations.

1. Results from $|\mathrm{JLSM_J}\rangle_\mathrm{c} \leftrightarrow |\mathrm{LMSM_S}\rangle_\mathrm{u}$ c = coupled, u = uncoupled

2. $\mathbf{H}^{\mathrm{Zeeman}}$ in coupled basis set.

3. Ways to deal with 2 incompatible terms.

   Easiest way is by Correlation Diagram (guided by non-crossing rule): states of same rigorous symmetry cannot cross. For coupled and uncoupled representations, the rigorous (conserved) symmetry is $\mathrm{M_J}$.

   We get a pattern without calculations. Gives guidance for what to expect in an intermediate case.

   War between 2 limiting cases.

   One term gives $\Delta E_{ij}^{(0)} \neq 0$ which tries to preserve one coupling case
   and one term gives $H_{ij}^{(1)} \neq 0$ which tries to destroy that limiting case.

4. Stepwise treatment
   A. Correlation Diagram
   B. Non-degenerate Perturbation Theory: $E^{(0)}$, $E^{(1)}$, $E^{(2)}$ using either basis set as framework [which term is $\mathbf{H}^{(0)}$]? The other term contributes to $E^{(1)}$ and $E^{(2)}$.

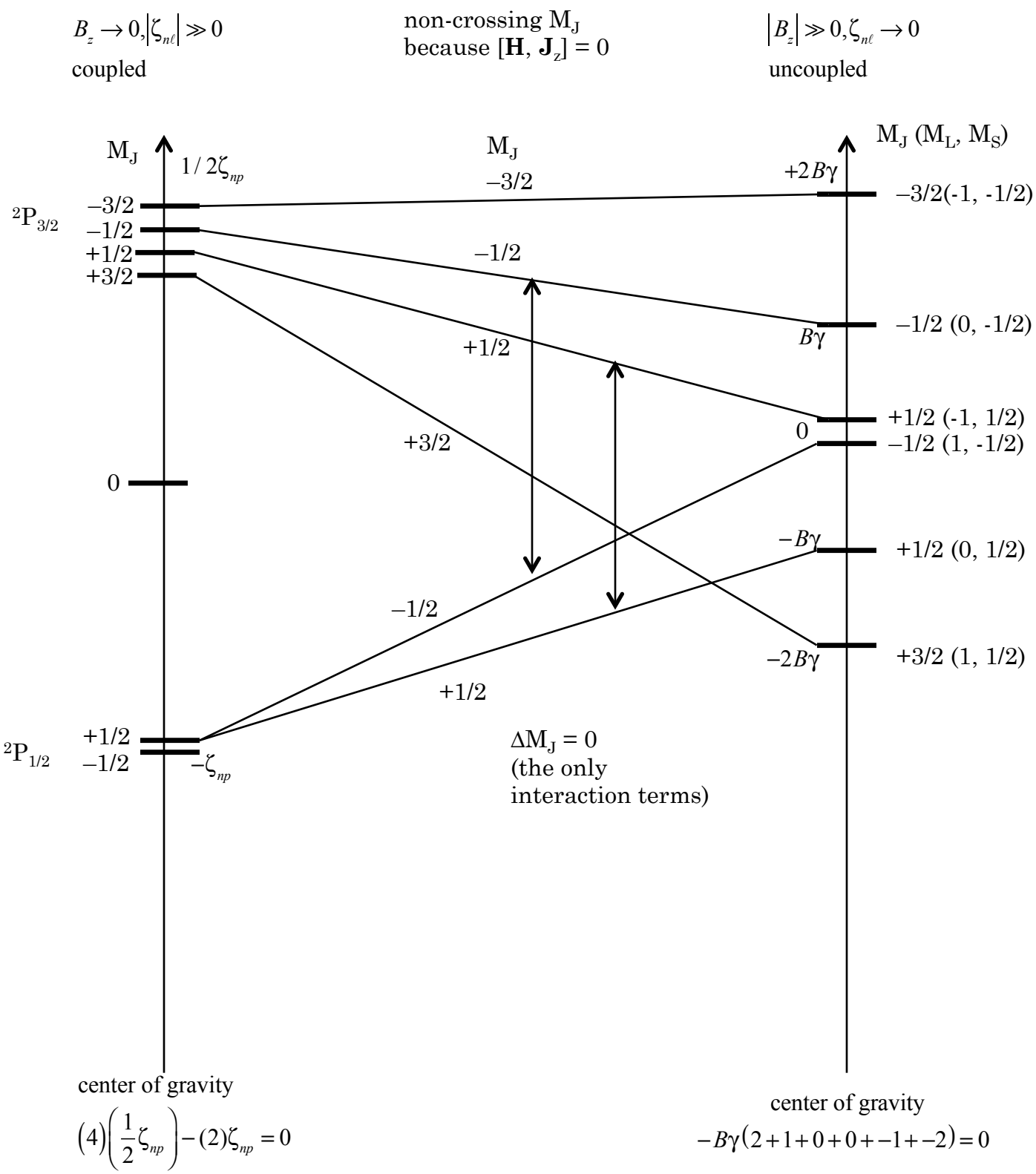
5. Limiting patterns and types of distortion from the simple pattern

   * energy levels
   * transition intensities

   Correlation diagram: left side is $\mathbf{H}^{\mathrm{SO}}$, right side is $\mathbf{H}^{\mathrm{Zeeman}}$.

Typically, there is a center-of-gravity rule for each limit.

^{2}P State Correlation Diagram



Transformation Uncoupled↔Coupled for $\mathbf{H}^{\text{Zeeman}}$

$$\mathbf{H}^{\text{SO}} + \mathbf{H}^{\text{Zeeman}} = \frac{\zeta_{n\ell}}{\hbar}\,\boldsymbol{\ell}\cdot\mathbf{s} - \gamma B_z\left(\mathbf{L}_z + 2\mathbf{S}_z\right)$$

In coupled basis set

$$\mathbf{H}^{\text{SO}} = \frac{\hbar\zeta_{n\ell}}{2}\left[J(J+1) - L(L+1) - S(S+1)\right]$$

$\mathbf{H}^{\text{Zeeman}}$ need to use coupled→uncoupled transformation

$$\begin{aligned}
|\text{JLSM}_\text{J}\rangle_\text{c} \quad &= \quad |\text{LM}_\text{L}\text{SM}_\text{S}\rangle_\text{u} \\
\left|\frac{3}{2}1\frac{1}{2}\frac{3}{2}\right\rangle_c &= \left|11\frac{1}{2}\frac{1}{2}\right\rangle_u \\
\left|\frac{3}{2}1\frac{1}{2}\frac{1}{2}\right\rangle_c &= \left(\frac{2}{3}\right)^{1/2}\left|10\frac{1}{2}\frac{1}{2}\right\rangle_u + \left(\frac{1}{3}\right)^{1/2}\left|11\frac{1}{2}-\frac{1}{2}\right\rangle_u \\
\left|\frac{1}{2}1\frac{1}{2}\frac{1}{2}\right\rangle_c &= -\left(\frac{1}{3}\right)^{1/2}\left|10\frac{1}{2}\frac{1}{2}\right\rangle_u + \left(\frac{2}{3}\right)^{1/2}\left|11\frac{1}{2}-\frac{1}{2}\right\rangle_u \\
\left|\frac{3}{2}1\frac{1}{2}-\frac{1}{2}\right\rangle_c &= \left(\frac{2}{3}\right)^{1/2}\left|10\frac{1}{2}-\frac{1}{2}\right\rangle_u + \left(\frac{1}{3}\right)^{1/2}\left|1-1\frac{1}{2}\frac{1}{2}\right\rangle_u \\
\left|\frac{1}{2}1\frac{1}{2}-\frac{1}{2}\right\rangle_c &= -\left(\frac{1}{3}\right)^{1/2}\left|10\frac{1}{2}-\frac{1}{2}\right\rangle_u + \left(\frac{2}{3}\right)^{1/2}\left|1-1\frac{1}{2}\frac{1}{2}\right\rangle_u \\
\left|\frac{3}{2}1\frac{1}{2}-\frac{3}{2}\right\rangle_c &= \left|1-1\frac{1}{2}-\frac{1}{2}\right\rangle_u
\end{aligned}$$

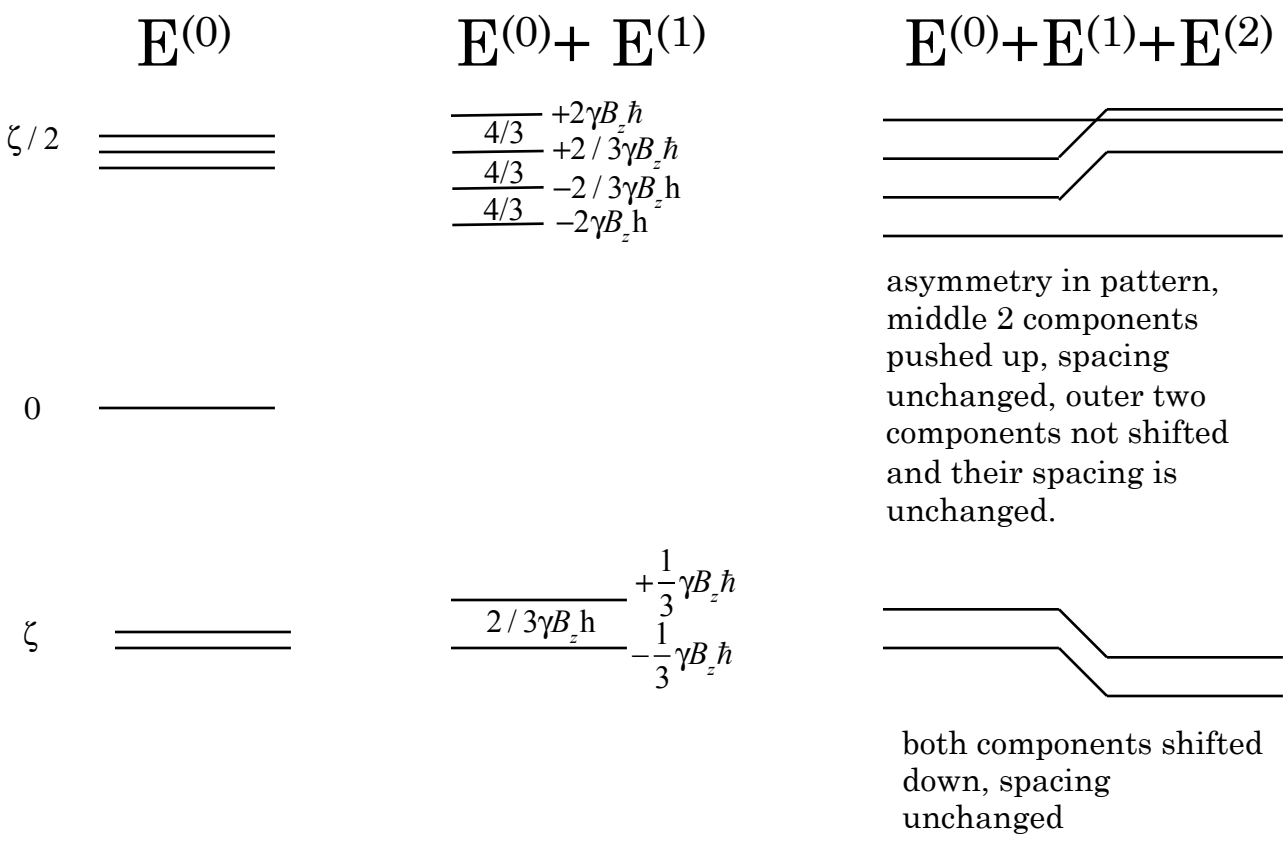
Matrix Elements of $\mathbf{H}^{\text{Zeeman}}$ in Coupled Basis

diagonal
$$\left\langle \frac{3}{2}1\frac{1}{2}\frac{3}{2}\middle|\mathbf{H}^{\text{Zeeman}}\middle|\frac{3}{2}1\frac{1}{2}\frac{3}{2}\right\rangle_c = -\gamma B_z \left\langle 11\frac{1}{2}\frac{1}{2}\middle|\mathbf{L}_z + 2\mathbf{S}_z\middle|11\frac{1}{2}\frac{1}{2}\right\rangle_u$$
$$= -\hbar\gamma B_z(1+1) = -2\hbar\gamma B_z$$
$$\left\langle \frac{3}{2}1\frac{1}{2}-\frac{3}{2}\middle|\mathbf{H}^{\text{Zeeman}}\middle|\frac{3}{2}1\frac{1}{2}-\frac{3}{2}\right\rangle_c = -\gamma B_z \left\langle 1-1\frac{1}{2}-\frac{1}{2}\middle|\mathbf{L}_z + 2\mathbf{S}_z\middle|1-1\frac{1}{2}-\frac{1}{2}\right\rangle_u$$
$$= -\hbar\gamma B_z(-1+-1) = 2\hbar\gamma B_z$$
$$\left\langle \frac{3}{2}1\frac{1}{2}\frac{1}{2}\middle|\mathbf{H}^{\text{Zeeman}}\middle|\frac{3}{2}1\frac{1}{2}\frac{1}{2}\right\rangle_c = -\gamma B_z \left[\left(\frac{2}{3}\right)_u \left\langle 10\frac{1}{2}\frac{1}{2}\middle|\mathbf{L}_z + 2\mathbf{S}_z\middle|10\frac{1}{2}\frac{1}{2}\right\rangle_u + \left(\frac{1}{3}\right)_u \left\langle 11\frac{1}{2}-\frac{1}{2}\middle|\mathbf{L}_z + 2\mathbf{S}_z\middle|11\frac{1}{2}-\frac{1}{2}\right\rangle_u\right]$$
$$= -\hbar\gamma B_z\left[\frac{2}{3}(0+1)+\frac{1}{3}(1-1)\right]$$
$$= -\frac{2}{3}\hbar\gamma B_z$$
$$\left\langle \frac{1}{2}1\frac{1}{2}\frac{1}{2}\middle|\mathbf{H}^{\text{Zeeman}}\middle|\frac{1}{2}1\frac{1}{2}\frac{1}{2}\right\rangle_c = -\gamma B_z \left[\left(\frac{1}{3}\right)_u \left\langle 10\frac{1}{2}\frac{1}{2}\middle|\mathbf{L}_z + 2\mathbf{S}_z\middle|10\frac{1}{2}\frac{1}{2}\right\rangle_u + \frac{2}{3}\left\langle 11\frac{1}{2}-\frac{1}{2}\middle|\mathbf{L}_z + 2\mathbf{S}_z\middle|11\frac{1}{2}-\frac{1}{2}\right\rangle\right]$$
$$= -\hbar\gamma B_z\left[\frac{1}{3}(0+1)+\frac{2}{3}(1-1)\right] = -\frac{1}{3}\hbar\gamma B_z$$

off-diagonal
$$\left\langle \frac{3}{2}1\frac{1}{2}\frac{1}{2}\middle|\mathbf{H}^{\text{Zeeman}}\middle|\frac{1}{2}1\frac{1}{2}\frac{1}{2}\right\rangle_c = -\hbar\gamma B_z\left[-\frac{2}{3}^{1/2}(0+1)+\frac{2}{3}^{1/2}(1-1)\right] = +\frac{2}{3}^{1/2}\hbar\gamma B_z$$
$$\left\langle \frac{3}{2}1\frac{1}{2}-\frac{1}{2}\middle|\mathbf{H}^{\text{Zeeman}}\middle|\frac{3}{2}1\frac{1}{2}-\frac{1}{2}\right\rangle_c = -\hbar\gamma B_z\left[\frac{2}{3}(0-1)+\frac{1}{3}(-1+1)\right] = +\frac{2}{3}\hbar\gamma B_z$$
$$\left\langle \frac{1}{2}1\frac{1}{2}-\frac{1}{2}\middle|\mathbf{H}^{\text{Zeeman}}\middle|\frac{1}{2}1\frac{1}{2}-\frac{1}{2}\right\rangle_c = -\hbar\gamma B_z\left[\frac{1}{3}(0-1)+\frac{2}{3}(-1+1)\right] = +\frac{1}{3}\hbar\gamma B_z$$
$$\left\langle \frac{3}{2}1\frac{1}{2}-\frac{1}{2}\middle|\mathbf{H}^{\text{Zeeman}}\middle|\frac{1}{2}1\frac{1}{2}-\frac{1}{2}\right\rangle_c = -\hbar\gamma B_z\left[-\frac{2}{3}(0-1)\right] = -\hbar\gamma B_z\left(\frac{2}{3}^{1/2}\right)$$
$$\left\langle \frac{3}{2}1\frac{1}{2}-\frac{3}{2}\middle|\mathbf{H}^{\text{Zeeman}}\middle|\frac{3}{2}1\frac{1}{2}-\frac{3}{2}\right\rangle_c = +2\hbar\gamma B_z$$

$$\mathbf{H}^{\text{Zeeman}} = \hbar\gamma B_z \begin{pmatrix} -2 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2}{3} & -\frac{2}{3}^{1/2} & 0 & 0 & 0 \\ 0 & -\frac{2}{3}^{1/2} & -\frac{1}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{3} & \frac{2}{3}^{1/2} & 0 \\ 0 & 0 & 0 & \frac{2}{3}^{1/2} & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & +2 \end{pmatrix}$$

Now that we have the full $\mathbf{H}^{\text{Zeeman}}$ matrix in the coupled basis set, we can analyze it by non-degenerate perturbation theory in the strong spin-orbit limit.



What about eigenvectors? Relative intensities of transitions into shifted components are altered.

You should draw a similar diagram for the strong Zeeman limit.

Coupled **H**

$m_J = 3/2$ $\quad \zeta/2 - 2\gamma B_z$

$J = 3/2$ $\quad \zeta/2 - \frac{2}{3}\gamma B_z \quad \frac{2^{1/2}}{3}\gamma B_z$

$m_J = 1/2$

$J = 1/2$ $\quad sym \quad -\zeta - \frac{1}{3}\gamma B_z$

$J = 3/2$ $\quad \zeta/2 + \frac{2\gamma}{3}B_z \quad -\frac{2^{1/2}}{3}\gamma B_z$

$m_J = -1/2$

$J = 1/2$ $\quad \text{sym} \quad -\zeta + \frac{1}{3}\gamma B_z$

$m_J = -3/2$ $\quad \zeta/2 + 2\gamma B_z$

Uncoupled **H**

$\zeta/2 - 2\gamma B_z$ $\quad m_L = 1, m_S = 1/2$

$-\zeta/2 \quad 2^{-1/2}\zeta$ $\quad m_L = 1, m_S = -1/2$

$\text{sym} \quad -\gamma B_z$ $\quad m_L = 0, m_S = 1/2$

$\gamma B_z \quad 2^{-1/2}\zeta$ $\quad m_L = 0, m_S = -1/2$

$\text{sym} \quad -\zeta/2$ $\quad m_L = -1, m_S = 1/2$

$\zeta/2 + 2\gamma B_z$ $\quad m_L = -1, m_S = -1/2$

$$E_{J,M_J} = E_{3/2,3/2} = \zeta/2 - 2\gamma B_z$$

$$E_{\pm,1/2} = \left(-\frac{\zeta}{4} - \frac{\gamma B_z}{2}\right) \pm \left[\frac{9}{16}\zeta^2 + \frac{\left(\gamma B_z\right)^2}{4} - \frac{\gamma B_z \zeta}{4}\right]^{1/2}$$

$$E_{\pm,-1/2} = \left(-\frac{\zeta}{4} + \frac{\gamma B_z}{2}\right) \pm \left[\frac{9}{16}\zeta^2 + \frac{\left(\gamma B_z\right)^2}{4} - \frac{\gamma B_z \zeta}{4}\right]^{1/2}$$

$$E_{3/2,-3/2} = \zeta/2 + 2\gamma B_z$$

Same energy levels as coupled.

Even though each of the matrices are different, when evaluated in the two basis sets, the eigen-energies must be identical.

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Wigner-Eckart Theorem

CTDL, pages 999 - 1085, esp. 1048-1053

Final lecture on $1e^-$ Angular Part
Next: 2 lectures are on $1e^-$ radial part
Many-e^- problems − 8 lectures!

Previous Lecture: $|JLSM_J\rangle$ vs. $|LM_LSM_S\rangle$
coupled uncoupled

Transformation between these basis sets is general and tabulated

Vector Coupling Coefficients
Clebsch-Gordan Coefficients
3-j Coefficients

Same information, increasingly convenient formats.

Correlation Diagrams between limiting cases

* non-crossing rule for states with the same value of rigorously good quantum numbers (zero calculation approach)

* non-degenerate perturbation theory
sequence of steps for inclusion of information about the opposite limit: $E^{(0)}$, $E^{(0)} + E^{(1)}$, $E^{(0)} + E^{(1)} + E^{(2)}$

* exact diagonalization

How does the pattern of energy levels for one limiting case morph into that for the other limiting case?

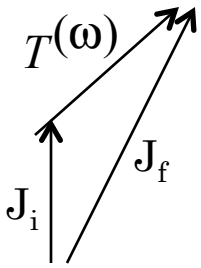
Note that $\mathbf{J}_i$, $\mathbf{L}_i$, $\mathbf{S}_i$ operators cannot cause off-diagonal matrix elements in $|JM_J\rangle$, $|LM_L\rangle$ or $|SM_S\rangle$ basis sets, respectively.

However $\mathbf{L}_z$ and $\mathbf{S}_z$ can cause $\Delta J = \pm 1$ matrix elements in the $|JM_J\rangle$ basis set.

Why? Because **L** and **S** are vectors with respect to **J**.

Triangle Rule: $|L - S| \leq J \leq L + S$

Maybe it is better to think about classification of operators as "like" an angular momentum $\rightarrow T^{(\omega)}_{\mu}$! Spherical tensor operators behave like angular momenta.



$$T^{(\omega)}_{\mu}\left|J_i M_i\right\rangle \text{ gives } J+\omega, J+\omega-1, \ldots \left|J-\omega\right|$$

This works! We construct operators classified by what they do to members of the $|JM_J\rangle$ basis set.

How? Commutation Rule definitions of the $T^{(\omega)}_{\mu}$ operators:

$$\left[\mathbf{J}_{\pm}, T^{(\omega)}_{\mu}\right] = \hbar\left[\omega(\omega+1) - \mu(\mu \pm 1)\right]^{1/2} T^{(\omega)}_{\mu\pm1}$$

$$\left[\mathbf{J}_{z}, T^{(\omega)}_{\mu}\right] = \hbar\mu T^{(\omega)}_{\mu}$$

All matrix elements $T^{(\omega)}_{\mu}$ in the $|JM_J\rangle$ basis set are derivable (and inter-related) from these commutation rules.

Do the above commutation rules look familiar? We see the same thing from $J_{\pm}|JM\rangle$ and $J_z|JM\rangle$.

This is a mixture of intuition plus rigor based on tabulated coupling constants.

The Wigner-Eckart Theorem gives us everything we need. The derivation of the W-E theorem from commutation rules is extremely tedious. The Herschbach handout illustrates some of the derivation. [Supplement #1]

Scalar, Vector, and Tensor Operators

Selection Rules

Scalar	**S**	$T_0^{(0)}$	matrix elements are $\Delta J = 0$, $\Delta M = 0$	M-independent
Vector	**V**	$T_\mu^{(1)}$	matrix elements are $\Delta J = 0, \pm 1$, $\Delta M = 0, \pm 1$	explicitly M-dependent
Tensor		$T_\mu^{(\omega)}$	ω = rank, μ = component	

We seldom see Tensors with $\omega > 2$.

Construct and classify operators via Commutation Rules

		Rank: ω	"Like"	components: μ
Scalar	1 component	0	J = 0	$\mu = 0$
Vector	3 components	1	J = 1	$\mu = 0 \leftrightarrow z$ $+1 \longrightarrow -(2)^{1/2}(x + iy)$ $-1 \longrightarrow +(2)^{1/2}(x - iy)$ (not quite like $\mathbf{J}_\pm$)
Tensor	$2\omega + 1$ components	ω	J = ω	+2, +1, 0, –1, –2, (for $\omega = 2$)

Example: $\mathbf{J} - \mathbf{L} + \mathbf{S}$

1. $[\vec{\mathbf{L}}, \vec{\mathbf{S}}] = 0$. **L** and **S** act as scalar operators with respect to each other (because they operate on different coordinates)
2. $\vec{\mathbf{L}}$ and $\vec{\mathbf{S}}$ act as vector operators with respect to $\vec{J}$
3. $\vec{\mathbf{L}} \cdot \vec{\mathbf{S}}$ acts like a scalar operator with respect to $\vec{J}$
4. $\vec{\mathbf{L}} \times \vec{\mathbf{S}}$ gives 3 components of a vector operator with respect to $\vec{J}$

We can use commutation rules to project out of any operator the part that acts like a $T_{\mu}^{(\omega)}$ or we can construct such $T_{\mu}^{(\omega)}$ operators explicitly!

Wigner-Eckart Theorem

$$\left\langle N'J'M'_J \left| T_{\mu}^{(\omega)} \right| NJM_J \right\rangle = A_{M\mu M'}^{J\omega J'} \delta_{M'M+\mu} \left\langle N'J' \left\| T^{(\omega)} \right\| NJ \right\rangle$$

N' and N are radial quantum numbers (i.e. everything that is not a specified angular momentum)

$A_{M\mu M'}^{J\omega J'}$ is a tabulated vector coupling coefficient (or Clebsch-Gordan or 3-j)

$\left\langle N'J' \left\| T^{(\omega)} \right\| NJ \right\rangle$ is a "reduced matrix element"

It is reduced in the sense that M', μ, and M are removed.

Often the reduced matrix elements of $T_{\mu}^{(\omega)}$ can be evaluated by looking at matrix elements of "stretched staes": $\vec{J} = \vec{J} + \vec{J}_2$, $J = J_1 + J_2$ and $\mu = \omega$.

Recall that *extreme members* of coupled and uncoupled basis sets are equal.

$$\left| J = L + S, L, S, M_J = L + S \right\rangle = \left| LM_L = L, SM_S = S \right\rangle$$

Major simplifications result for stretched states.

How to build specified $T_{\mu}^{(\omega)}$ operators out of components of specific angular momenta [denoted in square brackets].

$$T_{\pm 1}^{(\omega)}[L] = \mp 2^{-1.2}[L_x \pm iL_y]$$
$$T_0^{(1)}[L] = L_z$$

What about $T_{\pm 2}^{(2)}[L]? \rightarrow (L_{\pm})^2$, but what about $T_{\pm 1}^{(2)}$ and $T_{\pm 1}^{(1)}$?

Suppose we want tensor operators constructed from two vector operators.

$$T_0^{(0)}[A,B] = \sum_{k=-\omega}^{\omega} (-1)^k T_k^{(\omega)}[A] T_{-k}^{(\omega)}[B]$$

$$T_2^{(2)}[A,B] = T_1^{(1)}[A] T_1^{(1)}[B] \rightarrow A_+ B_+$$

$$T_1^{(2)}[A,B] = T_1^{(1)}[A] T_0^{(1)}[B] + T_0^{(1)}[A] T_1^{(1)}[B] \rightarrow A_+ B_0 + A_0 B_+$$

Display this scheme more clearly and compactly:

μ	$T_\mu^{(2)}[A,B]$	$T_\mu^{(1)}[A,B]$	$T_0^{(0)}[A,B]$
2	(+ +)		
1	(+ 0) + (0 +)	(+ 0) – (0 +)	
0	(+ –) + (– +)	(+ –) – (– +)	(0 0)
–1	(0 –) + (– 0)	(0 –) – (– 0)	
–2	(– –)		

Notice how we get $T^{(2)}$, $T^{(1)}$, and $T^{(0)}$ as orthogonal combinations of $A_i + B_i$.

Vector Coupling Coefficient

$$\underbrace{|J, J_1, J_2, M\rangle}_{\text{coupled}} = \sum_{\substack{M_2 = M - M_1 \\ M_2 = -J_2}}^{J_2} \underbrace{|J_1, M_1, J_2, M_2\rangle \langle J_1, M_1, J_2, M_2 | J, J_1, J_2, M\rangle}_{\text{completeness}}$$

vector coupling coefficient

$$\begin{pmatrix} J_1 & J_2 & J \\ M_1 & M_2 & -M \end{pmatrix} = (-1)^{J_1 - J_2 - M} (2J+1)^{-1/2} \quad \langle J_1, M_1, J_2, M_2 | J_1, J_2, J - M \rangle$$

$3-j$ Clebsch-Gordan

constraint $M_1 + M_2 - M = 0$

Vector Coupling:	no symmetry, no explicit constraints
Clebsch-Gordan	some symmetry, explicit constraint $M_1 + M_2 - M = 0$
3 – j:	maximum symmetry, maximum constraints (rules for permutation of columns)

Examples of Use of Commutation Rules to Reveal the Properties of Scalar Operators

Scalar Operator: $[\mathbf{J}_i, \mathbf{S}] = 0$ all i

1. $\Delta\mathbf{J} = \mathbf{0}$ selection rule from $[\mathbf{J}^2, \mathbf{S}] = 0$

$0 = [\mathbf{J}^2, \mathbf{S}]$ $0 = \langle J'M'|\mathbf{J}^2\mathbf{S} - \mathbf{S}\mathbf{J}^2|JM\rangle = \hbar[J'(J'+1) - J(J+1)]\langle J'M'|\mathbf{S}|JM\rangle$

either J = J′ or $\langle J'M'|\mathrm{S}|JM\rangle = 0$

ΔJ = 0 selection rule

2. $\Delta\mathbf{M} = \mathbf{0}$ selection rule from $[\mathbf{J}_z, \mathbf{S}] = 0$

$$0 = \langle JM'|J_zS - SJ_z|JM\rangle = \hbar(M' - M)\langle JM'|S|JM\rangle$$

either M′ = M or $\langle JM'|\mathrm{S}|JM\rangle = 0$

ΔM = 0 selection rule

3. **M independence of $\langle JM|\mathbf{S}|JM\rangle$ from $[\mathbf{J}_\pm, \mathbf{S}] = \mathbf{0}$**

$$0 = \langle JM'|\mathrm{J}_\pm\mathrm{S} - \mathrm{S}\mathrm{J}_\pm|JM\rangle = S_{JM}\langle JM'|\mathrm{J}_\pm|JM\rangle - S_{JM'}\langle JM'|\mathrm{J}_\pm|JM\rangle$$
$$= (S_{JM} - S_{JM'})\langle JM'|\mathrm{J}_\pm|JM\rangle$$

$\langle JM'|\mathbf{J}_\pm|JM\rangle \neq 0$ when $M' = M \pm 1$

Thus $S_{JM} = S_{JM\pm1}$, which means that S_{JM} is independent of M.

What is so great about Wigner-Eckart Theorem?

Massive reduction in number of independent matrix elements.

For example, J = 10, $\omega = 1$

$$\langle J'M'|T^{(1)}_\mu|JM\rangle$$

J′ limited to J ± 1 by triangle rule

J′	# of matrix elements			# of reduced matrix elements
9	$(2\cdot 9+1)(2\cdot 10+1)$	399	1	$c_-(10)=\left\langle 9\left\|T^{(1)}_{\mu}\right\|10\right\rangle$
10	$(2\cdot 10+1)(2\cdot 10+1)$	441	1	$c_0(10)=\left\langle 10\left\|T^{(1)}_{\mu}\right\|10\right\rangle$
11	$(2\cdot 11+1)(2\cdot 10+1)$	483	1	$c_+(10)=\left\langle 11\left\|T^{(1)}_{\mu}\right\|10\right\rangle$
	total	1323	3	only 3

(one might argue that 1323 is a factor of 7 too large)

$\frac{1323}{7} \leftrightarrow 3$ is a huge reduction of what we need to know!

Special case for $\Delta J = 0$ Matrix Elements of $\vec{\mathbf{V}}$!! Memorable!

$$\left\langle JM'\left|\vec{\mathbf{V}}\right|JM\right\rangle = \frac{\left\langle J\left\|\mathbf{J}\cdot\mathbf{V}\right\|J\right\rangle}{\hbar^2 J(J+1)}\left\langle JM'\left|\vec{\mathbf{J}}\right|JM\right\rangle = c_0(J)\left\langle JM'\left|\vec{\mathbf{J}}\right|JM\right\rangle$$

We can replace a $\Delta J = 0$ matrix element of $\vec{\mathbf{V}}$ by the corresponding matrix element of $\vec{J}$.

An extremely convenient (practical) operator replacement. Derive effective **H** by replacing $\vec{\mathbf{V}}$ by $\vec{J}$.

$c_0(J)$ can also be evaluated by reference to the easily derived matrix elements of stretched states.

Also can derive similar relationships via Commutation Rules

$$\left\langle J+1, M \middle| V_z \middle| JM \right\rangle = c_+(J)\left[(J+M+1)(J-M+1)\right]^{1/2}$$

$$\left\langle J+1, M\pm 1 \middle| V_\pm \middle| JM \right\rangle = c_+(J)\left[(J\pm M+2)(J\pm M+1)\right]^{1/2}$$

$$\left\langle JM \middle| V_z \middle| JM \right\rangle = c_0(J)\,M$$

$$\left\langle JM\pm 1 \middle| V_z \middle| JM \right\rangle = c_0(J)\left[J(J+1)-M(M\pm 1)\right]^{1/2}$$

$$\left\langle J-1, M \middle| V_z \middle| JM \right\rangle = c_-(J)\left[(J-M)(J+M)\right]^{1/2}$$

$$\left\langle J-1, M\pm 1 \middle| V_z \middle| JM \right\rangle = \pm c_-(J)\left[(J\mp M)(J\pm M+1)\right]^{1/2}$$

This has been just a taste of the power of spherical tensor algebra for problems with exact or approximate spherical symmetry.

3-j, 6-j, 9-j algebra too burdensome to learn and remember unless you are going to use it immediately.

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5.73 Quantum Mechanics I
Fall 2018

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Supplement #1 to Lecture #27

Angular Momentum Eigenvalues (from lecture notes by Professor Dudley Herschbach)

Consider any Hermitian operator $\underset{\sim}{J}$ whose components satisfy the following commutation rules

$$[J_x, J_y] = i\hbar J_z$$

and the cyclic permutations thereof. Equivalently, the rules may be written as

$$\underset{\sim}{J} \times \underset{\sim}{J} = i\hbar \underset{\sim}{J}$$

or as

$$[J_\ell, J_m] = i\hbar \sum_n \varepsilon_{\ell mn} J_n$$

where

$$\begin{aligned} \varepsilon_{\ell mn} &= +1 \text{ if } \ell, m, n \text{ are in cyclic order} \\ &= -1 \text{ if } \ell, m, n \text{ are in anti-cyclic order} \\ &= 0 \text{ if any two of } \ell, m, n \text{ are the same.} \end{aligned}$$

Seek to find eigenvalues λ for J^2 and μ for J_z such that

$$\begin{aligned} J^2 \left|\lambda\mu\right\rangle &= \lambda \left|\lambda\mu\right\rangle \\ J_z \left|\lambda\mu\right\rangle &= \mu \left|\lambda\mu\right\rangle . \end{aligned}$$

Since J^2 and J_z are Hermitian, λ and μ are real, and $|\lambda\mu\rangle$ are the simultaneous eigenvectors which render J^2 and J_z simultaneously diagonal.

First show $\lambda \geq \mu^2$

Proof: $\langle\lambda\mu\left|J^2 - J_z^2\right|\lambda\mu\rangle = \lambda - \mu^2$

But

$$J^2 - J_z^2 = J_x^2 + J_y^2 + \cancel{J_z^2} - \cancel{J_z^2}$$

$$\lambda\mu \;\; J_x^2 \;\; \lambda\mu \;\; = \sum_{\lambda'\mu'} \langle \lambda\mu \,|J_x|\, \lambda'\mu' \rangle \underbrace{\langle \lambda'\mu' \,|J_x|\, \lambda\mu \rangle}_{\lambda\mu \;\; J_x^\dagger \;\; \lambda'\mu' \quad \text{and } J_x^\dagger = J_x}$$

$$= \sum_{\lambda'\mu'} |\langle \lambda\mu \,|J_x|\, \lambda'\mu' \rangle|^2 \to 0 \text{ and similarly for } J_y^2 \text{ term.}$$

So

$$\lambda\mu \;\; J_x^2 + J_y^2 \;\; \lambda\mu \;\; = \lambda - \mu^2 \geq 0 \qquad \text{Q. E. D.}$$

Since $\mu^2 \geq 0$ this also implies $\lambda \geq 0$.
It is convenient to use the non-Hermitian operators

$$J_\pm = J_x \pm i J_y \qquad \text{Note } J_+^\dagger = J_-, J_-^\dagger = J_+.$$

These satisfy

$$[J_z, J_\pm] = \pm\hbar J_\pm \quad \text{since} \quad [J_z, J_x \pm i J_y] = i\hbar J_y \pm i(-i\hbar J_x)$$
$$= \hbar(J_x \pm i J_y) = \hbar J_\pm.$$

Apply this to $|\lambda\mu\rangle$ and find

$$(J_z J_\pm - J_\pm J_z)\,|\lambda\mu\rangle = \pm\hbar J_\pm \,|\lambda\mu\rangle$$

or

$$J_z(J_\pm \,|\lambda\mu\rangle) = (J_\pm J_z \pm \hbar J_\pm)\,|\lambda\mu\rangle$$
$$= (\mu \pm \hbar)(J_\pm \,|\lambda\mu\rangle) \qquad \text{since } J_z \,|\lambda\mu\rangle = \mu \,|\lambda\mu\rangle\,.$$

Thus $J_\pm \,|\lambda\mu\rangle$ is an eigenvector of J_z with eigenvalue $\mu \pm \hbar$. Hence J_+ "raises" the eigenvalue of μ to $\mu + \hbar$ and J_- "lowers" the eigenvalue of μ to $\mu - \hbar$. Now note

$$[J^2, J_\pm] = 0$$

since J^2 commutes with its components J_x and J_y. Thus

$$J^2(J_\pm \,|\lambda\mu\rangle) = J_\pm \underbrace{J^2 \,|\lambda\mu\rangle}_{\lambda \,|\lambda\mu\rangle} = \lambda(J_\pm \,|\lambda\mu\rangle).$$

Thus $J_\pm |\lambda\mu\rangle$ remains an eigenvector of J^2 with the same eigenvalue λ as $|\lambda\mu\rangle$.

By repeated application of J_+ we can get eigenvectors with J_z eigenvalues of $\mu+\hbar$, $\mu+2\hbar$, ... but the same eigenvalue λ of J^2. Since $\mu^2 \geq \lambda$, for a given λ there must be some highest value of μ, call it μ_h, such that $J_+ |\lambda\mu_h\rangle = 0$ rather than generating a new eigenvector of still higher J_z–eigenvalue. Similarly, repeated application of J_- gives $\mu-\hbar$, $\mu-2\hbar$, ... but would eventually violate $\mu^2 \leq \lambda$ unless there is some lowest value of μ, call it μ_ℓ, such that $J_- |\lambda\mu_\ell\rangle = 0$.

Now we use these conditions to show $\underline{\mu_n = -\mu_\ell}$. Consider applying J_- to $J_+ |\lambda\mu_h\rangle = 0$. Note the identity:

$$\begin{aligned} J_- J_+ &= (J_x - iJ_y)(J_x + iJ_y) \\ &= J_x^2 + J_y^2 + i[J_x, J_y] \\ &= J^2 - J_z^2 - \hbar J_z. \end{aligned}$$

Thus

$$J_- J_+ |\lambda\mu\rangle = (\lambda - \mu_h^2 - \hbar\mu_h) |\lambda\mu_h\rangle = 0.$$

Taking the matrix element with $\langle\lambda\mu_h|$ we find

$$\lambda - \mu_h^2 - \hbar\mu_h = 0.$$

Similarly,

$$J_+ J_- |\lambda\mu_\ell\rangle = (J^2 - J_z^2 + \hbar J_z) |\lambda\mu_\ell\rangle$$

leads to

$$\lambda - \mu_\ell^2 + \hbar\mu_\ell = 0.$$

Hence

$$\lambda = \underbrace{\mu_h(\mu_h+\hbar) = \mu_\ell(\mu_\ell - \hbar)}$$

Two solutions: $\underline{\mu_h = -\mu_\ell}$

or $\mu_h = \mu_\ell - \hbar$ but this second solution must be rejected since μ_h was assumed to be larger than μ_ℓ.

Now we can conclude also that $\mu_h = \mu_\ell + n\hbar$ where n is some integer. This follows since, if we start from $|\lambda\mu_\ell\rangle$ and apply J_+ repeatedly, we obtain the sequence of eigenvectors:

$$\underbrace{|\lambda\mu_\ell\rangle}_{\mu_\ell}, \quad \underbrace{J_+ |\lambda\mu_\ell\rangle}_{\mu_\ell+\hbar}, \quad \underbrace{J_+^2 |\lambda\mu_\ell\rangle}_{\mu_\ell+2\hbar}, \ \ldots \quad \underbrace{J_+^n |\lambda\mu_\ell\rangle}_{\mu_\ell+n\hbar=\mu_\hbar} = |\lambda\mu_n\rangle$$

Thus

$$\mu_h = -\mu_\ell = \mu_\ell + n\hbar$$

or

$$\underline{\mu_\ell = -\frac{n}{2}\hbar, \mu_h = +\frac{n}{2}\hbar}$$

where $n = 0, 1, 2, \ldots$ is some integer (related to the value of λ).
For convenience, we write

$$\mu = m\hbar, \quad m = -j, -j+1, \cdots + j$$

where $j = \frac{n}{2}$, with $j = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \ldots$
Then eigenvalues of J_z are $\underbrace{-j\hbar, (-j+1)\hbar, \ldots j\hbar}_{2j+1 \text{ different values}}$
Eigenvalues of J^2 are given by

$$\lambda = \mu_h(\mu_h + \hbar) = \mu_\ell(\mu_\ell - \hbar) = j\hbar(j\hbar + \hbar) = -j\hbar(-j\hbar - \hbar)$$
$$\lambda = j(j+1)\hbar^2$$

Also, it is convenient to label the eigenvectors by j, m rather than λ, μ, so

$$J^2 |jm\rangle = j(j+1)\hbar^2 |jm\rangle$$
$$j_z |jm\rangle = m\hbar |jm\rangle .$$

Comments

We derived the above eigenvalues using only the commutation property and the Hermitian property. We find that both integer and half-integer values of j and m are allowed.

Actually, we have solved a much more general problem than that posed by the orbital angular momentum of a particle. Thus, for several particles in the same central force field, the total angular momentum,

$$\sum_n \underset{\sim}{L}^{(n)},$$

also satisfies these relations, even if the particles interact with each other. Spin angular momenta likewise satisfy these relations.

For orbital angular momentum, $\underset{\sim}{L} = \underset{\sim}{q} \times \underset{\sim}{p}$ must require, in addition, that the system returns to its original state under a rotation by 2π. Such a rotation takes $\underset{\sim}{p} \to \underset{\sim}{p}$ and $\underset{\sim}{q} \to \underset{\sim}{q}$ so $\underset{\sim}{q} \times \underset{\sim}{p} \to \underset{\sim}{q} \times \underset{\sim}{p}$ and hence the eigenvectors of L^2 and L_z must be unchanged:

$$e^{-i2\pi J_z/\hbar} \left|jm\right\rangle = e^{-i2\pi m} \left|jm\right\rangle$$

$e^{-i2\pi m} = +1$ if m is integer and hence integer eigenvalues are acceptable for L^2, L_z. Half-integer values give $e^{-i2\pi m} = -1$ and hence are *not* acceptable for *orbital angular momentum*.

Half-integers do apply for *spin* angular momenta, which are not constructed from any $\underset{\sim}{q} \times \underset{\sim}{p}$ and thus can take on both integer and half-integer eigenvalues. This illustrates the power of operator derivation. A more general case would not have been included if we had used wave mechanical methods and representations by differential operators.

We have shown that, *for a fixed j value*,

$$J_+ \left|jm\right\rangle = a_m \left|j, m+1\right\rangle \text{ and } J_- \left|jm\right\rangle = b_m \left|j, m-1\right\rangle,$$

where a_m and b_m are constants, possibly complex numbers. The proportionality constants are simply related to each other, since

$$a_m = \langle j, m+1 | J_+ | jm \rangle = \left\langle jm \underbrace{|J_+^\dagger|}_{J_-} j, m+1 \right\rangle^* = \left(\int \psi_{jm}^* \underbrace{J_- \psi_{j,m+1}}_{b_{m+1}\psi_{j,m}} d\tau \right)^*$$

$$= b_{m+1}^*$$

Now, to evaluate a_m, consider the identity

$$J_- J_+ = J^2 - J_z^2 - \hbar J_z.$$

Apply this to $|jm\rangle$, then you have

$$J_- J_+ |jm\rangle = a_m J_- |j, m+1\rangle = a_m b_{m+1} |jm\rangle = |a_m|^2 |jm\rangle$$
$$J^2 - J_z^2 - \hbar J_z |jm\rangle = (j(j+1) - m^2 - m)\hbar^2 |jm\rangle .$$

Hence

$$a_m = \underbrace{[j(j+1) - m(m+1)}_{(j-m)(j+m+1) \leftarrow \text{ another common way of writing it,}}]^{1/2} \hbar e^{i\phi m}$$

where $e^{i\phi m}$ is an arbitrary phase factor. The usual convention is to take $\phi = 0$; this fixes the relative phases of the vectors $|jm\rangle$ having different values of m but the same j.

The only non-vanishing matrix elements of J_+ and J_- are:

$$\langle j, m+1 | J_+ | jm \rangle = \langle j, m | J_- | j, m+1 \rangle = [j(j+1) - m(m+1)]^{1/2} \hbar$$

↑ always the lower times the higher of the two m–values in the matrix element

Or you can write this alternatively as

$$\langle j', m' | J_+ | jm \rangle = [j(j+1) - m(m+1)]^{1/2} \hbar \delta_{j',j} \delta_{m',m+1}$$
$$\langle j', m' | J_- | jm \rangle = [j(j+1) - m(m-1)]^{1/2} \hbar \delta_{j',j} \delta_{m',m-1}$$

List of non-zero elements:

$$jm|J^2|jm = j(j+1)\hbar^2$$

$$\langle jm|J_z|jm\rangle = m\hbar$$

"add the bigger m to j and subtract the smaller"

$$\langle j, m\pm 1|J_\pm|jm\rangle = [j(j+1) - m(m\pm 1)]^{1/2}\hbar = [\overbrace{(j\pm m+1)(j\mp m)}]^{1/2}\hbar$$

$$J_x = \frac{1}{2}(J_+ + J_-), \quad J_y = \frac{1}{2i}(J_+ - J_-)$$

$$\langle j, m\pm 1|J_x|jm\rangle = \frac{1}{2}[j(j+1) - m(m\pm 1)]^{1/2}\hbar$$

$$\langle j, m\pm 1|J_y|jm\rangle = \pm\frac{1}{2i}[j(j+1) - m(m\pm 1)]^{1/2}\hbar$$

We can summarize elements of J_x, J_y, J_z by:

$$jm|\underset{\sim}{J}|jm = \hat{z}m\hbar$$

$$j, m\pm 1|\underset{\sim}{J}|jm = (\hat{x}\pm i\hat{y})\frac{1}{2}[j(j+1) - m(m\pm 1)]^{1/2}\hbar.$$

Comment

Thus we have found all matrix elements of J with eigenvectors $|jm\rangle$ of J^2, J_z. These eigenvectors and their properties are important, since any time we have a system of particles isolated in free space, their total angular momentum J^2, J_z commutes with the total Hamiltonian, no matter what kind of forces hold the system together (central or not). That is, the total angular momentum of an isolated system is a constant of the motion in quantum mechanics, just as in classical mechanics.

Hence it is important to be able to take matrix elements of other operators in the angular momentum states which characterize an isolated system.

Examples

$$j = 0: \quad J_+ = (0) \quad J_- = (0) \quad J_z = (0) \quad J^2 = (0)$$

$$j = \frac{1}{2} : J_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad J_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad J_z = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \quad J^2 = \begin{pmatrix} \frac{3}{4} & 0 \\ 0 & \frac{3}{4} \end{pmatrix}$$

$j = 1$:

$$J_+ = \begin{pmatrix} 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 \end{pmatrix} \quad J_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$J_- = \begin{pmatrix} 0 & 0 & 0 \\ \sqrt{2} & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{pmatrix} \quad J^2 = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$j = \frac{3}{2}$:

$$J_+ = \begin{pmatrix} 0 & \sqrt{3} & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad J_z = \begin{pmatrix} \frac{3}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & -\frac{3}{2} \end{pmatrix}$$

$$J_- = \begin{pmatrix} 0 & 0 & 0 & 0 \\ \sqrt{3} & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix} \quad J^2 = \begin{pmatrix} \frac{15}{4} & 0 & 0 & 0 \\ 0 & \frac{15}{4} & 0 & 0 \\ 0 & 0 & \frac{15}{4} & 0 \\ 0 & 0 & 0 & \frac{15}{4} \end{pmatrix}$$

MOMENTA AS DISPLACEMENT OPERATORS:
Geometrical Meaning of Commutation Rules

Linear Momentum

Let $|x_1\rangle$ be an eigenvector of the position operator X with eigenvalue x_1, i.e.

$$X\,|x_1\rangle = x_1\,|x_1\rangle\,.$$

Consider the new state vector defined by $e^{-iap_x/\hbar}\,|x_1\rangle$; we might ask whether it is also an eigenvector of X. To find out, evaluate

$$X\left(e^{-iap_x/\hbar}\,|x_1\rangle\right) = e^{-iap_x/\hbar}\underbrace{\left(X\,|x_1\rangle\right)}_{x_1|x_1\rangle} + \underbrace{\left[X, e^{-iap_x/\hbar}\right]}\,|x_1\rangle$$

Now

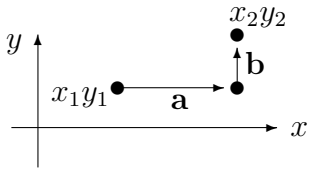
$$\begin{aligned}\left[X, e^{-iap_x/\hbar}\right] &= i\hbar\frac{d}{dp_x}\left(e^{-iap_x/\hbar}\right)\\ &= ae^{-iap_x/\hbar}\end{aligned}$$

Thus

$$\begin{aligned}X\left(e^{-iap_x/\hbar}\,|x_1\rangle\right) &= e^{-iap_x/\hbar}(X+a)\,|x_1\rangle = e^{-iap_x/\hbar}(x_1+a)\,|x_1\rangle\\ &= (x_1+a)\left(e^{-iap_x/\hbar}\,|x_1\rangle\right).\end{aligned}$$

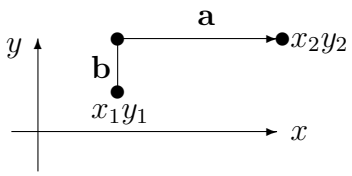
Hence $e^{-iap_x/\hbar}\,|x_1\rangle$ is indeed an eigenvector of X with eigenvalue x_1+a instead of x_1. The unitary operator $e^{-iap_x/\hbar}$ formed from the linear momentum operator p_x acts as a *displacement operator* for x position coordinates. Similarly, p_y generates displacements of the y coordinate and p_z of the z coordinate.
It is a geometrical fact that *linear displacements* of a point *commute*. For example:

$|x_2y_2\rangle = e^{-ibp_y/\hbar}e^{-iap_x/\hbar}|x_1y_1\rangle \Rightarrow$



first move along x by **a**, then along y by **b**

$= e^{-iap_x/\hbar}e^{-ibp_y/\hbar}|x_1y_1\rangle \Rightarrow$



reverse the order

The same result is obtained by applying displacements in either order. This agrees with $[p_x, p_y] = 0$ (and $\{p_x, p_y\} = 0$).

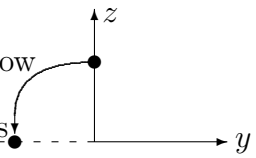
Angular momentum

L_x, L_y, L_z operators generate *angular* displacements or *rotations*; e.g.,

$$e^{-i\phi L_x/\hbar}$$

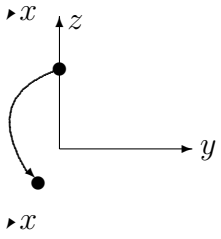
gives a rotation by angle ϕ about the x–axis, etc. However, geometrical *rotations about different axes do not commute.* For example, consider a state representing a particle on the z–axis, $|z_0\rangle$. Now

$$\underbrace{e^{-i\frac{\pi}{2}L_y/\hbar}}_{\substack{\text{rotation by } \pi/2 \\ \text{about } y\text{–axis}}} \underbrace{e^{-i\frac{\pi}{2}L_x/\hbar}|z_0\rangle}_{\substack{\text{rotation by } \pi/2 \\ \text{about } x\text{–axis}}} \Rightarrow \text{particle on } -y \text{ axis}$$



But

$$\underbrace{e^{-i\frac{\pi}{2}L_x/\hbar}}_{\substack{\text{rotation by } \pi/2 \\ \text{about } x\text{–axis}}} \underbrace{e^{-i\frac{\pi}{2}L_y/\hbar}|z_0\rangle}_{\substack{\text{rotation by } \pi/2 \\ \text{about } y\text{–axis}}} \Rightarrow \text{particle on } +x \text{ axis}$$



The results of these two rotations taken in opposite order differ by a rotation about the z–axis. Thus, because the rotations about different axes don't commute, we must expect the angular momentum operators, which generate these rotations, *not* to commute with each other. Indeed,

$$[L_x, L_y] = i\hbar L_z$$

corresponds to the above example, in which the commutator of rotations about the x and y axes depends on a z–axis rotation.

Rotational Transformation Properties and Selection Rules

The various observables of a dynamical system can be classified according to their transformation properties under rotations. This is of great value in determining the matrix elements of the corresponding operators and, in particular, leads to selection rules which limit the number of non–zero matrxix elements.

Under action of the rotation operator $U = e^{i\underset{\sim}{\phi}\cdot\underset{\sim}{J}/\hbar}$ an operator O is transformed according to

$$\underset{\sim}{Q}' = \underset{\sim}{U}\underset{\sim}{Q}\underset{\sim}{U}^{\dagger}.$$

A *scalar* operator $\underset{\sim}{S}$ is one which is invariant to this transformation (e.g., the Hamiltonian of an isolated system). Hence, for a scalar operator

$$\underset{\sim}{U}\underset{\sim}{S}\underset{\sim}{U}^{\dagger} = \underset{\sim}{S}$$

or

$$\underset{\sim}{U}\underset{\sim}{S} - \underset{\sim}{S}\underset{\sim}{U} = 0 \qquad \text{or } [\underset{\sim}{U}, \underset{\sim}{S}] = 0.$$

Thus, a scalar commutes with every rotation operator. Consider, in particular, an infinitesimal rotation $d\underset{\sim}{\phi}$, for which

$$\underset{\sim}{U} = 1 + \frac{i}{\hbar} d\underset{\sim}{\phi}\cdot\underset{\sim}{J}.$$

Since the direction of $d\underset{\sim}{\phi}$ is arbitrary, $\underset{\sim}{S}$ must commute with each component of $\underset{\sim}{J}$, or $[\underset{\sim}{S}, \underset{\sim}{J}] = 0$. As shown below, this property leads to the selection rules

$$\Delta j = 0, \quad \Delta m = 0$$

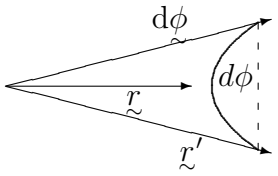
for the non-zero matrix elements of a scalar operator.

A *vector* operator $\underset{\sim}{V}$ is one with three components, V_x, V_y, V_z which transform under rotations like the coordinates of a point. For an infinitesimal rotation,

$$\underset{\sim}{V}' = \left(1 + \frac{i}{\hbar} d\underset{\sim}{\phi}\cdot\underset{\sim}{J}\right)\underset{\sim}{V}\left(1 - \frac{i}{\hbar} d\underset{\sim}{\phi}\cdot\underset{\sim}{J}\right).$$

Now note that if the position vector $\underset{\sim}{r}'$ is obtained from $\underset{\sim}{r}$ by rotation through a small angle $d\phi$ about an axis in the direction of the vector $d\underset{\sim}{\phi}$, we have, to first order in $d\phi$,

$$\underset{\sim}{r}' = \underset{\sim}{r} + d\underset{\sim}{\phi} \times \underset{\sim}{r}$$



and so

$$\underset{\sim}{V}' = \underset{\sim}{V} + d\underset{\sim}{\phi} \times \underset{\sim}{V}.$$

Hence, if terms in $(d\phi)^2$ are neglected, we obtain

$$d\underset{\sim}{\phi} \times \underset{\sim}{V} = \frac{i}{\hbar}\left[\left(d\underset{\sim}{\phi}\cdot\underset{\sim}{J}\right)\underset{\sim}{V} - \underset{\sim}{V}\left(d\underset{\sim}{\phi}\cdot\underset{\sim}{J}\right)\right].$$

Since $d\underset{\sim}{\phi}$ is arbitrary, this relation gives the commutator of $\underset{\sim}{V}$ with any component of $\underset{\sim}{J}$. Thus, if $d\underset{\sim}{\phi} = \varepsilon\hat{z}$ is a rotation about the z–axis, we find

$$\not{\varepsilon}(\hat{z}\times\underset{\sim}{V}) = \not{\varepsilon}\frac{i}{\hbar}\left[J_z\underset{\sim}{V} - \underset{\sim}{V}J_z\right]$$

or

$$\left[J_z, \underset{\sim}{V}\right] = -i\hbar(\hat{z}\times\underset{\sim}{V}) \qquad \text{or} \qquad \begin{aligned}[J_z, V_x] &= -i\hbar(-V_y) = i\hbar V_y\\ [J_z, V_y] &= -i\hbar(-V_x) = -i\hbar V_x\\ [J_z, V_z] &= 0\end{aligned}$$

etc.

In this way we obtain a set of nine commutation rules:

$$\begin{array}{lll} [J_x, V_x] = 0 & [J_y, V_x] = -i\hbar V_z & [J_z, V_x] = -i\hbar V_y \\ [J_x, V_y] = i\hbar V_z & [J_y, V_y] = 0 & [J_z, V_y] = -i\hbar V_y \\ [J_x, V_z] = -i\hbar V_y & [J_y, V_z] = i\hbar V_x & [J_z, V_z] = 0 \end{array}$$

The selection rules for non-zero matrix elements of a vector operator, i.e. an operator which satisfies the above rules (e.g., position $\underset{\sim}{r}$, linear momentum $\underset{\sim}{p}$, the angular momentum $\underset{\sim}{J}$ itself) are shown below to be given by

$$\Delta j = 0 \text{ and } \pm 1 \text{ for all components of } \underset{\sim}{V}$$

with

$$\Delta m = 0 \text{ for } V_z$$
$$\Delta m = \pm 1 \text{ for } V_\pm = V_x \pm iV_y.$$

Scalar Operators, S[1]

Defined by $[\underset{\sim}{S}, \underset{\sim}{J}] = 0$, for all three components of $\underset{\sim}{J}$. Corollary is $[S, J^2] = 0$ and $[S, J_z] = 0$. If we take the matrix elements, we have

$$\begin{aligned} j'm'|[S, J^2]|jm &= 0 \\ &= \ j'm'|SJ^2 - J^2S|jm \\ &= \hbar^2 \langle j'm'|Sj(j+1) - j'(j'+1)S|jm\rangle \\ &= \hbar^2[j(j+1) - j'(j'+1)] \langle j'm'|S|jm\rangle . \end{aligned}$$

Also,

$$\begin{aligned} \langle j'm'|[S, J_z]|jm\rangle &= 0 \\ &= \langle j'm'|SJ_z - J_zS|jm\rangle \\ &= \hbar \langle j'm'|Sm - m'S|jm\rangle \\ &= \hbar(m - m') \langle j'm'|S|jm\rangle . \end{aligned}$$

Therefore, $\langle j'm'|S|jm\rangle$ must vanish unless $j = j'$ and $m' = m'$. "Selection rules" for non-zero elements are: $\Delta j = 0$ and $\Delta m = 0$.
Let $s_{jm} \equiv \langle jm|S|jm\rangle$ denote the non-vanishing element. Since this is the only non-zero matrix element, $|jm\rangle$ is an eigenvector of S, i.e. $S|jm\rangle = s_{jm}|jm\rangle$.
Now we can show that the eigenvalues of the scalar operator S don't depend on m. Since S commutes with $J_\pm = J_x \pm iJ_y$, we have

$$S(J_+|jm\rangle) = J_+S|jm\rangle = s_{jm}(J_+|jm\rangle).$$

But $J_+|jm\rangle$ is proportional to $|j, m+1\rangle$ and still has some eigenvalue s_{jm} of S. We could continue this with $J_+^2 \to m+2, \ldots$ and with $J_- \to m-1$,

[1]These notes were prepared by Professor Dudley Herschbach of Harvard University

$J_-^2 \to m-2$, etc., and would get the same eigenvalue s_{jm} of S for all m states of a given j. Hence we would obtain

$$\langle j'm'|S|jm\rangle = \langle j\|S\|j\rangle\, \delta_{jj'}\delta_{mm'}$$

where $\langle j\|S\|j\rangle$ is called a *reduced* matrix element, a number that does *not* depend on m.

The above equation only describes the properties of S which are associated with its scalar character. In general, the states of the system will depend upon other quantum numbers in addition to j and m. If these are denoted collectively by α, the scalar operator need not be diagonal in α, so the general statement becomes

$$\langle \alpha' j'm'|S|\alpha jm\rangle = \langle \alpha' j\|S\|\alpha j\rangle\, \delta_{jj'}\delta_{mm'}$$

for

$$[S, \underset{\sim}{J}] = 0.$$

Vector Operators, V

Definition: A *vector* operator $\underset{\sim}{V}$ with respect to the angular momentum $\underset{\sim}{J}$ is any set of three operators V_x, V_y, V_z that satisfy the following commutation rules:

$$[J_i, V_j] = i\hbar \sum_k \varepsilon_{ijk} V_k \qquad \begin{aligned} \varepsilon_{ijk} &= 1, ijk \text{ cyclic} \\ &= -1, ijk \text{ anti-cyclic} \\ &= 0, \text{ any two subscripts the same} \end{aligned}$$

This is shorthand for

$$\begin{array}{lll} [J_x, V_x] = 0 & [J_y, V_x] = -i\hbar V_z & [J_z, V_x] = i\hbar V_y \\ [J_x, V_y] = i\hbar V_z & [J_y, V_y] = 0 & [J_z, V_y] = -i\hbar V_x \\ [J_x, V_z] = -i\hbar V_y & [J_y, V_z] = i\hbar V_x & [J_z, V_z] = 0. \end{array}$$

It is convenient to use

$$V_\pm = V_x \pm iV_y \qquad V_x = \frac{1}{2}(V_+ + V_-); \qquad V_y = \frac{1}{2i}(V_+ - V_-).$$

Selection Rules for m

Consider the commutators involving J_z, take matrix elements of the commutators:

a) $[J_z, V_z] = 0$

$\langle j'm'|J_zV_z - V_zJ_z|jm\rangle = \langle j'm'|m'\hbar V_z - V_zm\hbar|jm\rangle = \hbar(m'-m)\langle j'm'|V_z|jm\rangle$

Thus $\langle j'm'|V_z|jm\rangle = 0$ unless $m' = m$, $\underline{\Delta m = 0}$

b) $[J_z, V_+] = [J_z, V_x + iV_y] = i\hbar V_y + i(-i\hbar V_x) = \hbar V_+$

$\langle j'm'|J_zV_+ - V_+J_z|jm\rangle = \hbar\langle j'm'|V_+|jm\rangle$.

or

$\hbar(m' - m - 1)\langle j'm'|V_+|jm\rangle = 0$

$\langle j'm'|V_+|jm\rangle = 0$ unless $m' - m = +1$, $\underline{\Delta m = +1}$

c) Similarly, $[J_z, V_-] = -\hbar V_-$ and

$\langle j'm'|V_-|jm\rangle = 0$ unless $m' - m = -1$, $\underline{\Delta m = -1}$

Selection Rules for j

To find the selection rules for j, we want to examine commutators of $\underset{\sim}{V}$ with $\underset{\sim}{J}^2$. For this, some vector identities are useful. First we show

(1) $\underset{\sim}{J} \times \underset{\sim}{V} + \underset{\sim}{V} \times \underset{\sim}{J} = 2i\hbar\underset{\sim}{V}$.

This relation is another way to define a vector operator. It states that, because of the non-commuting algebra of quantum mechanics, $\underset{\sim}{J} \times \underset{\sim}{V} \neq -\underset{\sim}{V} \times \underset{\sim}{J}$ as would hold for ordinary vectors.

0.0.1 Proof:

$$(\underset{\sim}{J}\times\underset{\sim}{V}+\underset{\sim}{V}\times\underset{\sim}{J})_i=\sum_{j,k}(\varepsilon_{ijk}J_jV_k+\varepsilon_{ijk}V_jJ_k)$$ re-label via $j\leftrightarrow k$

Then use $\varepsilon_{ijk}=-\varepsilon_{ijk}$

$$=\sum_{j,k}\varepsilon_{ijk}(J_jV_k-V_kJ_j)$$

$$=\sum_{j,k}\varepsilon_{ijk}[J_j,V_k]=i\hbar\sum_{jk\ell}\varepsilon_{ijk}\varepsilon_{jk\ell}V_\ell$$ using the definition of a vector operator

Note $\varepsilon_{ijk}\varepsilon_{jk\ell}=\varepsilon_{ijk}\varepsilon_{\ell jk}$ as a cyclic permutation of subscripts leaves ε_{ijk} unchanged.

Then

$$\sum_{j,k}\varepsilon_{ijk}\varepsilon_{\ell jk}=2\delta_{i\ell}$$ factor 2 appears because both odd-odd and even-even permutations give a contribution

So

$$(\underset{\sim}{J}\times\underset{\sim}{V}+\underset{\sim}{V}\times\underset{\sim}{J})_i=2i\hbar\sum_\ell\delta_{i\ell}V_\ell=2i\hbar V_i\quad\text{Q.E.D.}$$

Now we show

(2) $[\underset{\sim}{J}^2,\underset{\sim}{V}]=i\hbar(\underset{\sim}{V}\times\underset{\sim}{J}-\underset{\sim}{J}\times\underset{\sim}{V})$.

Proof:

$$[J^2,V_j]=\sum_i[J_i^2,V_j]$$

$$=\sum_i\{J_i[J_i,V_j]+[J_i,V_j]J_i\}$$

$$[J^2,V_j]=i\hbar\sum_{i,k}\left\{J_i\underbrace{\varepsilon_{ijk}}V_k+\underbrace{\varepsilon_{ijk}}V_kJ_i\right\}$$

$$=i\hbar(-\underset{\sim}{J}\times\underset{\sim}{V}+\underset{\sim}{V}\times\underset{\sim}{J})_j\qquad\text{Q.E.D.}$$

It is convenient to define the operator

$$\underset{\sim}{K} \equiv \frac{1}{2}(\underset{\sim}{V} \times \underset{\sim}{J} - \underset{\sim}{J} \times \underset{\sim}{V}).$$

This is Hermitian (since $\underset{\sim}{V}$ and $\underset{\sim}{J}$ are) and is a vector operator if $\underset{\sim}{V}$ is.
Then Equation (2) states

$$[\underset{\sim}{J}^2, \underset{\sim}{V}] = 2i\hbar \underset{\sim}{K}. \tag{1}$$

However, we can't yet use this commutator to get selection rules on $\underset{\sim}{V}$, since the matrix elements of the commutator $\underset{\sim}{K}$ would seem to bear no simple relation to those of $\underset{\sim}{V}$. We will find that selection rules can be obtained from an identity involving the double commutator,

(3) $[\underset{\sim}{J}^2, [\underset{\sim}{J}^2, \underset{\sim}{V}]] = 2\hbar^2\{\underset{\sim}{J}^2\underset{\sim}{V} - 2(\underset{\sim}{J} \cdot \underset{\sim}{V})\underset{\sim}{J} + \underset{\sim}{V}\underset{\sim}{J}^2\}$.

This can be proven by examining further the properties of $\underset{\sim}{K}$.

$$[J^2, [\underset{\sim}{J}^2, \underset{\sim}{V}]] = 2i\hbar[\underset{\sim}{J}^2, \underset{\sim}{K}].$$

Since $\underset{\sim}{K}$ is a vector operator, we have from (2) that

$$[\underset{\sim}{J}^2, \underset{\sim}{K}] = i\hbar(\underset{\sim}{K} \times \underset{\sim}{J} - \underset{\sim}{J} \times \underset{\sim}{K}).$$

Also, from Equation (1) we have

$$\underset{\sim}{J} \times \underset{\sim}{K} + \underset{\sim}{K} \times \underset{\sim}{J} = 2i\hbar \underset{\sim}{K}.$$

Hence

$$\begin{aligned} \underset{\sim}{J} \times \underset{\sim}{K} - \underset{\sim}{K} \times \underset{\sim}{J} &= \underset{\sim}{J} \times \underset{\sim}{K} - (2i\hbar K - \underset{\sim}{J} \times \underset{\sim}{K}) \\ &= 2\underset{\sim}{J} \times \underset{\sim}{K} - 2i\hbar \underset{\sim}{K}. \end{aligned}$$

Also, from equation (1)

$$\underset{\sim}{K} \equiv \frac{1}{2}(\underset{\sim}{V} \times \underset{\sim}{J} - \underset{\sim}{J} \times \underset{\sim}{V}) = \underset{\sim}{V} \times \underset{\sim}{J} - i\hbar \underset{\sim}{V}.$$

Thus,

$$\underset{\sim}{J}\times\underset{\sim}{K}=J\times(\underset{\sim}{V}\times\underset{\sim}{J})-i\hbar(\underset{\sim}{J}\times\underset{\sim}{V})$$

$$(\underset{\sim}{J}\times\underset{\sim}{K})_i=\underbrace{\sum_{j,k}\varepsilon_{ijk}J_j(\underset{\sim}{V}\times\underset{\sim}{J})_k}-i\hbar(\underset{\sim}{J}\times\underset{\sim}{V})_i$$

$$\sum_{jk\ell m}\varepsilon_{ijk}J_j\varepsilon_{k\ell m}V_\ell J_m$$ shift to $\varepsilon_{\ell mk}=\varepsilon_{k\ell m}$ since cyclic permutation of subscripts leaves ε unchanged

$$\sum_{j\ell m}|(\delta_{i\ell}\delta_{jm}-\delta_{im}\delta_{j\ell})J_jV_\ell J_m=\sum_j(J_jV_iJ_j-J_jV_jJ_i)$$

$$=\sum_j(J_jJ_jV_i-J_j[J_j,V_i]-J_jV_jJ_i)$$

$$=J^2V_i-\underbrace{\sum_{j\ell}J_ji\hbar\varepsilon_{ji\ell}V_\ell}-(\underset{\sim}{J}-\underset{\sim}{V})J_i$$ replace by $-\varepsilon_{ij\ell}$, a non-cyclic permutation

$$+i\hbar\sum_{j\ell}J_j\varepsilon_{ij\ell}V_\ell=i\hbar(\underset{\sim}{J}\times\underset{\sim}{V})_i.$$

So we find

$$(\underset{\sim}{J}\times\underset{\sim}{K})_i=\underset{\sim}{J}^2V_i-(\underset{\sim}{J}\cdot\underset{\sim}{V})J_i+\cancel{i\hbar(J\times V)_i}-\cancel{i\hbar(J\times V)_i}$$

or

$$\underset{\sim}{J}\times\underset{\sim}{K}=\underset{\sim}{J}^2\underset{\sim}{V}-(\underset{\sim}{J}\cdot\underset{\sim}{V})\underset{\sim}{J}.$$

Now we can use these results to simplify the double commutator,

$$[\underset{\sim}{J}^2,[\underset{\sim}{J},\underset{\sim}{V}]]=2i\hbar[\underset{\sim}{J}^2,\underset{\sim}{K}]=(2i\hbar)(i\hbar)(-1)(2\underset{\sim}{J}\times\underset{\sim}{K}-2i\hbar\underset{\sim}{K})$$

$$=2\hbar^2(2\underset{\sim}{J}\times\underset{\sim}{K}-2i\hbar\underset{\sim}{K})$$

$$=2\hbar^2\{2J^2\underset{\sim}{V}-2(\underset{\sim}{J}\cdot\underset{\sim}{V})\underset{\sim}{J}-\underbrace{[\underset{\sim}{J}^2,\underset{\sim}{V}]}$$

$$J^2\underset{\sim}{V} - \underset{\sim}{V}J^2$$

and finally,

$$[\underset{\sim}{J},[\underset{\sim}{J}^2,\underset{\sim}{V}]] = 2\hbar^2\{J^2\underset{\sim}{V} - 2(\underset{\sim}{J}\cdot\underset{\sim}{V})\underset{\sim}{J} + \underset{\sim}{V}J^2\} \qquad \text{Q.E.D.}$$

Now we can obtain selection rules by taking matrix elements of this relation. Consider two cases:

Case I: Elements diagonal in j: Wigner-Eckart Theorem

$$jm'|[J^2,A]|jm \;\; = \langle jm'|j(j+1)A - Aj(j+1)|jm\rangle = 0$$

for any operator A. Thus,

$$jm'|[J^2,[J^2,V]]|jm \;\; = 0 = 2\hbar^2 \;\; jm'|J^2\underset{\sim}{V} - 2(\underset{\sim}{J}\cdot\underset{\sim}{V})\underset{\sim}{J} + \underset{\sim}{V}J^2|jm$$

or

$$\hbar^2 j(j+1) \;\; jm'|\underset{\sim}{V}|jm \;\; - \underbrace{\;jm'|(\underset{\sim}{J}\cdot\underset{\sim}{V})\underset{\sim}{J}|jm\;}\; = 0$$

$$= \sum_{j''m''} \;\; jm'|(\underset{\sim}{J}\cdot\underset{\sim}{V}|j''m'' \quad j''m''|\underset{\sim}{J}|jm \;\; .$$

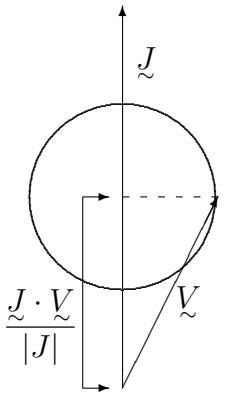
The operator $(\underset{\sim}{J}\cdot\underset{\sim}{V})$ is a *scalar* with respect to $\underset{\sim}{J}$ and therefore diagonal in both m and j, so that $j'' = j$ and $m'' = m'$, and its matrix elements are independent of m. Hence we find

$$\boxed{jm'|\underset{\sim}{V}|jm \;\; = \frac{j|\underset{\sim}{J}\cdot\underset{\sim}{V}|j}{\hbar^2 j(j+1)} \;\; jm'|\underset{\sim}{V}|jm}$$

This is the Wigner-Eckart theorem for a vector operator.

Powell & Craseman, page 371

$$jm'|\underset{\sim}{V}|jm \;=\frac{j|\underset{\sim}{J}\cdot\underset{\sim}{V}|j}{j(j+1)\hbar^2}\;\; jm'|\underset{\sim}{J}|jm$$



Suppose $\underset{\sim}{V}$ precesses around $\underset{\sim}{J}$. The time averaged value of the component normal to $\underset{\sim}{J}$ is zero. The time average of $\underset{\sim}{V}$ is therefore parallel to $\underset{\sim}{J}$ and has magnitude

$$\frac{J\cdot V}{|J|}.$$

Hence, on this model, the average is

$$\underset{\sim}{V} \;=\frac{\underset{\sim}{J}\cdot\underset{\sim}{V}}{J^2}\underset{\sim}{J}=\frac{(J\cdot V)}{|J|}\frac{\underset{\sim}{J}}{|J|}$$

The theorem is very useful, as it states that, for any vector operator $\underset{\sim}{V}$, the matrix elements diagonal in j are simply proportional to the corresponding matrix elements of $\underset{\sim}{J}$ itself. The proportionality constant, $c_0(j) = j|(\underset{\sim}{J}\cdot\underset{\sim}{V})|j\,/\,(\hbar j(j+1))$ is the same for all m–states. Therefore, we have via the Wigner-Eckart Theorem:

$$\begin{aligned}\langle j,m+1|V_+|jm\rangle &= c_0(j)[j(j+1)-m(m+1)]^{1/2}\\ \langle jm|V_z|jm\rangle &= c_0(j)m\\ \langle j,m-1|V_-|jm\rangle &= c_0(j)[j(j+1)-m(m-1)]^{1/2}\end{aligned}$$

with $c_0(j) = \langle\alpha' j\|V\|\alpha j\rangle$ a reduced matrix element. In particular, we note that all matrix elements of $\underset{\sim}{V}$ between $j = 0$ states vanish.

Case II: Elements non-diagonal in j

Now consider $j' \neq j$, again take matrix elements of Equation (3). LHS gives

$$\begin{aligned}j'm'|[J^2,[J^2,V]]|jm \;&=\; j'm'|J^2(J^2V-VJ^2)-(J^2V-VJ^2)J^2|jm\\ &=\{j'^2(j'+1)^2-2j(j+1)j(j+1)+j^2(j+1)^2\}\,\langle j'm'|V|jm\rangle\,.\end{aligned}$$

RHS gives

$$2h^2 \left\langle j'm'|J^2\underset{\sim}{V} - \underbrace{2(\underset{\sim}{J}\cdot\underset{\sim}{V})\underset{\sim}{J}} + \underset{\sim}{V}J^2|jm \right\rangle = 2\hbar^2\{j'(j'+1)+j(j+1)\}\ \ j'm'|\underset{\sim}{V}|jm$$

$$\text{drops out as} \quad j'm'|\underset{\sim}{J}|jm \ = 0, \ \text{because } j' \neq j.$$

Equating LHS = RHS and rearranging gives

$$\{(j'-j)^2-1\}\underbrace{\{(j'+1+1)^2}-1\}\ \ j'm'|\underset{\sim}{V}|jm \ = 0$$

$$\text{This factor } > 0 \text{ since } j' \neq j \text{ and } j' \geq 0, j \geq 0$$

Therefore

$$j'm'|\underset{\sim}{V}|jm \ = 0$$

unless $(j'-j)^2 - 1 = 0$ or $j' = j \pm 1$.
The *complete selection rules* for any vector operator thus are:

$$j'm'|\underset{\sim}{V}|jm \ = 0 \text{ unless}$$
$$j' = j \neq 0 \quad \text{or} \quad j' = j \pm 1$$

and, for any j', j

$$m' = m \text{ or } m' = m \pm 1.$$

We have already found (page 19) the matrix element for $j' = j$. Now we will do $j' = j + 1$.

$$\begin{aligned}
\text{Consider } [J_+, V_+] &= [J_x + iJ_y, V_x + iV_y] \\
&= [J_x, V_x] + i[J_x, V_y] + i[J_y, V_x] - [J_y, V_y] \\
&\quad\ \downarrow \qquad\quad \downarrow \qquad\quad \downarrow \qquad\quad \downarrow \\
&\quad\ 0 \qquad\quad i\hbar V_z \qquad -i\hbar V_z \qquad\quad 0 \\
&= 0.
\end{aligned}$$

Take matrix element and use $\langle j, m+1|J_+|j,m\rangle = \hbar[j(j+1) - m(m+1)]^{1/2}$
$= \hbar[(j+m+1)(j-1)]^{1/2}$

$$\begin{aligned} 0 &= \langle j+1, m+1|(J_+V_+ - V_+J_+|j, m-1\rangle \\ &= \langle j+1, m+1|J_+|j+1, m\rangle \langle j+1, m|V_+|j, m-1\rangle \\ &- \langle j+1, m+1|V_+|jm\rangle \langle jm|J_+|j, m-1\rangle \end{aligned}$$

where we use the $\Delta m = +1$ selection rule for V_+ and J_+.
This provides a recurrence relation for the matrix elements.

$$\langle j+1, m+1|J_+|j+1, m\rangle \langle j+1, m|V_+|j, m-1\rangle = \langle j+1, m+1|V_+|jm\rangle \times \langle jm|J_+|j, m-1\rangle$$

$\hbar[\cancel{(j-m+1)}(j+m+2)]^{1/2}$ $\hbar[\cancel{(j-m+1)}(j+m)]^{1/2}$

So
$$\frac{\langle j+1, m|V_+|j, m-1\rangle}{(j+m)^{1/2}} = \frac{\langle j+1, m+1|V_+|j, m\rangle}{(j+m+2)^{1/2}}.$$

This takes on a simple pattern if we divide both sides by $(j+m+1)^{1/2}$:

$$\begin{aligned} -c_+(j,m) \equiv \frac{\langle j+1, m|V_+|j, m-1\rangle}{[(j+m+1)(j+m)]^{1/2}} &= \frac{\langle j+1, m+1|V_+|j, m\rangle}{[(j+m+2)(j+m+1)]^{1/2}} \\ &= -c_+(j, m+1). \end{aligned}$$

Since m was arbitrary, $c_+(j,m) = c_+(j, m+1) = c_+(j,$ any other $m)$ so the ratio $c_+(j)$ must be independent of m. The m–independence of the matrix element is therefore given by

$$\langle j+1, m+1|V_+|j, m\rangle = -c_+(j)[(j+m+2)(j+m+1)]^{1/2},$$

with $c_+(j) = \ \alpha', j+1\|\underset{\sim}{V}\|\alpha, j$ a reduced matrix element that depends on the detailed nature of $\underset{\sim}{V}$, not merely on its vector character. However, it can be evaluated if the matrix element of $\underset{\sim}{V}$ can be evaluated for any single m value, e.g., $m = j$ or $m = 0$, for which the evaluation is often simpler than in the general case.

Now determine the $j' = j + 1$ elements of V_Z using the above result for V_+. Start with

$$-2\hbar V_Z = [J_-, V_+] \quad \text{which expresses } V_Z \text{ in terms of } J_- \text{ and } V_+, \text{ whose matrix elements we now know.}$$

$$\begin{aligned}
-2\hbar \langle j+1, m|V_Z|j, m\rangle &= \langle j+1, m|J_-|j+1, m+1\rangle \langle j+1, m+1|V_+|j, m\rangle \\
&\quad - \langle j+1, m|V_+|j, m-1\rangle \langle j, m-1|J_-|j, m\rangle \\
&= \hbar[(j+m+2)(j-m+1)]^{1/2}(-c_+(j))[(j+m+2)(j+m+1)]^{1/2} \\
&\quad - \hbar(-c_+(j))[(j+m+1)(j+m)]^{1/2}[(j+m)(j-m+1)]^{1/2} \\
&= -\hbar c_+(j)[(\cancel{j+m}+2) - \cancel{(j+m)}][(j+m+1)(j-m+1)]^{1/2} \\
&= -2\hbar c_+(j)[(j+m+1)(j-m+1)]^{1/2}
\end{aligned}$$

Thus,

$$\langle j+1, m|V_Z|j, m\rangle = c_+(j)[(j+m+1)(j-m+1)]^{1/2}.$$

Similarly, from

$$\hbar V_- = [J_-, V_Z]$$

we find

$$\langle j+1, m-1|V_-|j, m\rangle = c_+(j)[(j-m+2)(j-m+1)]^{1/2}.$$

Results for $j' = j - 1$ are derived in analogous fashion and involve a third reduced matrix element, $c_-(j) = \langle \alpha', j-1\|V\|\alpha, j\rangle$. Hence the m–dependence of a scalar or vector operator follows from its scalar or vector character *only*. Classification of operators by their transformation properties under rotation can be extended to tensors of any rank. In each case the form of the matrix elements is determined except for factors that depend on α and j.

SUMMARY: Non-zero Matrix Elements of a Vector Operator, $\underset{\sim}{V}$

$$\Delta j = +1 \qquad \langle j+1, m \pm 1 | V_\pm | jm \rangle = \mp c_+(j)[(j \pm m + 2)(j \pm m + 1)]^{1/2}$$
$$\langle j+1, m | V_Z | jm \rangle = c_+(j)[(j + m + 1)(j - m + 1)]^{1/2}$$

$$\Delta j = 0 \qquad \langle j, m \pm 1 | V_\pm | jm \rangle = c_0(j)[(j \pm m + 1)(j \mp m)]^{1/2}$$
$$\langle jm | V_Z | jm \rangle = c_0(j)m$$

$$\Delta j = -1 \qquad \langle j-1, m \pm 1 | V_\pm | jm \rangle = \pm c_-(j)[(j \mp m)(j \mp m - 1)]^{1/2}$$
$$\langle j-1, m | V_Z | jm \rangle = c_-(j)[(j - m)(j + m)]^{1/2}$$

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Supplement #2 to Lecture #27

Simplification of hyperfine $\mathbf{H}^{\rm hf}$ by Wigner-Eckart Theorem

There can be many angular momenta in an atom that has non–zero L, S, and I, where I is the nuclear spin. The individual angular momenta are

L	orbital angular momentum
S	electron spin angular momentum
I	nuclear spin angular momentum

and the coupled angular momenta are

$\vec{\mathbf{J}} = \vec{\mathbf{L}} + \vec{\mathbf{S}}$	total exclusive of nuclear spin
$\vec{\mathbf{F}} = \vec{\mathbf{J}} + \vec{\mathbf{I}}$	total angular momentum

and the laboratory frame projection of $\vec{\mathbf{F}}$ is

$$M_F.$$

The coupled basis states are

$$|FJLSIM_F\rangle\,.$$

If there are many electrons, we need to define the hyperfine term in the Hamiltonian as a sum over individual electron contributions

$$\mathbf{H}^{\rm hf} = \sum_i (a_i \mathbf{s}_i + b_i \boldsymbol{\ell}_i) \cdot \mathbf{I}.$$

A basis set like $|FJGLSIM_F\rangle$ would make life much simpler for evaluating matrix elements $I \cdot S$ and $I \cdot L$, but it is illegal because $[\mathbf{G}^2, \mathbf{J}^2] \neq 0$.

$$\vec{\mathbf{G}} = \vec{\mathbf{I}} + \vec{\mathbf{S}}$$
$$\vec{\mathbf{J}} = \vec{\mathbf{L}} + \vec{\mathbf{S}}$$
$$\vec{\mathbf{G}}^2 = I^2 + S^2 + 2I \cdot S$$
$$\vec{\mathbf{J}}^2 = L^2 + S^2 + 2L \cdot S$$

$$
\begin{aligned}
[\mathbf{G}^2, \mathbf{J}^2] &= 4[I \cdot S, L \cdot S] \\
[I \cdot S, L \cdot S] &= [I_x S_x + I_y S_y + I_z S_z, L_x S_x + L_y S_y + L_z S_z] \\
&= I_x [S_x, S_y] L_y + I_x [S_x, S_z] L_z \\
&\quad + I_y [S_y, S_x] L_x + I_y [S_y, S_z] L_z \\
&\quad + I_z [S_z, S_x] L_x + I_z [S_z, S_y] L_y \\
&= i\hbar [I_x L_y S_z - I_x L_z S_y - I_y L_x S_z + I_y L_z S_x + I_z L_x S_y - I_z L_y S_x] \\
&= i\hbar I \cdot (L \times S) \neq 0.
\end{aligned}
$$

We want to replace

$$\mathbf{H}^{\mathrm{hf}} = \sum_i (a_i \mathbf{s}_i + b_i \boldsymbol{\ell}_i) \cdot \mathbf{I}$$

by

$$\mathbf{H}^{\mathrm{hf}} = c_{\mathrm{JLS}} \mathbf{J} \cdot \mathbf{I}$$

and to evaluate c_{JLS}. To accomplish this we must first go to an uncoupled basis set

$$|FJLSIM_F\rangle = \sum_{M_J} |JLSM_J\rangle |IM_I = M_F - M_J\rangle \underbrace{\langle JLSM_J | \langle IM_F - M_J | FJLSIM_F \rangle}_{d_{M_J}}.$$

Call the vector-coupling coefficient d_{M_J} for simplicity (we are not actually going to look up and insert values for this mixing coefficient).

We seek only the diagonal matrix elements of $\mathbf{H}^{\mathrm{hf}}$

$$
\begin{aligned}
FJLSIM_F | \mathbf{H}^{\mathrm{hf}} | FJLSIM_F \; &= \sum_{M_J} \sum_{M_J'} d_{M_J} d_{M_J'} \times \\
&\left\langle JLSM_J \middle| \sum_i (a_i \mathbf{s}_i + b_i \boldsymbol{\ell}_i) \middle| JLSM_J' \right\rangle \langle IM_F - M_J | \mathbf{I} | IM_F - M_J' \rangle
\end{aligned}
$$

Since $\sum_i (a_i \mathbf{s}_i + b_i \boldsymbol{\ell}_i)$ is a vector with respect to $\mathbf{J}$ and we want only diagonal matrix elements with respect to J, L, and S, we can replace the microscopic

operator by the corresponding matrix element of **J** times the reduced matrix element

$$\left\langle JLSM_J|\sum_i(a_i\mathbf{s}_i+b_i\boldsymbol{\ell}_i)|JLSM'_J\right\rangle$$
$$=\left\langle JLS\ \ \sum_i(a_i\mathbf{s}_i+b_i\boldsymbol{\ell}_i)\ \ JLS\right\rangle\times\langle JLSM_J|\mathbf{J}|JLSM'_J\rangle$$

$$c_{\rm JLS}\equiv\left\langle JLS\ \ \sum_i(a_i\mathbf{s}_i+b_i\boldsymbol{\ell}_i)\ \ JLS\right\rangle$$
$$FJLSIM_F\ \ \mathbf{H}^{\rm hf}\ \ FJLSIM_F\ \ =\sum_{M_J}\sum_{M'_J}c_{\rm JLS}\,\langle JLSM_J|\mathbf{J}|JLSM'_J\rangle\cdot\times$$
$$\langle IM_F-M_J|\mathbf{I}|IM_F-M'_J\rangle$$

but now we can recognize completeness and contract the product of matrix elements to one of **J**·**I** in the fully coupled basis set

$$FJLSIM_F|\mathbf{H}^{\rm hf}|FJLSIM_F\ \ =c_{JLS}\,\langle FJLSIM_F|\mathbf{J}\cdot\mathbf{I}|FJLSIM_F\rangle$$

but

$$\vec{\boldsymbol{J}}+\vec{\boldsymbol{I}}=\vec{\boldsymbol{F}}$$
$$\vec{\boldsymbol{J}}^2+\vec{\boldsymbol{I}}^2+2\vec{\boldsymbol{I}}\cdot\vec{\boldsymbol{J}}=\vec{\boldsymbol{F}}^2.$$

Thus

$$FJLSIM_F|\mathbf{H}^{\rm hf}|FJLSIM_F\ \ =\frac{\hbar^2}{2}c_{JLS}[F(F+1)-J(J+1)-I(I+1)].$$

Now the problem for the $3d4p$ configuration reduces to expressing the 12 different c_{JLS} constants in terms of single-spin-orbital integrals

$$\ell\hbar b_{n\ell}=\langle n\ell\lambda=\ell|b\ell_z|n\ell\lambda=\ell\rangle\quad\text{for 3d and 4p}$$
$$\frac{1}{2}\hbar a_{n\ell}=\langle n\ell s\alpha|as_z|n\ell s\alpha\rangle\,.$$

To accomplish this, we start with the extreme states

$$\left\langle J = L+SLSM_J = L+S|\sum_i (a_i \mathbf{s}_{z_i} + b_i \boldsymbol{\ell}_{z_i})|L+SLSL+S \right\rangle =$$
$$c_{J=L+SLS}\,\langle J = L+SLSM_J = L+S|\mathbf{J}_z|L+SLSL+S\rangle$$

e.g. ${}^3F_4 M_J = 4 \;\; = |3d2\alpha 4p1\alpha\rangle$ (λ values)

$$\left\langle 4314 \;\; \sum_i (a_i \mathbf{s}_{z_i} + b_i \boldsymbol{\ell}_{z_i}) \;\; 4314 \right\rangle = c_{431}\hbar 4$$
$$= \hbar[a_{3d}/2 + a_{4p}/2 + 2b_{3d} + b_{4p}]$$

$$\boxed{c_{431} = \tfrac{1}{4}[a_{3d}/2 + a_{4p}/2 + 2b_{3d} + b_{4p}]}$$

to get the others, use the "sum rule." More on this in lecture #34. The basic idea is that the trace of a matrix is representation-invariant and the reduced matrix elements are independent of projection quantum numbers.

$$\mathbf{J}_- \;\; {}^3F_4 M_J = 4 \;\; = \sum_i (\ell_- + s_{-i})|3d2\alpha 4p1\alpha\rangle \quad (\lambda \text{ values})$$
$$\hbar[20-12]^{1/2} \;\; {}^3F_4 3 \;\; = \hbar\left\{[6-2]^{1/2}|3d1\alpha 4p1\alpha\rangle + [2-0]^{1/2}|3d2\alpha 4p0\alpha\rangle\right.$$
$$\left. +|3d2\beta 4p1\alpha\rangle + |3d2\alpha 4p1\beta\rangle\right\}$$
$$|{}^3F_4 3\rangle = 2^{-1/2}|3d1\alpha 4p1\alpha\rangle + \frac{1}{2}|3d2\alpha 4p0\alpha\rangle$$
$$+ 8^{-1/2}|3d2\beta 4p1\alpha\rangle + 8^{-1/2}|3d2\alpha 4p1\beta\rangle$$

$$3c_{431} = \frac{1}{2}\left[a_{3d}/2 + a_{4p}/2 + b_{3d} + b_{4p}\right] + \frac{1}{4}\left[a_{3d}/2 + a_{4p}/2 + 2b_{3d}\right]$$
$$+\frac{1}{8}\left[-a_{3d}/2 + a_{4p}/2 + 2b_{3d} + 1b_{4p}\right] + \frac{1}{8}\left[a_{3d}/2 - a_{4p}/2 + 2b_{3d} + b_{4p}\right]$$

$$\boxed{c_{431} = \tfrac{1}{8}a_{3d} + \tfrac{1}{8}a_{4p} + \tfrac{1}{2}b_{3d} + \tfrac{1}{4}b_{4p}} \quad \text{confirmed as expected}$$

$$
\begin{aligned}
{}^1F_3 3 \;\; &= 2^{1/2}\left[|3d2\alpha 4p1\beta\rangle - |3d2\beta 4p1\alpha\rangle\right] \\
3c_{330} &= \frac{1}{2}\left[a_{3d}/2 - a_{4p}/2 + 2b_{3d} + b_{4p}\right] \\
&\quad + \frac{1}{2}\left[-a_{3d}/2 + a_{4p}/2 + 2b_{3d} + b_{4p}\right]
\end{aligned}
$$

$$\boxed{c_{330} = \tfrac{2}{3}b_{3d} + \tfrac{1}{3}b_{4p}}$$

The $J = 3, M_J = 3$ box contains entries from

$${}^3F_4 3 \;\; , \;\; {}^3F_3 3 \;\; , \;\; {}^1F_3 3 \;\; , \text{and} \;\; {}^3D_3 3$$

$$
\begin{aligned}
&3c_{431} + 3c_{331} + 3c_{330} + 3c_{321} \\
&\quad = \langle 3d1\alpha 4p1\alpha\rangle + \langle 3d2\alpha 4p0\alpha\rangle + \langle 3d2\beta 4p1\alpha\rangle + \langle 3d2\alpha 4p1\beta\rangle \\
&\quad = a_{3d}/2 + a_{3d}/2 - a_{3d}/2 + a_{3d}/2 + a_{4p}/2 + a_{4p}/2 + a_{4p}/2 - a_{4p}/2 \\
&\quad + b_{3d} + b_{4p} + 2b_{3d} + 2b_{3d} + b_{4p} + 2b_{3d} + b_{4p} \\
&\quad = a_{3d} + a_{4p} + 7b_{3d} + 3b_{4p}
\end{aligned}
$$

We know c_{431} and c_{330}. Need to do a bit of algebra to get c_{331} and c_{321}.

$${}^3D_3 3 \;\; = \;\; {}^3DM_L = 2M_S = 1 \;\; = -\left(\frac{1}{3}\right)^{1/2}|3d1\alpha 4p1\alpha\rangle + \left(\frac{2}{3}\right)^{1/2}|3d2\alpha 4p0\alpha\rangle$$

$$
\begin{aligned}
3c_{321} &= \frac{1}{3}\left[a_{3d}/2 + a_{4p}/2 + b_{3d} + b_{4p}\right] \\
&\quad + \frac{2}{3}\left[a_{3d}/2 + a_{4p}/2 + 2b_{3d}\right]
\end{aligned}
$$

$$\boxed{c_{321} = \tfrac{1}{6}a_{3d} + \tfrac{1}{6}a_{4p} + \tfrac{5}{9}b_{3d} + \tfrac{1}{9}b_{4p}}$$

$$\therefore \boxed{c_{331} = \tfrac{1}{24}a_{3d} + \tfrac{1}{24}a_{4p} + \tfrac{11}{18}b_{3d} + \tfrac{11}{36}b_{4p}}$$

(steps omitted)

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Lecture 27 Supplement #3: A User's Guide to Angular Momentum Theory

The wavefunctions for both atoms and molecules in the gas phase may be factored into a *molecule-specific* radial part and a *universal* angular part. An enormously powerful one-size-fits-all theory exists for deriving the universal angular part of molecular basis-states and eigen-states, and calculation of the angular part of all matrix elements. This lecture is an attempt to provide a user's guide to angular momentum theory. It is adapted from pages 152-175 of Rotational Spectroscopy of Diatomic Molecules, John Brown and Alan Carrington, Cambridge University Press, 2003, and Angular Momentum, Richard N. Zare, Wiley, 1988, pages 180-200. The outline is:

- Commutation Rule definition of an angular momentum
- Matrix elements of the magnitude ($\mathbf{j}^2$) and components ($\mathbf{j}_x$, $\mathbf{j}_y$, $\mathbf{j}_z$, $\mathbf{j}_+$, $\mathbf{j}_-$) of a single angular momentum operator
- Uncoupled and Coupled representations for two angular momenta, $\mathbf{j}_1$ and $\mathbf{j}_2$, coupled to make a total angular momentum, $\mathbf{j}=\mathbf{j}_1+\mathbf{j}_2$
- Vector-coupling, Clebsch-Gordan, and 3-*j* coefficients for the transformation between coupled and uncoupled representations
- Commutation rule definition of a spherical tensor operator, T^k_μ (k rank, μ component, $-k \le \mu \le k$: scalar k=0, vector k=1, and k=2 2nd rank tensor)
- Combinations of three or more angular momenta to form composite basis states: 6-*j* and 9-*j* coefficients
- Combination of two or more angular momenta to form a composite spherical tensor operator, $\mathbf{T}^k_\mu(\mathbf{A},\mathbf{B})$
- Wigner-Eckart Theorem for matrix elements of spherical tensor operators
- Wigner-Eckart Theorem for matrix elements of composite spherical tensor operators.

I will illustrate each step with a worked example.

Commutation Rule Definition of an Angular Momentum

$$[\mathbf{J}_i, \mathbf{J}_j] = i\hbar \sum_k \varepsilon_{ijk} \mathrm{J}_k$$

is the universal signature of any operator that qualifies for the name "angular momentum."

* This angular momentum commutation rule can be derived from the most important commutation rule in Quantum Mechanics

$$\left[\hat{x}, \hat{p}_x\right] = i\hbar$$

* This angular momentum commutation rule is useful in deriving all of the properties of $\hat{J}^2, \hat{J}_x, \hat{J}_y, \hat{J}_z, \hat{J}_\pm = \hat{J}_x \pm i\hat{J}_y$ acting on any simultaneous eigenstate of $\hat{J}^2$ and $\hat{J}_z$, $\left|JM_J\right\rangle$:

$$\hat{J}^2\left|JM\right\rangle = \hbar^2 J(J+1)\left|JM\right\rangle$$
$$\hat{J}_z\left|JM\right\rangle = \hbar M\left|JM\right\rangle$$
$$\hat{J}_\pm\left|JM\right\rangle = \hbar\left[J(J+1) - M(M\pm 1)\right]^{1/2}\left|JM\pm 1\right\rangle$$

$$\hat{J}_x\left|JM\right\rangle = \frac{1}{2}\left(\hat{J}_+ + \hat{J}_-\right)\left|JM\right\rangle$$
$$= \frac{\hbar}{2}\left\{\left[J(J+1) - M(M+1)\right]^{1/2}\left|JM+1\right\rangle + \left[J(J+1) - M(M-1)\right]^{1/2}\left|JM-1\right\rangle\right\}$$

$$\hat{J}_y\left|JM\right\rangle = \frac{-i}{2}\left(\hat{J}_+ + \hat{J}_-\right)\left|JM\right\rangle$$
$$= \frac{\hbar}{2}(-i)\left\{\left[J(J+1) - M(M+1)\right]^{1/2}\left|JM+1\right\rangle - \left[J(J+1) - M(M-1)\right]^{1/2}\left|JM-1\right\rangle\right\}$$

* This commutation rule is also surprising because it requires that

$$\hat{J} \times \hat{J} = i\hbar\hat{J} \neq 0$$

because, classically, the cross product of any vector with itself is zero.

* It is more surprising that, while orthogonal linear translations commute, angular translations *do not* commute.

Transformation between Coupled and Uncoupled Representations

We used the $\hat{J}_\pm$ plus orthogonality method to derive the transformation between a coupled representation

$$\text{e.g.} \quad \hat{J} = \hat{L} + \hat{S} \qquad \left|JLSM_J\right\rangle$$

and an uncoupled representation: $|LM_LSM_S\rangle$. More generally, one knows that there must exist a unitary transformation between two complete same-dimension basis set representations, as is true for $|JLSM_J\rangle$ and $|LM_LSM_S\rangle$, where the same (L,S) dimension for each basis set is $(2L + 1)(2S + 1)$.

The ladders plus orthogonality method is extremely laborious and inelegant, but it is based on easily remembered matrix elements. There is a much better way, based on complete tables

of transformation coefficients between coupled and uncoupled representations. The relationship between the coupled and uncoupled representations is obtained via completeness:

$$\left|JLSM_J\right\rangle = \sum_{M_L,M_S} \left|LM_LSM_S\right\rangle\left\langle LM_LSM_S\right|JLSM_J\left\rangle.$$

Two simplifications may be made

$$\left|LM_LSM_S\right\rangle \equiv \left|LM_L\right\rangle\left|SM_S\right\rangle$$

and, based on $M_L + M_S = M_J$, the double-sum over M_L,M_S may be replaced by a single sum

$$\left|JLSM_J\right\rangle = \sum_{M_S=-S}^{S} \left|L, M_L = M_J - M_S, SM_J\right\rangle\left\langle L, M_L = M_J - M_S, SM_J\right|JLSM_J\left\rangle.$$

The factor on the RHS is called a **vector coupling coefficient**. Since the coupled↔uncoupled transformation is unitary [$\mathbf{U}^{-1} = \mathbf{U}^{\dagger}$], the inverse transformation is given by

$$\left|L, M_L = M_J - M_S, SM_S\right\rangle = \sum_{M_S=-S}^{S} \left\langle JLS, M_J = M_L + M_S\right|L, M_L = M_J - M_S, SM_S\left\rangle^{*}\right|JLS, M_J = M_L + M_S\left\rangle.$$

The transformation coefficient in the above equation is the conjugate transform of the transformation coefficient in the previous equation. These vector coupling coefficients are typically expressed (and tabulated) in a slightly more compact form as **Clebsch-Gordan coefficients**.

$$\left|JLSM_J\right\rangle = \sum_{M_S=-S}^{S} \left|LM_L\right\rangle\left|SM_S\right\rangle\left\langle LM_LSM_S\right|JM_J\left\rangle$$

and

$$\left|LM_L\right\rangle\left|SM_S\right\rangle = \sum_{\substack{J=|L-S| \\ M_J=M_L+M_S}}^{L+S} \left|JM_J\right\rangle\left\langle JM_J\right|LM_LSM_S\left\rangle.$$

The factor $\langle LM_L,SM_S|JM_J\rangle$ is the **Clebsch-Gordan coefficient**. It has been simplified by suppression of the redundant L,S quantum numbers in the ket. The M_J quantum number could have been suppressed as well because $M_J = M_L + M_S$.

A more compact and useful expression of the transformation between the coupled and uncoupled representations is given by the **Wigner 3-*j* symbols**. Formally, one can think of the 3-*j* symbol as expressing the coupling of three angular momenta { $|J,M_J\rangle$, $|LM_L\rangle$ and $|SM_S\rangle$} to make a "scalar" quantity, symbolized by $|0,0\rangle$ (see Brown and Carrington, page 154). One obtains a relationship between the 3-*j* symbol and a Clebsch-Gordan coefficient

$$\left\langle LM_L SM_S \middle| JM_J \right\rangle = (-1)^{L-S+M_J}(2J+1)^{1/2}\begin{pmatrix} L & S & J \\ M_L & M_S & M_J \end{pmatrix}.$$

CLEBSCH-GORDAN **3-*j***

Thus

$$\left| JM_J(LS) \right\rangle = (-1)^{L-S+M_J}(2J+1)^{1/2}\sum_{\substack{M_L,M_S \\ M_L+M_S=M_J}}\begin{pmatrix} L & S & J \\ M_L & M_S & -M_J \end{pmatrix}\left| LM_L SM_S \right\rangle.$$

The $-M_J$ in the 3-*j* coefficient is initially puzzling, but it expresses the requirement that $M_L + M_S = M_J$ or that the three M quantum numbers in the 3-*j* symbol must sum to zero:

$$M_L + M_S - M_J = 0.$$

Using a computer to compute and use 3-*j* coefficients requires significantly less time and effort than for you to look up a tabulated 3-*j* coefficient.

What is so great about 3-*j* coefficients? (i) the three angular momenta are treated symmetrically; (ii) the m_i and associated j_i appear together in a single column of the 3-*j* symbol; (iii) an odd interchange of columns (123 $\rightarrow$ 213 but not an even interchange 123 $\rightarrow$ 312) is obtained by

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & -m_3 \end{pmatrix} = (-1)^{j_1+j_2+j_3}\begin{pmatrix} j_2 & j_1 & j_3 \\ m_2 & m_1 & -m_3 \end{pmatrix}$$

and a sign reversal of all m in the bottom row is given by

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & -m_3 \end{pmatrix} = (-1)^{j_1+j_2+j_3}\begin{pmatrix} j_1 & j_2 & j_3 \\ -m_1 & -m_2 & m_3 \end{pmatrix}$$

and the general transformation is

$$\left| j_3 m_3 \left(j_1 j_2 \right) \right\rangle = (-1)^{j_1-j_2+m_3}\left(2j_3+1\right)^{1/2}\sum_{m_1,m_2=m_3-m_1}\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & -m_3 \end{pmatrix}\left| j_1 m_1 \right\rangle \left| j_2 m_2 \right\rangle.$$

The inverse transformation (from coupled to uncoupled) is given by

$$\left|j_1 m_1 j_2 m_2\right\rangle = \sum_{\substack{j_3=|j_1-j_2| \\ m_3=m_1+m_2}}^{j_1+j_2} (-1)^{j_2-j_1-m_3}\left(2j_3+1\right)^{1/2}\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & -m_3 \end{pmatrix}\left|j_1 j_2 j_3 m_3\right\rangle.$$

Notice that the same 3-*j* symbol appears in the coupled→uncoupled and uncoupled→coupled representations, but the sum is carried out over m_1 ($m_2 = m_3 - m_1$) in the uncoupled↔coupled transformation and over j_3 in the coupled→uncoupled transformation.

Example of coupled to uncoupled transformation

Suppose we are interested in the Zeeman effect for the J,M_J components of a ^3F ($S = 1$, $L = 3$) state. We know that, at zero-field, the ^3F state has $J = 4, 3, 2$ spin-orbit components and that

$$\mathbf{H}_{SO} = A\mathbf{L}\cdot\mathbf{S} = \frac{1}{2}A\left[J(J+1) - L(L+1) - S(S+1)\right]$$

because $\mathbf{J}^2 = (\mathbf{L}+\mathbf{S})^2$ thus $\mathbf{L}\cdot\mathbf{S} = \frac{1}{2}\left(\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2\right)$

$$E_{^3F_2} = E^{(0)}_{^3F} + \frac{1}{2}A[6-12-2] = E^{(0)}_{^3F} - 4A$$

$$E_{^3F_3} = E^{(0)}_{^3F} + \frac{1}{2}A[12-12-2] = E^{(0)}_{^3F} - A$$

$$E_{^3F_4} = E^{(0)}_{^3F} + \frac{1}{2}A[20-12-2] = E^{(0)}_{^3F} + 3A$$

We need the uncoupled representation to compute the Zeeman effect for each of the J,M_J levels. In the uncoupled representation $\mathbf{H}^{\text{Zeeman}} = -\hbar\mu_0 B_z(\mathbf{L}_z + 2\mathbf{S}_z)$ (μ_0 is the Bohr magneton). We want to know what $\mathbf{L}_z$ and $\mathbf{S}_z$ do to each $|\text{LSJM}_\text{J}\rangle$ coupled basis state. Here are a few examples of coupled→uncoupled $|\text{LSJM}_\text{J}\rangle \rightarrow |\text{LM}_\text{L}\text{SM}_\text{S}\rangle$ transformations:

$$\left|3144\right\rangle = a\left|3311\right\rangle$$
$$\left|3143\right\rangle = b\left|3310\right\rangle + c\left|3211\right\rangle$$
$$\left|3133\right\rangle = d\left|3310\right\rangle + e\left|3211\right\rangle$$
$$\left|3132\right\rangle = f\left|331-1\right\rangle + g\left|3210\right\rangle + h\left|3111\right\rangle$$
$$\left|3122\right\rangle = i\left|331-1\right\rangle + j\left|3210\right\rangle + k\left|3111\right\rangle$$
$$\left|3121\right\rangle = l\left|321-1\right\rangle + m\left|3110\right\rangle + n\left|3011\right\rangle$$

where the mixing coefficients a–n are evaluated using tabulated 3-*j* coefficients.

$$a = \begin{pmatrix} L & S & J \\ M_L & M_S & -M_J \end{pmatrix} = \begin{pmatrix} 3 & 1 & 4 \\ 3 & 1 & -4 \end{pmatrix} = 1.000$$

$\left\langle L_z + 2S_z \right\rangle = 5\hbar$ for ${}^3F_4 M_J = 4$

$$b = \begin{pmatrix} 3 & 1 & 4 \\ 3 & 0 & -3 \end{pmatrix} = 0.500$$

$$c = \begin{pmatrix} 3 & 1 & 4 \\ 2 & 1 & -3 \end{pmatrix} = 0.866$$

$\left\langle L_z + 2S_z \right\rangle = 3.75\hbar$ for ${}^4F_3 M_J = 3$

$$d = \begin{pmatrix} 3 & 1 & 3 \\ 3 & 0 & -3 \end{pmatrix} = 0.866$$

$$e = \begin{pmatrix} 3 & 1 & 3 \\ 2 & 1 & -3 \end{pmatrix} = -0.500$$

$\left\langle L_z + 2S_z \right\rangle = 3.25\hbar$ for ${}^3F_3 M_J = 2$

$$f = \begin{pmatrix} 3 & 1 & 3 \\ 3 & -1 & -2 \end{pmatrix} = 0.500$$

$$g = \begin{pmatrix} 3 & 1 & 3 \\ 2 & 0 & -2 \end{pmatrix} = 0.577$$

$$h = \begin{pmatrix} 3 & 1 & 3 \\ 1 & 1 & -2 \end{pmatrix} = -0.645$$

$\left\langle L_z + 2S_z \right\rangle = 2.16\hbar$ for ${}^3F_3 M_J = 2$

$$i = \begin{pmatrix} 3 & 1 & 2 \\ 3 & -1 & -2 \end{pmatrix} = 0.845$$

$$j = \begin{pmatrix} 3 & 1 & 2 \\ 2 & 0 & -2 \end{pmatrix} = -0.488$$

$$k = \begin{pmatrix} 3 & 1 & 2 \\ 1 & 1 & -2 \end{pmatrix} = 0.218$$

$\left\langle L_z + 2S_z \right\rangle = 1.333\hbar$ for ${}^3F_2 M_J = 2$

$$l = \begin{pmatrix} 3 & 1 & 2 \\ 2 & -1 & -1 \end{pmatrix} = 0.690$$

$$m = \begin{pmatrix} 3 & 1 & 2 \\ 1 & 0 & -1 \end{pmatrix} = -0.617$$

$$n = \begin{pmatrix} 3 & 1 & 2 \\ 0 & 1 & -1 \end{pmatrix} = 0.378$$

$\left\langle L_z + 2S_z \right\rangle = 0.666\hbar$ for ${}^3F_2 M_J = 1$

These results are consistent with a formula given on page 381 of Condon and Shortley.

$$\mathbf{H}^{\text{Zeeman}} = \gamma B_z g_{\text{JLS}} \hbar M_J$$

$$g_{\text{JLS}} = 1 + \frac{J(J+1) - L(L+1) + S(S+1)}{2J(J+1)}$$

Definition of Spherical Tensor Operators

We know how to use 3-*j* coefficients to couple two angular momentum operators to make a third angular momentum. We define an angular momentum operator by a commutation rule

$$\left[\hat{J}_i, \hat{J}_j\right] = i\hbar \sum_k \varepsilon_{ijk} J_k .$$

If we can re-express all angle-dependent operators as a sum of angular momenta-like operators (irreducible tensor operators), then we can use 3-*j*-like coefficients to evaluate the angle dependent factors of all matrix elements. There is also a special feature of the **H** matrix. In field-free space **H** acts like a scalar operator

$$\left[\hat{J}_\pm, \hat{T}^k_q\right] = \pm\hbar\left[(k \mp q)(k \pm q + 1)\right]^{1/2} \hat{T}^k_{q+1}$$

$$\left[\hat{J}_z, \hat{T}^k_q\right] = \hbar q \hat{T}^k_q$$

where k is the rank ($k = 0$ is Scalar, $k = 1$ is vector, $k = 2$ is 2nd rank tensor) and q is the component ($-k \le q \le k$). For a Scalar Operator ($q = 0$, $k = 0$) the commutation rule definition tells us that

$$\left[\hat{J}_\pm, \hat{T}^0_0\right] = 0$$

$$\left[\hat{J}_z, \hat{T}^0_0\right] = 0$$

which means that $\hat{T}^0_0$ acts like an M-independent constant

$$\left\langle JM \left| \hat{T}^0_0 \right| J'M' \right\rangle = c_J \delta_{JJ'} \delta_{MM'}$$

with exclusively diagonal matrix elements in the J,M_J basis set. For a Vector Operator,

$$\hat{V}_0 = \hat{V}_z$$
$$\hat{V}_\pm = \mp 2^{-1/2}\left(\hat{V}_x \pm i\hat{V}_y\right)$$
$$\left[J_z, V_0\right] = \hbar T_q^1 = \hbar V_q$$
$$\left[J_z, V_\pm\right] = \hbar V_{\pm 1}$$
$$\left[J_\pm, V_0\right] = \pm\hbar\left[\mp(1\pm 1)\right]^{1/2} V_\pm$$
$$\left[J_\pm, V_+\right] = \pm\hbar[(1\mp 1)(q\pm 1+1)]^{1/2} V_0$$
$$\left[J_\pm, V_-\right] = \pm\hbar\left[(1\pm 1)(1\mp 1+1)\right]^{1/2} V_0$$

Example

The spherical tensor classification of operators depends on the choice of the angular momentum operator used in the classification commutation rules. For example, the spin-orbit operator, $\mathbf{AL}\cdot\mathbf{S}$, is a scalar operator with respect to $\mathbf{J}$ but a vector operator with respect to both $\mathbf{L}$ and $\mathbf{S}$

$$\left[\mathbf{J}^2, \mathbf{AL}\cdot\mathbf{S}\right] = A\left[\mathbf{J}^2, \mathbf{L}\cdot\mathbf{S}\right]$$
$$\mathbf{L}\cdot\mathbf{S} = \frac{1}{2}\left(\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2\right)$$
$$\left[\mathbf{J}^2, \mathbf{AL}\cdot\mathbf{S}\right] = \frac{1}{2}A\left[\mathbf{J}^2, \mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2\right] = 0$$
$$\begin{aligned}\left[\mathbf{J}_z, A\mathbf{L}\cdot\mathbf{S}\right] &= A\left[\mathbf{J}_z, \mathbf{L}_x\mathbf{S}_x + \mathbf{L}_y\mathbf{S}_y + \mathbf{L}_z\mathbf{S}_z\right] \\ &= A\left[\mathbf{L}_z, \mathbf{L}_x\mathbf{S}_x + \mathbf{L}_y\mathbf{S}_y + \mathbf{L}_z\mathbf{S}_z\right] \\ &= A\left[\mathbf{S}_z, \mathbf{L}_x\mathbf{S}_x + \mathbf{L}_y\mathbf{S}_y + \mathbf{L}_z\mathbf{S}_z\right] \\ &= i\hbar A\left[\left(\mathbf{L}_y\mathbf{S}_x - \mathbf{L}_x\mathbf{S}_y\right) + \left(\mathbf{L}_x\mathbf{S}_y - \mathbf{L}_y\mathbf{S}_x\right)\right] \\ &= 0\end{aligned}$$

We knew this had to be true because $\mathbf{J}^2$, $\mathbf{J}_z$, $\mathbf{L}^2$, and $\mathbf{S}^2$ are a complete set of commuting operators. But now consider $[\mathbf{L}_z, A\mathbf{L}\cdot\mathbf{S}]$ and $[\mathbf{S}_z, A\mathbf{L}\cdot\mathbf{S}]$ individually.

$$\left[\mathbf{L}_z, A\mathbf{L}\cdot\mathbf{S}\right] = i\hbar A\left(\mathbf{L}_y\mathbf{S}_x - \mathbf{L}_x\mathbf{S}_y\right) \neq 0$$
$$\left[\mathbf{S}_z, A\mathbf{L}\cdot\mathbf{S}\right] = i\hbar A\left(\mathbf{L}_x\mathbf{S}_y - \mathbf{L}_y\mathbf{S}_x\right) \neq 0$$

It is also true that $\mathbf{L}$ and $\mathbf{S}$ are scalar operators with respect to each other, because they do not operate on the same coordinates.

Example: Tensor Operators are "like" Angular Momenta

We can use the tensor operator commutation rule definitions

$$\left[J_z, T_q^k\right] = \hbar q T_q^k$$
$$\left[J_\pm, T_q^k\right] = \pm\hbar\left[\left(k \mp q\right)\left(k \pm q + 1\right)\right]^{1/2} T_{q\pm1}^k$$

to construct operators that "look like" operators with specified angular momentum characteristics with respect to *any classifying angular momentum*. This means that we can use angular momentum coupling methods to calculate matrix elements in the classifying operator's basis set, e.g. $|j,m_j\rangle$. This is the basis for the Wigner-Eckart theorem

$$\left\langle n,j,m_j\left|\mathbf{T}_q^k(\mathbf{A})\right|n',j',m_j'\right\rangle = (-1)^{j-m_j}\begin{pmatrix} j & k & j' \\ -m_j & q & m_j' \end{pmatrix} = \left\langle n,m\left\|\mathbf{T}^k(\mathbf{A})\right\|n',j'\right\rangle.$$

The 3-j coefficient describes the angular momentum-like coupling of $\langle n,j,m_j|$, $\mathbf{T}_q^k(\mathbf{A})$, and $|n',j',m_j'\rangle$, and $\langle\, n,j\|\mathbf{T}^k(\mathbf{A})\|\, n'\,j'\rangle$ is called a "reduced matrix element" because in it, none of the projection quantum numbers are specified for the bra, the ket, or the tensor operator. *All* information about the projection quantum numbers is expressed in the 3-j coefficient! $\mathbf{T}_q^k(\mathbf{A})$ means that the components of operator **A** are arranged in spherical tensor form. For example, the relationships between Cartesian tensor and spherical tensor form are as follows:

$\mathbf{j}^2$ is a scalar with respect to **j**

$$T_0^0(\mathbf{j}) = \mathbf{j}^2$$

$\vec{\mathbf{j}} = \hat{i}\mathbf{j}_x + \hat{j}\mathbf{j}_y + \hat{k}\mathbf{j}_k$ is a vector with respect to **j**

$$T_1^1(\mathbf{j}) = -2^{-1/2}\left(\mathbf{j}_x + i\mathbf{j}_y\right) = -2^{-1/2}\mathbf{j}_+$$
$$T_0^1(\mathbf{j}) = \mathbf{j}_z$$
$$T_{-1}^1(\mathbf{j}) = 2^{-1/2}\left(\mathbf{j}_x - i\mathbf{j}_y\right) = 2^{-1/2}\mathbf{j}_-.$$

The 9 products $\mathbf{j}_i\mathbf{j}_j$ are components of a scalar, vector, and second rank tensor with respect to **j**.

$$T_0^0(jj) = -(3)^{-1/2}\left(j_x^2 + j_y^2 + j_z^2\right) = -(3)^{-1/2}\, j^2$$

$$T_0^1(jj) = i2^{-1/2}\left(j_x j_y - j_y j_x\right) = -\hbar 2^{-1/2}\, j_z$$

$$T_{\pm 1}^1(jj) = \mp(i/2)\left\{\left(j_y j_z - j_z j_y\right) \pm i\left(j_z j_x - j_x j_z\right)\right\}$$

$$= \mp(i/2)\left(i\hbar j_x \mp i\hbar j_y\right) = \pm\frac{\hbar}{2}\left(j_x \pm i j_y\right)$$

$$T_0^2(jj) = 6^{-1/2}\left\{2j_z^2 - j_x^2 - j_y^2\right\}$$

$$T_{\pm 1}^2(jj) = \mp\frac{1}{2}\left\{\left(j_x j_z + j_z j_x\right) \pm i\left(j_y j_z + j_z j_y\right)\right\}$$

$$T_{\pm 2}^2(jj) = \frac{1}{2}\left\{\left(j_x^2 + j_y^2\right) \pm i\left(j_x j_y + j_y j_x\right)\right\}$$

It is also possible to construct spherical tensor operators out of two vector or tensor operators

$$T_q^k(\mathbf{u},\mathbf{v})$$

$$T_Q^K\left(\mathbf{R}^{k_1},\mathbf{R}^{k_2}\right) = \sum_{q_1,q_2}\left\langle k_1,k_2,q_1,q_2 \middle| KQ\right\rangle \mathbf{T}_{q_1}^{k_1}(R)\mathbf{T}_{q_2}^{k_2}(R)$$

3-j, 6-j, and 9-j coefficients

Our goal is to be able to evaluate the angular factor of matrix elements of operators composed of products and sums of angular momentum operators. These operators operate on basis states that are simultaneously eigenstates of many angular momenta ($\mathbf{j}^2$) and angular momentum z-components ($\mathbf{j}_z$). The defining angular momentum commutation rule

$$\left[\mathbf{j}_i,\mathbf{j}_j\right] = i\hbar\sum_k \mathbf{j}_k$$

tells us how to define (or construct) angular momentum operators and a similar commutation rule tells us how to define (or construct) spherical tensor operators, $\mathbf{T}_\mu^k(\mathbf{A})$, of rank k and component μ that act like angular momentum operators. The Wigner-Eckart Theorem tells us how to use angular momentum algebra to couple the bra, ket, and tensor operator to make the numerical value of the matrix element, which is a scalar quantity.

We need to define the weapons of our angular momentum arsenal. 3-j coefficients tell us how to relate the coupled $|j_1,j_2,j,m_j\rangle$ and uncoupled $|j,m_1,j_2,m_2\rangle$ basis states so that we can evaluate matrix elements for either the coupled or uncoupled representation, whichever is more convenient. For example, it is more convenient to deal with the spin-orbit operator, $\mathbf{A}(\mathbf{R})\mathbf{L}\cdot\mathbf{S}$, in the coupled representation, whereas the uncoupled representation is more convenient for the Zeeman Hamiltonian

$$\mathbf{H}^{\text{Zeeman}} = -\gamma B_z(\mathbf{L}_z + 2\mathbf{S}_z).$$

Another property of the 3-j coefficients is that they relate schemes where

$$j_1 + j_2 = j$$
$$j - j_1 = j_2$$

and

$$j - j_2 = j_1.$$

This property is explored by permuting two columns of the 3-j coefficient,

$$\begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & -m_j \end{pmatrix}.$$

We are going to need to deal with more than two angular momenta. For example, there is the nuclear spin I of an atom or the separate nuclear spins of N nuclei in an N-atom molecule. There is also the problem of coupling the ℓ_i and s_i of individual electrons into the L and S of a many-electron atom or molecule. 3-j coefficients guide the coupling of the magnitude and the z-component of three angular momenta. 6-j coefficients, guide the coupling of 3 angular momenta $\mathbf{j}_1$, $\mathbf{j}_2$, and $\mathbf{j}_3$ into a total angular momentum,

$$\mathbf{j} = \mathbf{j}_1 + \mathbf{j}_2 + \mathbf{j}_3.$$

There are 3 ways to accomplish this:

$$\left|\left(j_1, j_2\right) j_{12}, j_3, j\right\rangle$$
$$\left|\left(j_1, j_3\right) j_{13}, j_2, j\right\rangle$$
$$\left|\left(j_2, j_3\right) j_{23}, j_1, j\right\rangle.$$

So we need j_{12}, j_{23}, and j_{13}. There are six angular momenta (d_1, d_2, d_3, j_{12}, j_{23}, j_{13}) to couple to form the total angular momentum, j. It is easily proved that the dimension of each of the three coupled basis sets is conserved. It is also possible to show that operation by $\mathbf{j}_\pm$ on the various basis sets does not depend on any of the m quantum numbers. So we have tables of 6-j coefficients that describe the relationships between the three coupled representations

$$\left|j_1, j_2, j_3\left(j_{23}\right); j\right\rangle = \sum_{j_{12}} \left|j_1, j_2\left(j_{12}\right) j_3; j\right\rangle \left\langle j_1, j_2\left(j_{12}\right) j_3; j \right| j_1, j_2, j_3\left(j_{23}\right); j\right\rangle$$

where

$$\left\langle j_1, j_2\left(j_{12}\right) j_3; j \middle| j_1, j_2, j_3\left(j_{23}\right); j \right\rangle = (-1)^{j_1+j_2+j_3+j}\left[\left(2j_{12}+1\right)\left(2j_{23}+1\right)\right]^{1/2}\left\{ \begin{array}{ccc} j_1 & j_2 & j_{12} \\ j_3 & j & j_{23} \end{array} \right\}.$$

The 6-j coefficients are invariant under any interchange of columns and an interchange of any 2 numbers in the bottom row with the corresponding 2 numbers in the top row

$$\left\{ \begin{array}{ccc} j_1 & j_2 & j_{12} \\ j_3 & j & j_{23} \end{array} \right\} = \left\{ \begin{array}{ccc} j_1 & j_2 & j_3 \\ \ell_1 & \ell_2 & \ell_3 \end{array} \right\}$$

where the second 6-j is how the tables of 6-j coefficients are arranged. The values of the 6-j coefficients are unchanged by permutations of any two columns

$$\left\{ \begin{array}{ccc} j_1 & j_2 & j_3 \\ \ell_1 & \ell_2 & \ell_3 \end{array} \right\} = \left\{ \begin{array}{ccc} j_2 & j_1 & j_3 \\ \ell_2 & \ell_1 & \ell_3 \end{array} \right\} = \text{etc.}$$

or interchange of any two upper and lower pairs

$$= \left\{ \begin{array}{ccc} \ell_1 & \ell_2 & j_3 \\ j_1 & j_2 & \ell_3 \end{array} \right\} = \text{etc.}$$

When there are four angular momenta, the basis set transformations are given by 9-j coefficients

$$\left| \left[j_1, j_2\left(j_{12}\right), \left(j_3, j_4\right) j_{34} \right] j \right\rangle = \sum_{j_{13}, j_{24}} \left[\left(2j_{13}+1\right)\left(2j_{24}+1\right)\left(2j_{12}+1\right)\left(2j_{34}+1\right)\right]^{1/2} \left\{ \begin{array}{ccc} j_1 & j_3 & j_{13} \\ j_2 & j_4 & j_{24} \\ j_{12} & j_{34} & j \end{array} \right\}$$

where the value of the 9-j symbol is multiplied by

$$(-1)^{j_1+j_2+j_3+j_4+j_{12}+j_{13}+j_{24}+j_{34}+j}$$

for exchange of any two rows or columns, unchanged for even permutations of rows or columns, multiplied by –1 for odd permutations of rows or columns, and unchanged by reflection about either diagonal. When one argument of a 9-j symbol is zero, we get reduction to a 6-j symbol

$$\left\{\begin{array}{ccc} j_1 & j_3 & j_{13} \\ j_2 & j_4 & j_{24} \\ j_{12} & j_{34} & 0 \end{array}\right\} = \delta_{j_{13},j_{24}}\delta_{j_{12},j_{34}}\left[\left(2j_{13}+1\right)\left(2j_{12}+1\right)\right]^{-1/2}(-1)^{j_3+j_{13}+j_2+j_{12}}\left\{\begin{array}{ccc} j_1 & j_3 & j_{13} \\ j_4 & j_2 & j_{12} \end{array}\right\}.$$

The 6-j and 9-j symbols are both independent of all *m* quantum numbers.

The Wigner-Eckart Theorem

The Wigner-Eckart Theorem is a powerful tool. It is *reductionistic* in the sense that it reduces an enormous number of angle-dependent matrix elements to a vastly smaller number of *reduced matrix elements* (independent of all projection quantum numbers). The Wigner-Eckart Theorem often guides further reduction of these reduced matrix elements into a much smaller number of structural parameters. It provides a super-highway from *description* to *insight*. It serves as a roadmap to the assembly of a complete and rigorous description of a complex web of interactions between sub-systems: a molecule and an external field, one molecule and another molecule, electrons and nuclei, inter-electronic interactions, intra-electronic interactions (e.g. spin-orbit), inter-nuclear-spin interactions,… The Wigner-Eckart Theorem suggests interpretive shortcuts. It legitimizes the "vector model". Armed with the Wigner-Eckhart Theorem, the scientist is able to recognize and exploit "regular" patterns to "assign" a spectrum and also to recognize and detect local irregularities in a pattern in order to uncover, identify, and quantitate the physical nature of a previously neglected interaction mechanism. The Wigner-Eckart Theorem is a subject worthy of a full semester graduate course as well as one that invites a lifetime of discovery of new uses for it.

Matrix Elements of a Vector Operator

In spherical tensor form a vector operator (a rank-one tensor operator) is expressed as

$$\mathbf{V}_{+1}(\mathbf{J}) = -2^{-1/2}\left(\mathbf{J}_x + i\mathbf{J}_y\right) = -2^{-1/2}\mathbf{J}_+$$
$$\mathbf{V}_0(\mathbf{J}) = \mathbf{J}_z$$
$$\mathbf{V}_{-1}(\mathbf{J}) = 2^{-1/2}\left(\mathbf{J}_x - i\mathbf{J}_y\right) = -2^{-1/2}\mathbf{J}_-$$

and the scalar product of two vector operators **A** and **B** is

$$\mathrm{V(A)}\cdot\mathrm{V(B)} = \sum_{q=-1}^{+1}(-1)^q \mathrm{V}_q(\mathrm{A})\mathrm{V}_q(\mathrm{B}) = \mathrm{A}\cdot\mathrm{B}$$

The Wigner-Eckart Theorem for a vector operator, **V**, that is defined by its commutation rules with components of **J**, as a vector with respect to **J**, can be reduced to

$$\left\langle \eta jm\middle|\mathbf{V}_\pm\middle|\eta jm'\right\rangle = \alpha(\eta, j)\left\langle \eta jm\middle|\mathbf{J}_\pm\middle|\eta jm'\right\rangle$$
$$\left\langle \eta jm\middle|\mathbf{V}_0\middle|\eta jm'\right\rangle = \alpha(\eta, j)\left\langle \eta jm\middle|\mathbf{J}_z\middle|\eta jm'\right\rangle$$

or

$$\left\langle \eta jm \pm 1 \middle| \mathbf{V}_{\pm} \middle| \eta jm' \right\rangle = \alpha(\eta, j)\hbar\left[j(j+1) - m(m \pm 1) \right]^{1/2}$$
$$\left\langle \eta jm \middle| \mathbf{V}_0 \middle| \eta jm' \right\rangle = \alpha(\eta, j)\hbar m \delta_{mm'}$$

η symbolizes a collection of other identifiers of the state. The above equations are the $\Delta j = 0$ reduced set of matrix elements of $\mathbf{V}$. Non-zero $\Delta j = \pm 1$ matrix elements are often between basis states for which $\left| E_j^{(0)} - E_{j\pm1}^{(0)} \right| \gg V_{jm, j\pm1m}$ (e.g. hyperfine components, a single spin-orbit component of an L–S–J state) and the effects of these $\Delta j \neq 0$ interactions are often small enough to be ignored.

One special value of this reduction to a $\Delta j = 0$ state space is the existence of "operator replacements." For example, for the Zeeman effect

$$\mathbf{H}^{\text{Zeeman}} = -\boldsymbol{\mu}_L \cdot \mathbf{B} - \boldsymbol{\mu}_S \cdot \mathbf{B}$$
$$\boldsymbol{\mu}_L = -\beta \mathbf{L}, \quad \boldsymbol{\mu}_S = -2\beta \mathbf{S}$$
$$\boldsymbol{\mu}_L = -\beta \left\langle \ell m_\ell = \ell \middle| \mathbf{L}_z \middle| \ell m_\ell = \ell \right\rangle = -\beta\hbar\ell$$

Similarly for $\boldsymbol{\mu}_S$ (s = 1/2 for an electron)

$$\boldsymbol{\mu}_S = -2\beta\mathbf{S} = -\beta\hbar \text{ (for s = 1/2).}$$

If the magnetic field is along the laboratory Z direction

$$\mathbf{H}^{\text{Zeeman}} = \beta B_Z(\mathbf{L}_Z + 2\mathbf{S}_Z).$$

Here is where the operator replacement trick comes in

$$\mathbf{J} = \mathbf{L} + \mathbf{S} \equiv \mathbf{V}$$
$$\left\langle \alpha jm \middle| \mathbf{V} \middle| \alpha jm' \right\rangle = C \left\langle \alpha jm \middle| \mathbf{J} \middle| \alpha jm' \right\rangle$$

The constant C is evaluated by the following series of tricks

$$\left\langle \alpha jm \middle| \mathbf{V} \cdot \mathbf{J} \middle| \alpha jm \right\rangle = \sum_{m'} \left\langle \alpha jm \middle| \mathbf{V} \middle| \alpha jm' \right\rangle \left\langle \alpha jm' \middle| \mathbf{J} \middle| \alpha jm \right\rangle$$

completeness

$$= C \sum_{m'} \left\langle \alpha jm \middle| \mathbf{J} \middle| \alpha jm' \right\rangle \left\langle \alpha jm' \middle| \mathbf{J} \middle| \alpha jm \right\rangle$$
$$= C\hbar^2 j(j+1)$$
$$C = \frac{\left\langle \alpha jm \middle| \mathbf{V} \cdot \mathbf{J} \middle| \alpha jm \right\rangle}{\hbar^2 j(j+1)}$$

thus

$$\left\langle \alpha jm \middle| \mathbf{V} \middle| \alpha jm' \right\rangle = \frac{\left\langle \alpha jm \middle| \mathbf{V} \cdot \mathbf{J} \middle| \alpha jm \right\rangle}{\hbar^2 j(j+1)} \left\langle \alpha jm \middle| \mathbf{J} \middle| \alpha jm' \right\rangle$$

This formula, called the Landé formula, is a special case of the Wigner-Eckart Theorem. Now we use this formula to evaluate matrix elements of $\mathbf{H}^{\text{Zeeman}}$ in the coupled basis

$$\left\langle \ell sjm\left|\mathbf{L}_Z+2\mathbf{S}_Z\right|\ell sjm\right\rangle=\frac{\left\langle \ell sjm\left|(\mathbf{L}+2\mathbf{S})\bullet\mathbf{J}\right|\ell sjm\right\rangle}{\hbar^2 j(j+1)}\underbrace{\left\langle \ell sjm'\left|\mathbf{J}_Z\right|\ell sjm\right\rangle}_{\hbar m\delta_{mm'}}$$

$$(\mathbf{L}+2\mathbf{S})\cdot\mathbf{J}=(\mathbf{J}+\mathbf{S})\cdot\mathbf{J}=\mathbf{J}^2+\mathbf{S}\cdot\mathbf{J}$$

$$\mathbf{L}=\mathbf{J}-\mathbf{S}$$

$$\mathbf{L}^2=\mathbf{J}^2+\mathbf{S}^2-2\mathbf{S}\cdot\mathbf{J}$$

$$\mathbf{S}\cdot\mathbf{J}=1/2(\mathbf{J}^2+\mathbf{S}^2-\mathbf{L}^2)$$

$$(\mathbf{L}+2\mathbf{S})\cdot\mathbf{J}=\mathbf{J}^2+\mathbf{S}\cdot\mathbf{J}=1/2[3\mathbf{J}^2+(\mathbf{S}^2-\mathbf{L}^2)]$$

and, at last,

$$\left\langle \ell sjm\left|\mathbf{L}_Z+2\mathbf{S}_S\right|\ell sjm\right\rangle=\frac{\left\langle \ell sjm\left|\left(\mathbf{L}+2\mathbf{S}\right)\cdot\mathbf{J}\right|\ell sjm\right\rangle}{\hbar^2 j(j+1)}\hbar m\delta_{mm'}$$

$$=\frac{\hbar m}{j(j+1)}\left[\frac{3}{2}J(J+1)+\frac{1}{2}S(S+1)-\frac{1}{2}L(L+1)\right].$$

This gives the Landé g-factor formula

$$\mathbf{H}^{\text{Zeeman}}=B_Z\hbar M_J g_{\mathbf{LSJ}}$$

$$g_{\mathbf{LSJ}}=1+\frac{J(J+1)-L(L+1)+S(S+1)}{2J(J+1)}.$$

Page 1054 of CTDL derives a similar operator replacement formula called the Projection Theorem. For any vector quantity, **v**, in the body-fixed coordinate system, the projection of **v** on **j**, $\mathbf{v}_{\|}$ is conserved

$$\mathbf{v}_{\|}=\frac{\mathbf{j}\cdot\mathbf{v}}{\mathbf{j}^2}\mathbf{j}$$

or, in operator terminology

$$\mathbf{V}=\frac{\left\langle \eta j|\mathbf{J}\cdot\mathbf{V}|\eta j\right\rangle}{\hbar^2 j(j+1)}\mathbf{J}$$

This is the basis for the **vector model**, which tells us how body frame quantities, averaged over body-rotation, are communicated to the laboratory frame. The vector model is an amazingly insightful interpretive tool!

The Wigner-Eckart Theorem reduces matrix elements of a commutation-rule-classified spherical tensor operator to a reduced matrix element times a 3-*j* coefficient

$$\left\langle \eta' j' m' \middle| \mathbf{T}_q^k(\mathbf{A},\mathbf{B}) \middle| \eta j m \right\rangle = (-1)^{j'-m'} \begin{pmatrix} j' & k & j \\ -m' & q & m \end{pmatrix} \left\langle \alpha' j' \middle\| \mathbf{T}^k(\mathbf{A},\mathbf{B}) \middle\| \alpha j \right\rangle$$

There are several ways to evaluate the reduced matrix elements. One way, illustrated by the treatment of $\mathbf{H}^{\text{Zeeman}}$, involves tricks with reduced-validity (e.g. $\Delta j = 0$ only) replacement operators. Another more general way is to evaluate the matrix element directly for a specific choice of m, m', and q (usually "stretched state", $m = m' = j$ and $q = 0$) and then exploit the m, m', q-independence of the reduced matrix element to obtain its value from the single explicitly evaluated m, m', q-dependent matrix element.

In order to evaluate one specific matrix element of $\mathbf{T}_q^k(\mathbf{A},\mathbf{B})$ it is necessary to know how to construct "compound irreducible tensor operators." Once this is done, matrix elements of simpler $\mathbf{T}_{q'}^{k'}(\mathbf{A})$ and $\mathbf{T}_{q''}^{k''}(\mathbf{B})$ operators may be evaluated in the most convenient basis sets (obtained using 3-*j*, 6-*j*, and 9-*j* coefficients.

Compound Irreducible Tensor Operators

Let **F** be the total of all angular momenta in a system, for example in an AB diatomic molecule

$$\mathbf{F} = \mathbf{J} + \mathbf{I}_{\text{tot}} = \mathbf{R} + \mathbf{L} + \mathbf{S} + \mathbf{I}_A + \mathbf{I}_B$$

One very important insight is that, although **J**, $\mathbf{I}_A$, $\mathbf{I}_B$, **R**, **L**, and **S** are all classified as vector operators (first rank tensor) with respect to **F**, because each operates on a different set of coordinates, **R**, **L**, **S**, $\mathbf{I}_A$, $\mathbf{I}_B$ are all scalar operators with respect to each other. This means that a basis set $|R,L,S,I_A,I_B\rangle$ exists which is a simultaneous eigenstate of $\mathbf{R}^2$, $\mathbf{L}^2$, $\mathbf{S}^2$, $\mathbf{I}_A^2$, and $\mathbf{I}_B^2$. There are many choices of intermediate quantum numbers

$$\mathbf{Q} = \mathbf{S} + \mathbf{I}_{\text{tot}}, \quad \mathbf{F} = \mathbf{Q} + \mathbf{N}, \quad \mathbf{N} = \mathbf{R} + \mathbf{L}$$

and we need 6-*j* coefficients to transform among basis sets based on different sets of intermediate quantum numbers. The key point is that we have all of the tools to evaluate matrix elements for any chosen set of intermediate quantum numbers.

If we have products of two commuting angular momenta **A** and **B**, how do we construct spherical tensor operators of rank k = 2, 1, and 0? Rank k = 0 is special because the field-free Hamiltonian is a scalar (k = 0) operator with respect to **F**.

Suppose we have two non-communicating state spaces, 1 and 2, such as those associated with two commuting angular momenta L↔1 and S↔2.

$$\left|j_1, j_2, j, m\right\rangle = \sum_{m_1, m_2} (-1)^{j_1 - j_2 + m} \begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & m \end{pmatrix} \left|j_1, m_1\right\rangle \left|j_2, m_2\right\rangle$$

We are interested in evaluating matrix elements of a compound spherical tensor operator. Suppose that we have a compound spherical tensor operator, $\mathbf{X}^k_q$, that is composed of a product of spherical tensor operators that operate on independent sets of variables in spaces 1 and 2:

$$\mathbf{X}^k_q\left(\mathbf{T}^{k_1}_{q_1}, \mathbf{U}^{k_2}_{q_2}\right) = \sum_{q_1, q_2} (-1)^{k_1 - k_2 + q} (2k+1)^{1/2} \begin{pmatrix} k_1 & k_2 & k \\ q_1 & q_2 & -q \end{pmatrix} \mathbf{T}^{k_1}_{q_1} \mathbf{U}^{k_2}_{q_2}$$

and we want to evaluate the general matrix element

$$\left\langle \eta, j_1, j_2, j, m \middle| \mathbf{X}^k_q \middle| \eta', j_1', j_2', j', m' \right\rangle = (-1)^{j-m} \begin{pmatrix} j & k & j' \\ -m & q & m' \end{pmatrix} \left\langle \eta, j_1, j_2, j \middle\| \mathbf{X}^k \middle\| \eta', j_1', j_2', j' \right\rangle.$$

After some algebra we get, for the reduced matrix element:

$$\left\langle \eta, j_1, j_2, j \middle\| \mathrm{X}^k \middle\| \eta', j_1', j_2', j' \right\rangle = \left[(2j+1)(2j'+1)(2k+1)\right]^{1/2} \begin{Bmatrix} j_1 & j_1' & k_1 \\ j_2 & j_2' & k_2 \\ j & j' & k \end{Bmatrix}$$

$$\times \sum_{\eta''} \left\langle \eta, j_1 \middle\| \mathrm{T}^{k_1} \middle\| \eta', j_1' \right\rangle \left\langle \eta'', j_2 \middle\| \mathrm{U}^{k_2} \middle\| \eta', j_2' \right\rangle.$$

We have many extremely useful special cases of this general matrix element. Since the field–free Hamiltonian is a spherical tensor of rank 0, it is a scalar operator. Massive simplifications ensue for the matrix element of $\mathbf{X}^0_0$

$$\mathbf{X}^0_0 \equiv \left[\mathbf{T}^k \otimes \mathbf{U}^k\right]^0_0$$

$$\left\langle \eta, j_1, j_2, j \middle\| \mathbf{X}^0 \middle\| \eta', j_1', j_2', j' \right\rangle = \left\langle \eta, j_1, j_2, j \middle\| \left[\mathbf{T}^k \otimes \mathbf{U}^k \right]_0^0 \middle\| \eta', j_1', j_2', j' \right\rangle$$

$$= (-1)^{k+j_2+j+j_1'} (2k+1)^{-1/2} (2j+1)^{1/2} \times \begin{Bmatrix} j_1 & j_2 & j \\ j_2' & j_1' & k \end{Bmatrix} \sum_{\eta''} \left\langle \eta, j_1 \middle\| \mathbf{T}^k \middle\| \eta'', j_1' \right\rangle \left\langle \eta'', j_2 \middle\| \mathbf{U}^k \middle\| \eta', j'_2 \right\rangle.$$

Rather than express **X** as $\left[\mathbf{T}^k \otimes \mathbf{U}^k \right]_0^0$, it is more transparent to express it as $\mathbf{T}^k \cdot \mathbf{U}^k$

$$\left\langle \eta, j_1, j_2, j, m \middle| \mathbf{T}^k \cdot \mathbf{U}^k \middle| \eta', j_1', j_2', j', m' \right\rangle = \delta_{j,j'} \delta_{m,m'} (-1)^{j_1'+j_2+j} \begin{Bmatrix} j_1 & j_2 & j \\ j_2' & j_1' & k \end{Bmatrix}$$

$$\times \sum_{\eta''} \left\langle \eta, j_1 \middle\| \mathbf{T}^k \middle\| \eta'', j_1' \right\rangle \left\langle \eta'', j_2 \middle\| \mathbf{U}^k \middle\| \eta', j_2' \right\rangle.$$

A major simplification occurs when $\mathbf{X}_q^k$ operates only on the variables in *either* space 1 or space 2. If **X** operates only on variables in space 1, **X** behaves as $\mathbf{T}_{q_1}^{k_1}$, $\mathbf{X}_q^k = \mathbf{T}_{q_1}^{k_1}$, and $\mathbf{U}_{q_2}^{k_2}$ is a constant with respect to **X**, thus

$$\mathbf{U}_{q_2}^{k_2} = \mathbf{U}_0^0$$

$$\left\langle \eta, j_1, j_2, j \middle\| \mathrm{T}^{k_1} \middle\| \eta', j_1', j_2', j' \right\rangle$$

$$= \delta_{j_2,j_2'} (-1)^{j_1+j_2+j'+k_1} \left[(2j'+1)(2j+1) \right]^{1/2} \begin{Bmatrix} j_1 & j & j_2 \\ j' & j_1' & k_1 \end{Bmatrix} \left\langle \eta, j \middle\| \mathrm{T}^{k_1} \middle\| \eta', j_1' \right\rangle,$$

or, if **X** operates only on variables in space 2, $\mathbf{X}_q^k = \mathbf{U}_{q_2}^{k_2}$, then

$$\left\langle \eta, j_1, j_2, j \middle\| \mathbf{U}^{k_2} \middle\| \eta', j_1', j_2', j' \right\rangle = \delta_{j_1,j_1'} (-1)^{j_1+j_2'+j+k} \left[(2j'+1)(2j+1) \right]^{1/2} \begin{Bmatrix} j_2 & j & j_1 \\ j' & j_2' & k_2 \end{Bmatrix} \left\langle \eta, j_2 \middle\| \mathbf{U}^{k_2} \middle\| \eta', j_2' \right\rangle.$$

An Example:
Consider an example, the spin-orbit interaction for a two-electron configuration:

$$\mathbf{H}^{\mathrm{SO}} = \xi(\mathrm{r}_1) \boldsymbol{\ell}_1 \cdot \mathbf{s}_1 + \xi(\mathrm{r}_2) \boldsymbol{\ell}_2 \cdot \mathbf{s}_2$$

Let

$$\zeta_{n\ell;n'\ell'} \equiv \int_0^\infty R_{n\ell}(r) \xi(r) R_{n'\ell'}(r) r^2 dr$$

$$\left\langle {}^{2S+1}L_J M_J \left| \mathbf{H}^{\mathrm{SO}} \right| {}^{2S'+1}L'_{J'} M'_J \right\rangle = \zeta_{n_1\ell_1;n'_1\ell'_1}\delta_{JJ'}\delta_{M_J,M'_J}(-1)^{L'+S+J}\begin{Bmatrix} L & S & J \\ S' & L' & 1 \end{Bmatrix}\left\langle L\left\|\ell_1\right\|L'\right\rangle\left\langle S\left\|s_1\right\|S'\right\rangle$$

$$+\zeta_{n_2\ell_2;n'_2\ell'_2}\delta_{JJ'}\delta_{M_J,M'_J}(-1)^{L'+S+J}\begin{Bmatrix} L & S & J \\ S' & L' & 1 \end{Bmatrix}\left\langle L\left\|\ell_2\right\|L'\right\rangle\left\langle S\left\|s_2\right\|S'\right\rangle.$$

Now to evaluate the reduced matrix elements:

$$\left\langle \ell_1\ell_2 L\left\|\ell_1\right\|\ell'_1\ell'_1 L'\right\rangle = \left\langle L\left\|\ell_1\right\|L'\right\rangle \equiv \delta_{\ell_2,\ell'_2}(-1)^{\ell_1+\ell_2+L'+1}\left[(2L'+1)(2L+1)\right]^{1/2}\begin{Bmatrix} \ell_1 & L & \ell_2 \\ L' & \ell'_1 & 1 \end{Bmatrix}\left\langle \ell_1\left\|\ell_1\right\|\ell_1{}'\right\rangle$$

and similarly for $\left\langle L\left\|\ell_2\right\|L'\right\rangle$.

$$\left\langle s_1 s_2 S\left\|s_1\right\|s'_1 s'_2 S'\right\rangle = \left\langle S\left\|s_1\right\|S'\right\rangle \equiv \delta_{s_2,s'_2}(-1)^{S_1+S_2+S'+1}\left[(2S'+1)(2S+1)\right]^{1/2}\begin{Bmatrix} s_1 & \delta & s_2 \\ s' & s'_1 & 1 \end{Bmatrix}\left\langle s_1\left\|S_1\right\|s'\right\rangle$$

and similarly for $\left\langle S\left\|s_2\right\|S'\right\rangle$.

We know how to evaluate the reduced matrix elements

$$\left\langle L'\left\|L^{(1)}\right\|L\right\rangle = \left[(2L+1)L(L+1)\right]^{1/2}\delta_{L'L}$$
$$\left\langle S'\left\|S^{(1)}\right\|S\right\rangle = \left[(2S+1)S(S+1)\right]^{1/2}\delta_{S'S}$$

This enables us to evaluate the $\mathbf{H}^{\mathrm{SO}}$ matrix element

$$\left\langle {}^{2S+1}L_J, M_J \left\|\mathbf{H}^{\mathrm{SO}}\right\| {}^{2S'+1}L'_{J'} M'_{J'}\right\rangle = \zeta_{n_1,\ell_1,n'_1\ell'_1}\delta_{J,J'}\delta_{M_J,M'_J}\delta_{\ell_1,\ell'_1}\delta_{\ell_2,\ell'_2}(-1)^{S+S'+J+\ell_1+\ell_2+1}$$
$$\times\left[(2L'+1)(2L+1)(2S'+1)(2S+1)(2\ell_1+1)\ell_1(\ell_1+1)(3/2)\right]^{1/2}$$
$$\times\begin{Bmatrix} L & S & J \\ S' & L' & 1 \end{Bmatrix}\begin{Bmatrix} \ell_1 & L & \ell_2 \\ L' & \ell_1 & 1 \end{Bmatrix}\begin{Bmatrix} \frac{1}{2} & S & \frac{1}{2} \\ S' & \frac{1}{2} & 1 \end{Bmatrix}$$

and a similar term for $\eta_2\ell_2 s_2$.

Many important conclusions may be drawn for $\mathbf{H}^{\mathrm{SO}}$ expressed in this rigorous and universal way:

1. Selection rules: $\Delta S = 0, \pm 1$, $\Delta L = 0, \pm 1$

2. Within a ^{2S+1}L multiplet state

$$E(^{2S+1}L_J) - E(^{2S+1}L_{J-1}) = AJ \text{ (A is the spin-orbit constant),}$$

which is the Landé interval rule.

3. When either L = 0 or S = 0, there is no spin-orbit splitting
4. Spin-orbit perturbations between two L–S–J states follow a $\Delta J = 0$ selection rule.
5. A single $\zeta_{n\ell}$ parameter characterizes all of the spin-orbit splittings and off-diagonal matrix elements within a given $(n\ell)^p$ configuration.

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